Marked groups, kernels and intersections

Abstract

Many constructions of incidence geometries require the same two tasks: control the intersections of subgroups generated by selected markers, and identify the permutation group acting on the resulting objects. We give a common treatment in which a quotient supplies the visible geometry and its kernel records the remaining intersection obstructions. The main construction assembles local extensions over specified sets of markers. Injectivity on every proper local parabolic is enough to isolate the entire local kernels, even when their supports are nested. It follows that the full fiber product is generated by the prescribed markers and has an exact kernel profile on every parabolic. We derive local recognition, deletion maps, lower-central identities and separator factorizations from the same proof. Exact coordinate profiles also transfer typed coset Helly; with the necessary involutory local markings this gives regular geometries. A finite nilpotent quotient and an invariant subgroup ladder then give a complete parameter space for marked finite two-group quotients with a prescribed class-two quotient. Further results treat exact quotient intersections, comparison through one exceptional parabolic, faithful actions on retained objects, relative generation and Frattini coordinates. These results are intended to be cited with explicit hypothesis bindings; local presentations, geometric identifications and finite classifications remain the responsibility of their applications.

1. Markings, images and kernels

A marked group is a group G together with a finite indexed generating family $(r_i)_{i\in I}$. Except where stated, the markers need not be involutions or distinct. For $S \subseteq I$, put

$$G_S = \langle r_i : i \in S\rangle, \qquad G_\emptyset = 1.$$

We call these subgroups parabolics, retaining their index sets even when two of them happen to coincide. The marking has the intersection property, or IP, if

$$G_S \cap G_T = G_{S\cap T} \quad (S, T \subseteq I). \tag{1.1}$$

A marked homomorphism sends each source marker to the corresponding target marker. Its image on G_S is therefore the target parabolic with index set S.

All subgroup products below are products of sets unless a subgroup assertion is made. We write $\gamma_1(G) = G$ and $\gamma_{j+1}(G) = [\gamma_j(G), G]$. We use the elementary correspondence and isomorphism theorems for groups. In the finite p-group arguments we retain Burnside's basis theorem, the nilpotence of finite p-groups, and the usual multilinear commutator description of the lower-central graded Lie ring. Precisely, its degree-one part generates the graded Lie ring, and degree j is generated by left-normed commutators of weight j in any generating set. In Section 8 we explicitly retain the normal-form theorem for an amalgamated free product. None of these classical inputs is presented as a result originating in this manuscript.

Lemma U.K0 (lifting one intersection). Let $q : H \to Q$ be a homomorphism and let $C \leq A \cap B$, where $A, B, C \leq H$. If

$$q(A) \cap q(B) = q(C),$$

then

$$A \cap B = C \quad \iff \quad A \cap B \cap \ker q \leq C. \tag{1.2}$$

Proof. The forward implication is immediate. For the converse, let $x \in A \cap B$. Its image belongs to $q(C)$, so choose $c \in C$ with $q(c) = q(x)$. Then $xc^{-1} \in A \cap B \cap \ker q$, which is contained in C by hypothesis. Multiplying by c gives $x \in C$. The inclusion $C \leq A \cap B$ gives equality. □

Theorem U.K (kernel profiles). Let $q : H \twoheadrightarrow Q$ be a marked epimorphism and suppose that the marking of Q has IP. Write $K = \ker q$ and $K_S = K \cap H_S$. Then H has IP if and only if

$$K_S \cap K_T = K_{S\cap T} \quad (S, T \subseteq I). \tag{1.3}$$

Proof. Apply U.K0 to $A = H_S$, $B = H_T$, and $C = H_{S\cap T}$. The image hypothesis is exactly IP in Q. The subgroup on the left of the containment in (1.2) is $K_S \cap K_T$. Its containment in $H_{S\cap T}$ is equivalent to its containment in $K_{S\cap T}$; the reverse inclusion always holds. This proves each required intersection and its converse. □

The next result concerns the opposite direction. Passing to a quotient can create new intersections, and this defect is a product condition in its kernel.

Theorem U.Q (exact intersections of images). Let $q : H \to Q$ be a homomorphism with kernel N, and let $A, B \leq H$. Then

$$q(A) \cap q(B) = q(A \cap B) \quad \iff \quad N \cap AB = (N \cap A)(N \cap B). \tag{1.4}$$

If q is injective on A and B, the right side reduces to $N \cap AB = \{1\}$. If $N \leq A$, the equality always holds.

Proof. Suppose the image intersection is exact, and take $ab \in N$, where $a \in A$ and $b \in B$. Since $q(a) = q(b)^{-1}$, choose $c \in A \cap B$ with this image. Then

$$ab = (ac^{-1})(cb) \in (N \cap A)(N \cap B).$$

The reverse inclusion requires no hypothesis. Conversely, if $q(a) = q(b)$, then $ab^{-1} \in N \cap AB$. Write $ab^{-1} = uv$, with $u \in N \cap A$ and $v \in N \cap B$. The element $u^{-1}a = vb$ lies in $A \cap B$ and has image $q(a)$, proving the required image equality. Under injectivity both factors on the right of (1.4) are trivial. If $N \leq A$, both sides of that equality are N. □

Proposition U.Coset (exact coset transport). Let $(H_i)_{i\in I}$ be subgroups of G, let $B = \bigcap_i H_i$, and let $N \trianglelefteq G$ satisfy $N \leq B$. Quotienting by N gives a bijection

$$H_i g \longmapsto (H_i/N)(gN) \tag{1.5}$$

on the typed right cosets. It preserves the intersections of arbitrary families of those cosets, including the existence of a common element. It consequently preserves incidence, flags, residues and all properties determined by the resulting incidence system. The actions of G and G/N have the same permutation image. Maximality of any inclusion between intersections of the H_i is preserved.

Proof. The inverse image of the coset on the right of (1.5) is exactly $H_i g$. Thus two source cosets give the same target coset precisely when they were equal, and taking inverse images commutes with all intersections. Surjectivity of the quotient map supplies a source representative of each target coset and a common lift whenever a target intersection is nonempty. This proves the assertions about incidence and common elements. The quotient action is visibly the same action on the identified coset set. Every subgroup in any interval under consideration contains N; the subgroup correspondence therefore preserves that interval and its maximal inclusions. □

2. Deletion maps and supported extensions

A deletion for $S \subseteq I$ is an endomorphism $e_S : G \to G$ satisfying $e_S(r_i) = r_i$ for $i \in S$ and $e_S(r_i) = 1$ otherwise. Requiring these values matters: a general retraction onto G_S need not be a deletion.

Proposition U.D (Boolean deletions). If a marked group has a deletion e_S for every $S \subseteq I$, then

$$e_S e_T = e_{S\cap T}, \qquad \operatorname{im} e_S = \operatorname{Fix}(e_S) = G_S. \tag{2.1}$$

The marking has IP, and

$$\ker e_S = \langle\langle r_i : i \notin S \rangle\rangle_G, \qquad G = (\ker e_S) \rtimes G_S. \tag{2.2}$$

Proof. The two maps in the first equality agree on every generator. The image is generated by precisely the retained markers, and each of those markers is fixed; this proves the image and fixed-point statements. An element of $G_S \cap G_T$ is fixed by $e_S e_T = e_{S\cap T}$, so belongs to $G_{S\cap T}$. The reverse inclusion is immediate.

Let M be the normal closure in (2.2). Clearly $M \le \ker e_S$. In G/M, every omitted marker is trivial, so the image of any word equals the image of its deletion. A word in $\ker e_S$ is therefore trivial in G/M, proving equality. Finally, for $g \in G$, the factorization $g = (g e_S(g)^{-1}) e_S(g)$ has factors in $\ker e_S$ and G_S. Their intersection is trivial because e_S fixes G_S. This is the asserted semidirect product. □

We now assemble extensions whose extra kernel is invisible on each proper part of its support. The groups and kernels may be infinite and nonabelian.

Theorem U.SK (supported-kernel realization). Let I be finite and let $\mathcal{F}$ be a finite family of distinct subsets of I. Let $B = \langle b_i : i \in I \rangle$ have IP. For every $S \in \mathcal{F}$, suppose the following data are given:

1. A deletion homomorphism $p_S : B \to B_S$, fixing b_i for $i \in S$ and killing b_i otherwise.
2. A group $E_S = \langle e_{S,i} : i \in S \rangle$ and a marked epimorphism $q_S : E_S \to B_S$.
3. Injectivity of q_S on $(E_S)_T = \langle e_{S,i} : i \in T \rangle$ for every $T \subsetneq S$.

Put $N_S = \ker q_S$, and form the group

$$H = \left\{ (b, (x_S)_{S\in\mathcal{F}}) \in B \times \prod_{S\in\mathcal{F}} E_S : q_S(x_S) = p_S(b) \text{ for every } S \right\}. \tag{2.3}$$

For $i \in I$, define h_i to have base coordinate b_i and S-coordinate $e_{S,i}$ when $i \in S$, and 1 otherwise. These elements generate all of H. If $\pi : H \to B$ is the base projection and $H_L = \langle h_i : i \in L \rangle$, then, for every $L \subseteq I$,

$$\pi(H_L) = B_L, \qquad \ker\pi = \prod_{S\in\mathcal{F}} N_S, \qquad \ker\pi \cap H_L = \prod_{\substack{S\in\mathcal{F}\\ S\subseteq L}} N_S. \tag{2.4}$$

The factors on the right are embedded as pure coordinate subgroups. The marking of H has IP. If B and all E_S are finite, then

$$|H_L| = |B_L| \prod_{\substack{S\in\mathcal{F}\\ S\subseteq L}} |N_S|. \tag{2.5}$$

Proof. The compatibility equations in (2.3) are preserved by products and inverses, so H is a subgroup of the displayed direct product. The prescribed tuple h_i satisfies those equations by the literal deletion rule. Let H' be the subgroup generated by these tuples. Its base image is B, and the projection of $H'_S = \langle h_i : i \in S \rangle$ onto E_S is onto.

We first prove, by induction on inclusion among the finitely many supports, that every pure coordinate element of N_S belongs to H'_S. Fix $n \in N_S$. Choose a word on the markers indexed by S which evaluates to n in E_S, and evaluate the same word in H', obtaining $w \in H'_S$. Its base coordinate lies in B_S and is $q_S(n) = 1$.

Consider its coordinate at another support T. Because the word uses only markers from S, that coordinate lies in $(E_T)_{S\cap T}$. Its image under q_T is $p_T(1) = 1$. If $T \not\subseteq S$, then $S \cap T \subsetneq T$, and the proper-parabolic injectivity hypothesis forces this coordinate to be 1. The only possible nonidentity coordinates of w are consequently at supports $T \subseteq S$. Its S-coordinate is the prescribed n, while every coordinate at a proper support $T \subsetneq S$ belongs to N_T.

For each such proper support, the induction hypothesis supplies the pure coordinate element having exactly that value in N_T, as an element of $H'_T \leq H'_S$. Multiply w by the inverses of those elements. Distinct pure coordinate subgroups commute in the ambient direct product, so these operations remove the unwanted coordinates without changing the S-coordinate. The result is the pure element n in coordinate S, and remains in H'_S. A minimal support has no smaller coordinates to remove, so this argument includes the initial step. No commutativity within an individual kernel has been used.

The kernel of π in the full group H is exactly the direct product of the N_S, by (2.3). The induction places that entire kernel in H'. Since H' already maps onto B, every $h \in H$ has the same base coordinate as some $h' \in H'$; then $h(h')^{-1} \in \ker \pi \leq H'$. Thus $H' = H$.

The image equality $\pi(H_L) = B_L$ follows on generators. If $x \in H_L \cap \ker \pi$, its S-coordinate lies in $(E_S)_{L \cap S}$ and in N_S. This coordinate is trivial whenever $S \not\subseteq L$, again by proper-parabolic injectivity. Conversely, when $S \subseteq L$, the isolation argument placed the entire pure N_S in $H_S \leq H_L$. This proves all three assertions in (2.4).

The coordinate products in (2.4) satisfy

$$\left(\prod_{S \subseteq L} N_S\right) \cap \left(\prod_{S \subseteq M} N_S\right) = \prod_{S \subseteq L \cap M} N_S.$$

Theorem U.K now proves IP in H. More explicitly, an element of $H_L \cap H_M$ has base in $B_{L \cap M}$; subtracting a lift from $H_{L \cap M}$ leaves an element in both displayed kernel products, hence in the kernel product for $L \cap M$. This also shows exactly where IP of the base is used. Finally (2.5) is the order formula for the surjection $H_L \to B_L$ with the kernel just computed. □

Corollary U.SK.Local (recognition on an antichain). Under U.SK, if $\mathcal{F}$ is an antichain under inclusion, then the coordinate projection $H_S \to E_S$ is a marked isomorphism for every $S \in \mathcal{F}$.

Proof. It is onto on generators. If an element maps to 1, its base coordinate also vanishes, since the base value of a word on S is the q_S-image of its S-coordinate. Formula (2.4) says that the remaining kernel in H_S has only the factor N_S: the antichain excludes every other support contained in S. The S-coordinate detects that factor, so the element is 1. □

Corollary U.SK.Delete (derived deletions). If, in addition, B has deletions for every subset of I, then every E_S and H has all its Boolean deletions.

Proof. For $T \subsetneq S$, the restriction $q_S : (E_S)_T \to B_T$ is a marked isomorphism: it is onto on generators and injective by hypothesis. Compose q_S, the deletion $B_S \to B_T$, and the inverse of this isomorphism. This gives a homomorphism $E_S \to (E_S)_T$ that fixes exactly the T-markers and kills the others. For $T = S$ use the identity. Proposition U.D supplies the Boolean identities.

For $L \subseteq I$, apply the base deletion e_L and, in coordinate S, the local deletion onto $S \cap L$. The two sides of each compatibility equation agree on the local markers; the base deletions commute by U.D. Thus the coordinatewise operation preserves (2.3). On h_i it fixes h_i if $i \in L$, and kills it otherwise. It is the required deletion of H. □

Corollary U.SK.Central (lower-central kernel). Under U.SK, suppose $c \geq 1$, $\gamma_c(B) = 1$, and $N_S = \gamma_c(E_S)$ for every support. Then

$$\gamma_c(H) = \ker \pi. \tag{2.6}$$

Proof. Applying π to a commutator of weight c gives 1, so $\gamma_c(H) \leq \ker \pi$. For the converse, repeat the inclusion induction in the proof of U.SK. The projection $H_S \to E_S$ is onto; therefore its image on $\gamma_c(H_S)$ is $\gamma_c(E_S) = N_S$. Given $n \in N_S$, choose a lift $w \in \gamma_c(H_S)$. Its base is 1, since $\gamma_c(B) = 1$. The same proper-support argument makes all coordinates at $T \not\subseteq S$ trivial. The remaining proper-support

coordinates lie in N_T; by induction their pure copies belong to $\gamma_c(H)$. Removing them leaves the pure n, still in $\gamma_c(H)$. All pure kernels are obtained, proving (2.6). □

Corollary U.SK.Separator (factorization across a separator). Under U.SK, let $I = L \cup M$. Suppose $B = B_L B_M$ as a product of sets, and every support in $\mathcal{F}$ is contained in L or in M. Then $H = H_L H_M$.

Proof. For $h \in H$, factor $\pi(h) = b_L b_M$ and choose lifts $u_L \in H_L$, $u_M \in H_M$. Then $h = n u_L u_M$ for some $n \in \ker \pi$. Assign every pure coordinate factor of n to one of the two sides containing its support. Formula (2.4) gives $n = n_L n_M$, with $n_L \in H_L$, $n_M \in H_M$. Each pure N_S is normal in H, because conjugation in the ambient product acts coordinatewise and $N_S \trianglelefteq E_S$. Hence the product of the assigned M-factors is normal in H. Moving n_M past u_L gives

$$h = (n_L u_L)(u_L^{-1} n_M u_L \, u_M) \in H_L H_M.$$

The opposite inclusion is automatic. □

The hypotheses of U.SK do not identify H with a universal amalgam of the E_S. Additional cross-relations and a presentation-recognition argument may be necessary. Nor can injectivity on every proper parabolic be replaced by injectivity on individual markers alone.

Proposition U.Fiber (two compatible split factors). Let A and B be marked groups on finite label sets L and M, and put $J = L \cap M$. Suppose their J-parabolics are identified with the same marked group C, and that $p_A : A \to C$ and $p_B : B \to C$ are deletions. Then

$$A \times_C B \cong (\ker p_A \times \ker p_B) \rtimes C. \tag{2.7}$$

This full fiber product is generated by the tuples $(a_i, 1)$ for $i \in L \setminus J$, $(1, b_i)$ for $i \in M \setminus J$, and (c_i, c_i) for $i \in J$. The two original groups embed through $a \mapsto (a, p_A(a))$ and $b \mapsto (p_B(b), b)$. If finite, its order is $|A||B|/|C|$.

Proof. Every compatible pair has a unique form $(n_A c, n_B c)$, giving the split decomposition and the order formula. The stated copies of A and B are generated by subsets of the stated tuple marking. They contain $(n_A, 1)$ and $(1, n_B)$, respectively, while either contains the diagonal copy of C. These subgroups generate the split product, proving the assertion. The embedding formulas are injective because their first and second projections, respectively, are identities. □

3. Finite two-group quotients with a fixed class-two shadow

The next theorem separates a universal finite search space from the task of proving that a particular family has the required class-two quotient. That identification is an application hypothesis, not a consequence of the search.

Theorem U.Nilpotent (complete marked kernel parameter space). Let $F = \langle x_i : i \in I \rangle$, where $|I| = d$ and $x_i^2 = 1$. Suppose F has all Boolean deletions. Put $Q = F/\gamma_3(F)$, with its image marking. For $k \geq 0$, set

$$U = F/\gamma_{k+3}(F), \qquad K = \gamma_3(U), \qquad K_S = K \cap U_S. \tag{3.1}$$

Then U is a finite two-group,

$$|U| \leq 2^{\sum_{j=1}^{k+2} d^j}, \tag{3.2}$$

the markings of U and Q have Boolean deletions and IP, and U/K is canonically the marked group Q. For every $S \subseteq I$,

$$K_S = \gamma_3(U_S). \tag{3.3}$$

There is a bijection between the following two sets:

1. Ordered marked isomorphism classes of finite two-group quotients $f : F \twoheadrightarrow G$ for which the induced marked map $Q \to G/\gamma_3(G)$ is an isomorphism, $|G| = |Q|2^k$, and G has IP.

2. Normal subgroups $N \trianglelefteq U$ satisfying $N \le K$, $[K : N] = 2^k$, and

$$\frac{NK_S}{N} \cap \frac{NK_T}{N} = \frac{NK_{S\cap T}}{N} \quad (S, T \subseteq I). \tag{3.4}$$

The correspondence sends N to U/N with the original image markers. If $D \le \mathrm{Sym}(I)$ is specified and each $\sigma \in D$ induces an automorphism $x_i \mapsto x_{\sigma(i)}$ of F, then equivalence allowing precisely these label permutations is the D-orbit relation on the kernels in item 2.

Proof. The degree-one part of the lower-central graded Lie ring of F is generated by elements of exponent dividing two. By multilinearity, every iterated commutator in those generators also has exponent dividing two in its homogeneous component. In degree j, at most d^j left-normed commutators suffice to generate that component. Consequently

$$|\gamma_j(F)/\gamma_{j+1}(F)| \le 2^{d^j}.$$

Multiplying these bounds for $1 \le j \le k+2$ proves finiteness, the two-power order and (3.2). Each deletion preserves every lower-central term, and therefore descends both to U and to Q. Proposition U.D gives IP for both image markings; no separate IP hypothesis on Q is needed. The epimorphic-image property of the lower central series gives $U/K \cong F/\gamma_3(F) = Q$.

For (3.3), one inclusion follows from $U_S \le U$. If $z \in \gamma_3(U) \cap U_S$, the deletion e_S fixes z and sends $\gamma_3(U)$ into $\gamma_3(U_S)$. This proves the reverse inclusion.

Let G satisfy item 1. Its subgroup $\gamma_3(G)$ has order 2^k. In a nilpotent group, equality between two successive nontrivial lower-central terms would persist at every later term, contradicting termination of that series. Thus each nontrivial step beginning at $\gamma_3(G)$ strictly decreases its order. There are at most k such steps, and $\gamma_{k+3}(G) = 1$. The map $F \to G$ therefore factors through U. Its kernel N lies in K, since the induced map $U/K \to G/\gamma_3(G)$ is an isomorphism, and the orders give $[K : N] = 2^k$.

Conversely, let $N \trianglelefteq U$ with $N \le K$ and $[K : N] = 2^k$, and write $G = U/N$. Its third lower-central subgroup is K/N, its canonical class-two quotient is Q, and its order is $|Q|2^k$. If $q : U \to G$, then

$$(K/N) \cap G_S = q(K \cap U_S) = NK_S/N. \tag{3.5}$$

Indeed, $q(u) \in K/N$ with $u \in U_S$ implies $u \in K$, because $q^{-1}(K/N) = K$. The reverse inclusion is immediate. Applying U.K to $G \to Q$ proves that IP in G is equivalent to precisely (3.4).

An isomorphism preserving each ordered marker commutes with the two maps from U, because it agrees with them on a generating family. Their kernels are consequently equal. Equal kernels plainly give the same marked quotient. Finally, every specified label permutation acts on F, preserves its characteristic lower-central terms, and hence acts on U and on the displayed kernels. A marker-permuting isomorphism induces the corresponding kernel equality after this automorphism, and the converse descends that automorphism to the quotients. This proves the orbit assertion at exactly the specified equivalence scope. □

Theorem U.Ladder (invariant index-two descent). Let U be a finite two-group and $M \trianglelefteq U$. The U-normal subgroups $L \le M$ of index two are exactly the inverse images of the hyperplanes of

$$M/(\Phi(M)[M, U]). \tag{3.6}$$

For $K \trianglelefteq U$, every U-normal subgroup $N \le K$ of index 2^k is obtained by k successive such steps.

Proof. An index-two subgroup L of M contains $\Phi(M)$ and has quotient C_2. If L is U-normal, conjugation by U on M/L is trivial, since $\mathrm{Aut}(C_2) = 1$. Thus $[M, U] \le L$. Conversely, a hyperplane inverse image has index two and contains $[M, U]$; for $l \in L$ and $u \in U$, the conjugate of l differs from l by an element of $[M, U]$, so lies in L. The quotient in (3.6) is elementary abelian by Burnside's basis theorem, proving the first assertion.

Suppose $N \trianglelefteq U$ and $N < M$. The nontrivial finite two-group M/N has a nonzero Frattini quotient V. Conjugation by U/N acts on the nonzero linear maps $V \to \mathbb{F}_2$. If $\dim V = a$, there are $2^a - 1$ such maps, an odd number. Every orbit has two-power size, so at least one orbit has size one. The kernel of a fixed nonzero map lifts to a U-normal subgroup between N and M, of index two in M. Repeating with the new subgroup gives a chain terminating at N; its length is k, since the total index is 2^k. □

Lemma U.Order (presentation recognition by a sharp bound). Let $A_1, \dots, A_t$ be finite normal subgroups of G, with $|A_j| \le a_j$. Suppose $[G : A_1 \cdots A_t] \le b$. Then $|G| \le b \prod_j a_j$. If a finite quotient of G has this latter order, its quotient map is an isomorphism.

Proof. Normality makes each partial product a subgroup. The multiplication map from $A_1 \times \cdots \times A_t$ onto the product is surjective as a map of sets, so that product has order at most $\prod_j a_j$. Multiplication by the quotient index proves the bound and finiteness of G. A quotient attaining the bound forces equality of the finite orders and hence a trivial kernel. □

4. Coset comparison and coordinate-kernel geometry

For subgroups $P_i \le G$, typed coset Helly means that every pairwise intersecting family of right cosets $P_i g_i$, with distinct indices, has a common element. This property is separate from IP. It supplies a common representative for an incident flag, whereas IP determines the stabilizer of a flag already represented in this way.

Lemma U.Helly (anchored criterion). A finite indexed subgroup family $(P_i)_{i \in I}$ has typed coset Helly if and only if the following condition holds. For every $i \in I$, every $J \subseteq I \setminus \{i\}$ with $|J| \ge 2$, and any selected $j_0 \in J$, each $g \in P_{j_0}$ satisfying $g \in P_i P_j$ for all $j \in J$ admits $k \in \bigcap_{j \in J} P_j$ with $P_i g = P_i k$.

Proof. The cosets $P_i g$ and P_j intersect precisely when $g \in P_i P_j$. Thus Helly gives a common element k, proving necessity. Conversely, induct on the number of cosets in a pairwise intersecting family. A family of at most two already has a common element. For a larger family, the induction hypothesis gives a common element of all but one coset. Right multiplication by its inverse turns those cosets into the base cosets P_j. The remaining coset meets P_{j_0}, so it has a representative $g \in P_{j_0}$. Its intersections with the other base cosets are precisely the product conditions in the statement. The criterion supplies k, and undoing the right translation gives a common element of the original family. □

Theorem U.Compare (all-parabolic comparison). Let $G = \langle r_i : i \in I \rangle$, where I is finite and $|I| \ge 3$. Put $P_i = G_{I \setminus \{i\}}$, fix $e \in I$, and set

$$\mathcal{B} = \bigcup_{i \ne e} P_i, \qquad \mathcal{W} = \mathcal{B}\mathcal{B}. \tag{4.1}$$

Let $q : G \twoheadrightarrow H$ be a marked epimorphism. Suppose q is injective on the set $\mathcal{W}$, and, for every $w \in \mathcal{W}$,

$$q(w) \in q(P_e) \quad \Longleftrightarrow \quad w \in P_e. \tag{4.2}$$

Then the marking of G has IP if and only if that of H has IP. Independently, the maximal-parabolic cosets of G have typed coset Helly if and only if those of H do.

Proof. The set $\mathcal{B}$ contains 1, is closed under inverses, and is contained in $\mathcal{W}$. If $S \subsetneq I$ and $S \ne I \setminus \{e\}$, then S omits a nonexceptional index, so $G_S \subseteq \mathcal{B}$. It follows that, for $b \in \mathcal{B}$ and every proper S,

$$q(b) \in q(G_S) \quad \Longleftrightarrow \quad b \in G_S. \tag{4.3}$$

For the exceptional index set this is (4.2). Otherwise, lift the image membership to $a \in G_S \subseteq \mathcal{B}$; injectivity on $\mathcal{W}$ and $q(a) = q(b)$ imply $a = b$.

For two proper index sets S, T, at least one corresponding subgroup is contained in $\mathcal{B}$, unless both index sets equal $I \setminus \{e\}$. In the former case, lift an element of $q(G_S) \cap q(G_T)$ through that protected

subgroup, and use (4.3) to put the lift in the other subgroup. Thus q identifies the actual intersection bijectively with the intersection of the images, and also identifies $G_{S\cap T}$ with its image. Equality of these two subgroups is equivalent upstairs and downstairs. The remaining self-intersection is tautological, as is any pair involving $G_I = G$. This proves the IP equivalence without making any assumption about distinct index sets generating distinct subgroups.

The same argument gives two facts for Helly. First, whenever J contains a protected index $j_0 \neq e$, restriction of q is a bijection

$$\bigcap_{j\in J} P_j \longrightarrow \bigcap_{j\in J} q(P_j). \tag{4.4}$$

Lift through P_{j_0} and apply (4.3) for each remaining membership. Second, if $g \in \mathcal{B}$ and $i \neq j$, then

$$q(g) \in q(P_i)q(P_j) \quad \Longleftrightarrow \quad g \in P_iP_j. \tag{4.5}$$

When both indices are protected, a factorization downstairs lifts to $ab \in \mathcal{W}$; comparison with $g \in \mathcal{W}$ gives equality. When $i = e$, choose the protected second factor $b \in P_j$. The remaining membership test is $q(gb^{-1}) \in q(P_e)$, and $gb^{-1} \in \mathcal{W}$, so (4.2) applies. When $j = e$, multiply by the inverse of a protected first factor on the left. The reverse implication in all cases follows simply by applying q.

Apply U.Helly in the two groups. Every set J appearing there has at least two indices, hence contains a protected j_0. Every candidate representative in $q(P_{j_0})$ lifts uniquely to $g \in P_{j_0}$. Equation (4.5) preserves all its eligibility conditions, while (4.4) identifies the possible common representatives k. Finally,

$$P_ig = P_ik \quad \Longleftrightarrow \quad gk^{-1} \in P_i.$$

Here $g, k \in P_{j_0}$, so $gk^{-1} \in \mathcal{B}$, and (4.3) preserves this final test. Thus the anchored criterion has the same truth value upstairs and downstairs, proving the Helly equivalence. □

Corollary U.Compare.Matrix (bounded integral certificate). Suppose $G \leq \mathrm{GL}_m(\mathbb{Z})$, and let q_n be reduction modulo n. If all entries of all matrices in $\mathcal{W}$ have absolute value at most C, then q_n is injective on $\mathcal{W}$ for $n > 2C$. Suppose also that, for every $w \in \mathcal{W}$, a finite specified collection of integer tests has entries bounded by D, membership in $q_n(P_e)$ forces at least one complete test to vanish modulo n, and every integral passing case has a proved witness placing w in P_e. Then (4.2) holds for $n > D$.

Proof. Equal reductions of two matrices in $\mathcal{W}$ make every entry of their difference divisible by n and bounded by $2C$, hence zero. Similarly, a test vanishing modulo $n > D$ vanishes integrally. Its witness gives the forward implication in (4.2); the reverse implication is automatic. □

If several quotient maps satisfy U.Compare against the same source, no maps between the quotient groups are needed. One checked quotient determines IP and Helly in the source and in every such quotient. A finite protected product set still requires an exact membership test against the whole exceptional image, not a truncated search.

Proposition U.Geometry (coset recognition and faithful chamber action). Suppose the marking of G consists of nonidentity involutions, has IP, and has typed coset Helly for its maximal parabolics. The coset incidence system is then a thin, residually connected geometry, and G is its full type-preserving automorphism group, acting regularly on chambers. More generally, the full type-preserving automorphism group of any thin, residually connected geometry of finite rank acts freely on chambers. A chamber-transitive subgroup is consequently the full group.

Proof. In the coset system, a common coset point extends every flag to a chamber by choosing the coset of each missing type through that point. Right translation sends every such flag to its base flag. For a base flag of type set S, IP gives stabilizer $G_{I\setminus S}$. In particular the full chamber stabilizer is 1, while a panel stabilizer is a group generated by one nonidentity involution and has order two. There are therefore exactly two chambers through each panel.

Consider a residue with remaining type set T, where $|T| \geq 2$. Its stabilizer is G_T, and its maximal object stabilizers are $G_{T\setminus\{i\}}$, $i \in T$. Every marker indexed by T lies in at least one of these proper parabolics. A word in those subgroups supplies a walk between their translated base objects in the residue incidence graph; all objects are translates of base objects and all base objects lie in a common flag. The residue is therefore connected.

For the general assertion, first observe that every residually connected finite-rank geometry has a connected chamber graph. This follows by induction on rank: chambers through a fixed object are connected through galleries in its residue, and a path in the incidence graph, extending each consecutive incident pair to a chamber, connects arbitrary chambers through these subgraphs. The rank-one case holds because all chambers share the empty panel. In a thin geometry an automorphism fixing a chamber fixes its unique neighbor across each panel. It therefore fixes every chamber along every gallery, and hence every object. The chamber action is free. If a subgroup is chamber-transitive, any full automorphism agrees on a base chamber with one subgroup element; their quotient fixes that chamber and is the identity. Apply this to the coset geometry to obtain the stated full automorphism group. □

4.1. Helly transfer through coordinate kernels

Write right cosets as Pg, with the subgroup on the left. Let $q : H \to B$ be an epimorphism of groups generated by corresponding marked families $(h_i)_{i\in I}$ and $(b_i)_{i\in I}$, where I is finite. For $J \subseteq I$, write $H_J = \langle h_j : j \in J\rangle$ and $B_J = \langle b_j : j \in J\rangle$. In particular, $q(H_J) = B_J$ for every J.

Fix any set $\mathcal{J}$ of subsets of I as the permitted types. The maximal-parabolic case is $\mathcal{J} = \{I\setminus\{i\} : i \in I\}$.

Theorem U.SK.Helly (coordinate-kernel transfer). Suppose B has IP. Suppose $N = \ker q$ is an internal direct product $\prod_{\alpha\in A} N_\alpha$, with a finite factor set A, and for every $J \in \mathcal{J}$,

$$N \cap H_J = \prod_{\alpha\in A_J} N_\alpha \qquad \text{for some } A_J \subseteq A. \tag{4.6}$$

If the family $(B_J)_{J\in\mathcal{J}}$ has typed coset Helly, then so does $(H_J)_{J\in\mathcal{J}}$.

No commutativity within the factors, finiteness of H or B, or faithfulness of an action of B on N is assumed. The required intersection-image fact is $q(H_{J\cap K}) = B_J \cap B_K$; it follows here from the marked map and IP of B. IP of H is not an additional assumption for this transfer theorem.

Proof. Consider pairwise intersecting cosets $H_{J_i}g_i$. Their images $B_{J_i}q(g_i)$ intersect pairwise. Choose a common base element b by the assumed Helly property, and choose u in H with $q(u) = b$. Each set

$$F_i = H_{J_i}g_i \cap q^{-1}(b)$$

is nonempty: because $bq(g_i)^{-1} \in B_{J_i}$, a preimage of that element in H_{J_i}, multiplied on the right by g_i, belongs to F_i.

We first show that $F_i \cap F_j \neq \emptyset$. Choose $x \in H_{J_i}g_i \cap H_{J_j}g_j$ and put z=q(x). Both b and z belong to each of the two base cosets, so

$$bz^{-1} \in B_{J_i} \cap B_{J_j} = B_{J_i\cap J_j}.$$

The marked map is surjective on this intersection parabolic. Choose $h \in H_{J_i\cap J_j}$ with $q(h) = bz^{-1}$. Since h belongs to both H_{J_i} and H_{J_j}, left multiplication by h preserves both right cosets. Thus hx belongs to their intersection and has image b. This proves the assertion without assuming that q carries an arbitrary intersection onto the intersection of its images.

Choose $x_i \in F_i$ and put $k_i = x_iu^{-1} \in N$. If $y \in F_i$, then $yx_i^{-1} \in N \cap H_{J_i}$; the converse also holds. Hence

$$F_iu^{-1} = (N \cap H_{J_i})k_i. \tag{4.7}$$

Under the direct product coordinates of N, (4.6) makes (4.7) unrestricted in coordinates $\alpha \in A_{J_i}$, and fixes coordinate $\alpha \notin A_{J_i}$ to $(k_i)_\alpha$. Pairwise intersections imply that any two constraints fixing the same coordinate fix it to the same value. Choose that value in every constrained coordinate and the identity in every unconstrained coordinate. The resulting element belongs to all sets in (4.7). Multiplication on the right by u supplies a common element of the original cosets. This proves the theorem. □

Corollary U.SK.Geometry. In the supported-kernel realization U.SK, assume the base marking gives a regular hypertope and every marked generator in every attached group is an involution. Then the resulting marking also gives a regular hypertope, and its full type-preserving automorphism group is the constructed group.

Proof. U.SK gives IP and the exact profile

$$N \cap H_J = \prod_{S \subseteq J} N_S.$$

The base maximal-parabolic cosets have typed Helly. Theorem U.SK.Helly applies with factors indexed by the supports S. Each constructed generator is a tuple whose base coordinate and every nontrivial attached coordinate are involutions; hence its square is the identity. Its base image is nonidentity, so its order is exactly two. U.Geometry now gives thinness, residual connectedness, chamber regularity and equality with the full type-preserving automorphism group. In particular, this conclusion uses typed Helly in addition to IP. □

A sufficient alternative to local involutivity is that every support has size at least two: each singleton local parabolic is then proper, and injectivity into its nonidentity involutory base image forces the local marker to have order two. This is a sufficient condition, not an equivalent reformulation. Without either condition the geometric corollary fails. With a singleton type set and support, the quotient $C_4 \to C_2$ is injective on its only proper parabolic, the trivial subgroup. U.SK then constructs C_4, whose distinguished generator has order four. In rank three, adjoining two independent C_2 markers gives $C_4 \times C_2^2$ with the same defective singleton attachment and a four-element panel.

5. Retained objects and orbit multiplicities

The representation of a group by permutation matrices is exact: its regular action is faithful, and replacing permutations by their matrices preserves multiplication. For geometric applications one must also retain the distinguished parabolic partitions. A matrix realization alone does not establish the intersection or common-point properties of those partitions.

Theorem U.Action (faithful action after retaining objects). Let G act on sets X and Y. Suppose a G-invariant subset $Y_0 \subseteq Y$ has an equivariant surjection onto X. Suppose also that Y has an injective equivariant encoding in a tagged disjoint union of finite products, each factor of which is X or its power set, and that the tags are fixed by G. Then

$$\ker(G \curvearrowright Y) = \ker(G \curvearrowright X) = \bigcap_{x \in X} G_x. \tag{5.1}$$

If X is a disjoint union of transitive orbits with selected stabilizers H_j, the common kernel is

$$N = \bigcap_j \operatorname{core}_G(H_j) = \operatorname{core}_G\left(\bigcap_j H_j\right). \tag{5.2}$$

The faithful permutation image is G/N. If its action on Y is chamber-transitive on a thin, residually connected finite-rank geometry, then G/N is the full type-preserving automorphism group. The preimages in G of a chamber stabilizer and of a panel stabilizer are respectively N and a subgroup of index two over N.

Proof. An element fixing Y pointwise fixes X through the equivariant surjection. Conversely, an element fixing X fixes all subsets of X, every finite tuple formed from these objects, and all fixed tags. Injectivity of the encoding then forces it to fix Y pointwise. This proves (5.1). In a transitive orbit all point stabilizers are conjugates of the selected one, so intersection over all points gives the first equality in (5.2). The second is the intersection of the same conjugate subgroups in the opposite order. The first isomorphism theorem identifies the faithful image.

By U.Geometry that image has trivial chamber stabilizers and is the full type-preserving group. Its panel stabilizer acts transitively on the two chambers through the panel, with trivial stabilizer of either chamber, so has order two. Taking preimages proves the final assertion. □

Independently labeled copies of the same subset need extra data in their encoding; they cannot be silently treated as the subset itself.

Theorem U.Orbits (orbit data and normalizers). Let a finite group A act on a finite set Ω, and choose one representative H of each conjugacy class of subgroups that occurs as a point stabilizer. Let m_H be the number of orbits isomorphic to A/H, with $m_H = 0$ for all omitted classes. Then the numbers m_H determine the A-set up to equivariant isomorphism, and

$$\ker(A \curvearrowright \Omega) = \bigcap_{m_H>0} \operatorname{core}_A(H). \tag{5.3}$$

Its equivariant permutation group is abstractly

$$C_{\mathrm{Sym}(\Omega)}(A) \cong \prod_H ((N_A(H)/H) \wr \mathrm{Sym}(m_H)). \tag{5.4}$$

If the action $\rho : A \hookrightarrow \mathrm{Sym}(\Omega)$ is faithful, conjugation gives an exact sequence

$$1 \longrightarrow C_{\mathrm{Sym}(\Omega)}(\rho(A)) \longrightarrow N_{\mathrm{Sym}(\Omega)}(\rho(A)) \longrightarrow \{\alpha \in \mathrm{Aut}(A) : m_{\alpha(H)} = m_H \text{ for all } H\} \longrightarrow 1. \tag{5.5}$$

No splitting of this sequence is asserted.

Proof. Every orbit is equivariantly A/H, where H is the stabilizer of a selected point, and two such transitive sets are isomorphic exactly when the stabilizers are conjugate. This proves the classification by multiplicities and the kernel formula.

An equivariant permutation preserves the isomorphism type of each orbit. For one orbit A/H, an equivariant map is determined by the image of H. That image nH must be fixed by H, and bijectivity makes its stabilizer equal to H; hence $n \in N_A(H)$. Conversely each such n gives a permutation $aH \mapsto anH$. Composition corresponds to multiplication in the opposite order, so inversion identifies this permutation group with $N_A(H)/H$. For m_H copies, one may independently choose such a map in every copy and permute the copies, yielding the wreath product. Different isomorphism types are independent, proving (5.4).

Faithfulness makes conjugation by a normalizing permutation τ induce a unique automorphism α of A, with

$$\tau\rho(a)\tau^{-1} = \rho(\alpha(a)).$$

If a point has stabilizer H, its image under τ has stabilizer $\alpha(H)$; therefore the multiplicities must be invariant. The kernel of this conjugation map is the centralizer. Conversely, suppose α preserves the multiplicities. Pair each orbit with a target orbit whose stabilizer class is $\alpha(H)$, and choose base points x, y with actual stabilizers $H, \alpha(H)$, adjusting within the target orbit if needed. Define $\tau(ax) = \alpha(a)y$. The stabilizer equality makes this well-defined and bijective on that pair of orbits, and the paired definitions combine to a permutation of Ω. It satisfies the displayed intertwining equation, proving surjectivity and exactness. □

Proposition U.ExtensionOrbits (classification above a retained base). Let a group B act on sets Z, E, and let $p : Z \to E$ be equivariant. For one representative e of each orbit in $B\backslash E$, there is a natural bijection

$$B\backslash Z \;\cong\; \coprod_{e\in B\backslash E} B_e\backslash p^{-1}(e). \tag{5.6}$$

Proof. Translate any $z \in Z$ so that $p(z)$ is the selected representative of its orbit. If two translated points over that representative are B-equivalent, an element taking one to the other fixes their image e, so belongs to B_e. Conversely a B_e-equivalence is already a B-equivalence. These two observations prove surjectivity and injectivity of (5.6). □

6. Relative generation and Frattini coordinates

A finite indexed family $(s_i)_{i\in I}$ is independent relative to $B \le G$ if

$$s_i \notin \langle B, s_j : j \neq i\rangle \quad (i \in I).$$

It is a relative generating family if it also generates G together with B. For a finite group G, let $m(G)$ be the largest size of an irredundant generating family and let $i(G)$ be the largest size of an independent family, without the generation requirement. If G acts on a subgroup N, independence under this action means that no member is generated by all G-conjugates of the other members; write $i_G(N)$ for the corresponding maximum.

Proposition U.RelativeBound (normal quotient bound). If G is finite and $N \trianglelefteq G$, then

$$m(G) \le m(G/N) + i(N). \tag{6.1}$$

If N is abelian, then $i(N)$ can be replaced by $i_G(N)$.

Proof. Take an irredundant generating family S of G, and a subfamily $T \subseteq S$ whose images minimally generate G/N. For every $s \in S\backslash T$, choose a word w_s in T with $sN = w_sN$, and put $v_s = sw_s^{-1} \in N$. If one v_s belonged to the subgroup generated by the other v_t, substituting $v_t = tw_t^{-1}$ would express s using only $S \setminus \{s\}$. Thus the v_s are independent. Since $|T| \le m(G/N)$, (6.1) follows.

If N is abelian, the equality $G = N\langle T\rangle$ says that conjugation by G on N is already conjugation by $\langle T\rangle$. An expression for v_s using conjugates of the other v_t would again substitute to an expression for s without s. The family is therefore G-independent, proving the sharper bound. □

For a finite indexed subgroup family $H_i \le G$, write $H_J = \bigcap_{j\in J} H_j$, with $H_\emptyset = G$ and $B = H_I$.

Theorem U.Relative (generation from residual maximality). Suppose all H_J, $J \subseteq I$, are distinct, and for every $J \subsetneq I$ there is $k \notin J$ such that $H_{J\cup\{k\}}$ is maximal in H_J. Choose

$$s_i \in H_{I\setminus\{i\}} \setminus H_i. \tag{6.2}$$

Then

$$H_J = \langle B, s_i : i \notin J\rangle \quad (J \subseteq I). \tag{6.3}$$

In particular these elements generate G irredundantly relative to B, and their relative parabolics satisfy

$$\langle B, s_i : i \in S\rangle \cap \langle B, s_i : i \in T\rangle = \langle B, s_i : i \in S \cap T\rangle. \tag{6.4}$$

The group need not be finite.

Proof. Distinctness of $H_{I\setminus\{i\}}$ and B makes (6.2) possible, since its intersection with H_i is B. Every s_j with $j \neq i$, and every element of B, belongs to H_i, while s_i does not. This proves relative independence.

Prove (6.3) by downward induction from $J = I$. Choose k as in the hypothesis. By induction, $H_{J\cup\{k\}}$ is generated by B and the s_i with $i \notin J \cup \{k\}$. The element s_k belongs to H_J, because $J \subseteq I \setminus \{k\}$, and lies outside $H_{J\cup\{k\}}$, because it lies outside H_k. Maximality therefore says that adjoining s_k generates H_J. This is (6.3). Finally substitute $J = I\setminus S$ and $I\setminus T$ into the elementary equality $H_J \cap H_{J'} = H_{J\cup J'}$ to obtain (6.4). □

Lemma U.Maximal (relative finite generation). If $G = \langle B, t_1, \dots, t_r\rangle$ with r finite, then every proper subgroup containing B is contained in a maximal subgroup of G.

Proof. Order the proper overgroups of the given subgroup by inclusion. The union of a chain is a subgroup. If that union were G, a single member of the chain would contain all the finitely many t_j, and hence all of G, a contradiction. Thus every chain has a proper upper bound. Zorn's lemma gives a maximal proper overgroup. □

Define $\Phi(G)$ to be the intersection of all maximal subgroups, with intersection G if there are none. This subgroup is characteristic. Whenever every proper overgroup of B is contained in a maximal subgroup, a subgroup $H \geq B$ satisfies

$$\Phi(G)H = G \quad \Longleftrightarrow \quad H = G : \tag{6.5}$$

a maximal overgroup of a proper H would contain both factors on the left.

Theorem U.FrattiniTransport (relative generating families). Under the maximal-overgroup hypothesis of (6.5), a finite indexed family is an irredundant relative generating family of G over B if and only if its image is one of $G/\Phi(G)$ over $B\Phi(G)/\Phi(G)$.

Proof. Generation in the quotient says $\Phi(G)\langle B, s_i : i \in I\rangle = G$, which is equivalent to generation in G by (6.5). Once generation holds, irredundancy says that each deletion subgroup $\langle B, s_j : j \neq i\rangle$ is proper. Equation (6.5), applied to each of those subgroups, says exactly that its image is proper in the quotient. These are precisely the quotient irredundancy conditions. □

The maximal-overgroup hypothesis holds for finite G and, by U.Maximal, for relative finite generation. It cannot simply be omitted for arbitrary infinite groups: in $C_{p^\infty} \times C_2$, the divisible factor is the Frattini subgroup, and a generator of the quotient does not generate the whole group.

Theorem U.FrattiniLadder (containment in the common stabilizer). Suppose the H_J are all distinct and some ordering of $I = \{1, \dots, r\}$ gives a chain

$$G = K_0 > K_1 > \cdots > K_r = B, \qquad K_j = H_{\{1,\dots,j\}}, \tag{6.6}$$

in which every inclusion is maximal. If $N \trianglelefteq G$ lies in every maximal subgroup of G, then $N \leq B$.

Proof. Choose $t_j \in K_{j-1} \setminus K_j$. Maximality gives $K_{j-1} = \langle K_j, t_j\rangle$, and iteration gives $G = \langle B, t_1, \dots, t_r\rangle$. Every proper H_i therefore lies in a maximal subgroup by U.Maximal. Such a maximal subgroup contains N, so $NH_i \neq G$.

Suppose $N \not\leq B$, and choose the first k with $N \not\leq K_k$. Then $N \leq K_{k-1}$, and normality and maximality give

$$K_{k-1} = N(K_{k-1} \cap H_k).$$

Suppose at an upward step that $K_j = N(K_j \cap H_k)$, where $1 \leq j < k$. Since $N \leq K_{j-1}$,

$$K_j \leq N(K_{j-1} \cap H_k) \leq K_{j-1}.$$

If this middle group were K_j, then $K_{j-1} \cap H_k \leq K_j$, forcing

$$K_{j-1} \cap H_k = K_j \cap H_k.$$

The corresponding index sets differ because $j < k$, contradicting the assumed distinctness. Maximality therefore makes the middle group K_{j-1}. Continuing upward reaches $G = NH_k$, contradicting the first paragraph. Hence $N \leq B$. □

Theorem U.Phi (primitive saturation and coordinates). Let P be a finite p-group with an irredundant generating marking $(r_i)_{i\in I}$, $|I| = d$, and let $F = \Phi(P)$. Then, for every $S \subseteq I$,

$$\Phi(P_S) = P_S \cap F, \qquad P_S F = \pi^{-1}\,\mathrm{span}(\bar{r}_i : i \in S), \qquad [P : P_S F] = p^{d-|S|}. \tag{6.7}$$

The saturated parabolics $P_S F$ intersect according to intersection of index sets. Each omission subgroup $P_{I\setminus\{i\}}$ has exactly one maximal overgroup, namely $P_{I\setminus\{i\}}F$.

If $B \geq F$ and $(s_i)_{i\in J}$ is an irredundant relative generating family over B, then $B \trianglelefteq P$, P/B has basis $(s_i B)_{i\in J}$ over $\mathbb{F}_p$, and

$$\langle B, s_i : i \in T\rangle/B = \mathrm{span}(s_i B : i \in T). \tag{6.8}$$

If $p = 2$ and all original r_i are involutions, then $F = P'$ and $\Phi(P_S) = P'_S$ for every S.

Proof. Burnside's basis theorem makes the $\bar{r}_i$ a basis of P/F. The image of P_S therefore has order $p^{|S|}$, with kernel $P_S \cap F$. Since this image is elementary abelian, $\Phi(P_S) \leq P_S \cap F$. On the other hand P_S is generated by $|S|$ elements, so Burnside gives $[P_S : \Phi(P_S)] \leq p^{|S|}$. The quotient by $P_S \cap F$ already has that order. Equality follows, proving the first formula. The remaining formulas and the intersection assertion are inverse-image statements for coordinate subspaces.

Every maximal subgroup of a finite p-group contains F and corresponds to a hyperplane of P/F. A hyperplane containing a specified coordinate hyperplane equals it, proving uniqueness of the maximal overgroup of an omission subgroup.

For the relative assertion, every subgroup containing F is normal, because P/F is abelian. The quotient P/B is elementary abelian. The indicated images span it, and any nontrivial linear relation would express one s_i using B and the other markers, contradicting irredundancy. They form a basis, proving (6.8). Finally, an abelian group generated by involutions is elementary abelian of exponent two. Thus $P' \leq \Phi(P)$, while the elementary abelian quotient P/P' gives the reverse inclusion. The same argument applies to each P_S. □

These coordinates also describe the corresponding omission cosets: a coset prescribes one coordinate value, and every simultaneous assignment gives a chamber. The geometry is complete multipartite, with p objects in each type, and the quotient translation group acts regularly on chambers. For $p > 2$ it need not be the full type-preserving automorphism group; the thin case $p = 2$ is the one covered by U.Geometry.

7. How the statements are used

The coordinate-kernel theorem U.SK.Helly additionally requires exact kernel profiles and base IP to move pairwise intersections to a common fiber. Its geometric corollary requires actual involutory local markers, as well as a regular base geometry. These obligations are checked for the three dihedral pair attachments in Module A.

The application interface is mathematical rather than merely notational. An application of U.SK supplies the marked base, every actual deletion, each local marked epimorphism, and injectivity on every proper local parabolic. Its order formula then needs only the orders of the base and local kernels. Its universal-presentation claim, if any, still requires a presentation map and an independent sharp bound such as U.Order. An application of U.Nilpotent must first identify the canonical class-two quotient of every target group. An application of U.Compare must prove injectivity on the whole protected product set and exact membership in the exceptional image. An application of U.Action supplies the actual encoding and retained surjection, not an assumed equality of kernels.

These requirements permit the same complete proof to be cited several times without repeating its induction or quotient argument. They also identify the portions that cannot be omitted from an application: local models, presentation recognition, geometric realization, source-specific parameter ranges, and any

exhaustive finite computation on which its result depends. The companion contract ledger records these bindings separately from the historical attribution of each original theorem.

8. Genuine group pushouts and primary torsion

Fix injective homomorphisms $i_A : C \to A$ and $i_B : C \to B$. Let $A * B$ be the free product and let

$$P = (A * B)/\langle\langle i_A(c) i_B(c)^{-1} : c \in C \rangle\rangle. \tag{8.1}$$

We use the amalgam normal-form theorem in the following precise form: the canonical maps of A and B into P are injective; an alternating reduced word with factors outside the common copy of C has nonzero reduced length and is nonidentity; every element is conjugate either into a factor or to a cyclically reduced alternating word of length at least two, whose positive powers remain reduced with proportionally increasing length. This is a retained classical theorem.

Theorem U.Amalgam (universal property and torsion). The group P in (8.1), with its canonical maps j_A, j_B, is a pushout of $A \leftarrow C \to B$ in the category of groups: for every group D and homomorphisms $f_A : A \to D$, $f_B : B \to D$ satisfying $f_A i_A = f_B i_B$, there is a unique homomorphism $f : P \to D$ with $f j_A = f_A$ and $f j_B = f_B$.

Let p be a prime. Say that a group has p-primary finite torsion when every finite-order element has p-power order. This property passes to subgroups, to extensions with finite p-group quotient, and to amalgamated free products of groups having the property. If $i_A(C) \neq A$ and $i_B(C) \neq B$, then P is infinite. If A and B are finitely generated, then so is P.

Proof. The free-product universal property gives a unique homomorphism $A * B \to D$ extending f_A, f_B. Agreement on C puts every defining relator in its kernel, so it factors through P. The images of A and B generate P; hence this factorization is unique. The equality $j_A i_A = j_B i_B$ follows from the defining relations, completing the pushout property.

The torsion assertion for subgroups is immediate. For an extension $1 \to K \to E \to R \to 1$, with R a finite p-group and K having p-primary finite torsion, take any finite cyclic subgroup $\langle x \rangle$ of E. Its intersection with K is a cyclic p-group, and its image in R is a p-group. The order formula for this restricted exact sequence makes $|x|$ a power of p.

For an amalgam, the normal-form theorem places a torsion element, up to conjugacy, in a factor: the alternative cyclically reduced word has powers of unbounded reduced length and cannot have finite order. The factor hypothesis therefore proves the torsion assertion. If the common subgroup is proper in both factors, choose $a \in A \setminus i_A(C)$ and $b \in B \setminus i_B(C)$. The words $(ab)^n$ are reduced of length $2n$, so are all distinct; this proves infinitude. Finally, the images of finite generating sets of A and B generate their free product and its quotient P, proving finite generation. □

The simultaneous fiber product of U.SK has its displayed compatibility equations and its proved generation theorem. It is not the pushout in U.Amalgam. A graph of proof dependencies is neither of these group constructions.

Attribution and references

The elementary image, kernel, orbit and amalgam arguments above use established group theory. The relative-generation and Frattini applications have their source context in Cameron–Cara [CC02] and Archer–Cara–Krempa [ACK05]; their original results retain that attribution. The constructions motivating the supported-kernel and finite-quotient interfaces are those of [HFLa], [HFLb], [HS26] and [CFHS25], together with the order-1024 study [GLP18] and the three-generator hypertope construction [HFLh]. Their source-specific results are stated and credited in Application A, where the canonical forest quotient, square model, classification branches and computations are proved or expressly retained at their actual scope.

Within this project, U.K, U.Q, U.D, U.Compare, U.Action and the relative and Frattini arguments reuse earlier written derivations rather than starting their proofs again. U.SK abstracts the common supported-extension construction. Its inclusion-induction proof removes the earlier separate normal-generator-word hypothesis, and its deletion corollary derives a hypothesis that had previously been assumed. U.Nilpotent separates the previously forest-specific search theorem into a general exact-quotient statement and an application-specific proof identifying that quotient. These descriptions concern the organization and development of the present proof library; they are not claims of priority over the literature. The separate provenance ledger binds the exact predecessor proofs, the new interfaces and the limited existing Lean certificates. The coordinate-kernel Helly proof was developed as the common geometric step for the three-generator hypertope construction and is integrated here with its necessary local-involutivity hypothesis. The affine and source-presentation arguments remain in Module A. These additions make no historical novelty or open-conjecture claim.

[CC02] P. J. Cameron and P. Cara, *Independent generating sets and geometries for symmetric groups*, Journal of Algebra **258** (2002), 641–650. DOI: 10.1016/S0021-8693(02)00550-1. The retained target edition is the twelve-page author manuscript.

[ACK05] C. Archer, P. Cara and J. Krempa, *Using the Frattini Subgroup and Independent Generating Sets to Study RWPRI Geometries*, Beiträge zur Algebra und Geometrie **46** (2005), 169–177. The retained target edition is the published nine-page text.

[HFLa] D.-D. Hou, Y.-Q. Feng and D. Leemans, *On regular polytopes of 2-power order*, revised author manuscript dated 8 November 2019; journal publication 2020. DOI: 10.1007/s00454-019-00119-5.

[HFLb] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-polytopes of order* 2^n, arXiv:1802.08947v2, 22 January 2019; journal title *2-groups behaving as automorphism groups of regular 3-polytopes*.

[HS26] D.-D. Hou and E. Schulte, *String C-groups of 2-power order project onto a common string C-group*, arXiv:2605.20691v1, 20 May 2026.

[CFHS25] G. Cunningham, Y.-Q. Feng, D.-D. Hou and E. Schulte, *Symmetric polytopes whose automorphism groups are 2-groups*, arXiv:2512.15511v1, 17 December 2025.

[GLP18] Y. Gomi, M. L. Loyola and M. L. A. N. De Las Peñas, *String C-groups of order 1024*, Contributions to Discrete Mathematics **13**(1) (2018), 1–22. DOI: 10.55016/ojs/cdm.v13i1.62475. The exact published edition and presentation corrections are specified in Application A's source contract; this umbrella does not claim to reproduce that census on its own.

[HFLh] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-hypertopes with* 2^n *chambers*, arXiv:1901.09034v1, 23 January 2019. The exact presentation and corrected structural displays are treated in Module A.

Module A: Construction and classification of two-group polytopes and hypertopes

Abstract

We develop construction and classification methods for regular polytopes and hypertopes whose automorphism groups have two-power order. A forest of noncommuting generators determines a canonical class-two quotient. Its square-lattice sections provide local kernels that can be attached independently by the supported-kernel theorem of *Marked groups, kernels and intersections*. This yields prescribed edge orders, square sections and three-generator hypertope types, with exact orders and complete presentations. Conversely, a fixed class-two quotient reduces bounded-order classification to an exhaustive invariant-kernel problem. We combine this reduction with cyclic-core and central-descent arguments to

recover the high-type rank-three families and the full order-1024 classification, including named facets and corrected presentations. The same local models also support regular, chiral and alternating extensions. Source conclusions, retained geometric inputs and corrections are stated explicitly.

Introduction

Local sections serve two complementary purposes in finite incidence geometry. One may prescribe them and construct a marked group realizing those prescriptions, or constrain their orders and classify every group that can occur. The same kernel coordinates can organize both tasks. In the construction direction, a local quotient is attached to a subset of the marker labels. In the classification direction, a canonical nilpotent quotient confines the search to normal subgroups of one finite group.

The starting point is a vertex-and-edge model for a forest diagram. The intersection property forces independent edge commutators in degree two, so every finite marked two-group with that diagram has the same class-two quotient (§2). The affine square group supplies exact local kernels and two useful quotient decompositions (§3). The supported-kernel theorem of the companion paper [U] then assembles prescribed edge orders and consecutive square sections. Its coordinate-kernel Helly theorem upgrades three dihedral attachments to an actual regular hypertope. A separate sharp order bound identifies the source's complete presentation (§4).

The reverse direction begins with a bounded cyclic core, reducing the high-type rank-three problem to eight explicit models (§5). Central descent and complete binary lifting link the bounded quotients (§6). One invariant-kernel census, followed by complete presentation and marked-isomorphism checks, yields the order-1024 classification (§7). Finally, free geometric extensions use the amalgam theorem of [U] and precisely stated earlier constructions (§8).

The six applications are the prescribed-order theorem of Hou, Feng and Leemans [HFLa], their high-type rank-three classification [HFLb], their three-generator hypertope construction [HFLh], the common-quotient theorem of Hou and Schulte [HS26], the square-section and extension theorems of Cunningham, Feng, Hou and Schulte [CFHS25], and the order-1024 classification of Gomi, Loyola and De Las Peñas [GLP18]. Their original results retain that attribution. The general algebraic proofs occur once in [U]; this paper supplies the actual marked maps, local injectivity, geometric hypotheses and source-presentation identifications needed to use them. Earlier cyclic-core, geometric-assembly and infinite-decomposition theorems remain explicit retained inputs. All finite classification data and the complete certificate interface are specified in Appendix C.

1. Markings, geometric inputs and the interface to U

A marking is an ordered generating family of nonidentity involutions r_i, indexed by a finite set I. Write $G_J = \langle r_j : j \in J \rangle$. The intersection property is $G_J \cap G_K = G_{J \cap K}$ for every pair of subsets. A string C-group also has $[r_i, r_j] = 1$ for $|i - j| > 1$; nondegeneracy means that consecutive generators do not commute. The type is their consecutive product orders. A regular hypertope is a thin, residually connected incidence geometry with a chamber-transitive type-preserving automorphism group. It need not have a string diagram; its rank-three type here records all three pair-product orders. Use $[x, y] = x^{-1}y^{-1}xy$, $x^y = y^{-1}xy$, and D_8 for the dihedral group of order eight. Ordered marked isomorphisms preserve every generator. The [GLP18] census instead identifies generator sets; for a nondegenerate string this adds precisely reversal.

We retain the regular-polytope/string-C-group correspondence, the marked epimorphism/cover correspondence and the coset description of faces and stabilizers. A deletion sends selected markers to themselves and all others to one. Its geometric interpretation gives the flat amalgamation property (FAP) at the corresponding initial or final cut. Flatness itself is a separate incidence conclusion: a factorization of the group by two relevant face stabilizers makes every face of the first kind incident with every face of the second kind. It is never inferred merely from IP.

The general facts used from U are listed with their exact interfaces. Their complete proofs occur once in U; each application below verifies the stated hypotheses.

- **U.K**: if a marked epimorphism $G \to B$ has kernel N and the base has IP, then G has IP exactly when the profiles $N \cap G_J$ have the same subset-intersection property.
- **U.Q**: for a quotient $q : G \to G/N$, $q(A) \cap q(B) = q(A \cap B)$ holds exactly when $N \cap BA = (N \cap B)(N \cap A)$. In particular $N \cap BA = 1$ preserves the intersection and embeds both ends; if $N \leq A$, then $q(A) \cap q(B) = q(A \cap B)$. The latter still requires checking that the displayed quotient marking on A is valid.
- **U.D**: a full Boolean family of genuine deletions implies IP and split parabolic retractions, including infinite groups.
- **U.SK**: exact base deletions and epimorphisms $E_S \to B_S$ injective on every proper-support parabolic give the full supported fiber group, with kernel profile $\ker \pi \cap H_J = \prod_{S \subseteq J} \ker(E_S \to B_S)$. It has IP, inherits Boolean deletions, and has the resulting parabolic order formula. Antichain supports embed their full local groups. If every local kernel is $\gamma_c(E_S)$ and B has class below c, the global kernel is $\gamma_c(H)$. A supported separator gives $H = H_J H_K$ when $B = B_J B_K$.
- **U.SK.Helly/U.SK.Geometry**: a marked quotient with base IP and an exact direct-product coordinate profile transfers typed coset Helly from the base. If the base is a regular hypertope and every local marker is an involution, U.SK therefore gives a regular hypertope with the constructed full type-preserving group. In the pair-support case local involutivity also follows from proper-singleton injectivity.
- **U.Nilpotent/U.Ladder**: for an involutory finitely generated presentation F with deletions, its class-two quotient Q inherits deletions and IP. Finite marked quotients with that same Q and order $|Q|2^k$ are classified by suitable kernels inside $\gamma_3(F/\gamma_{k+3}(F))$. Their IP is the exact all-parabolic kernel-profile condition. Every invariant index-two step is a hyperplane preimage in $M/(\Phi(M)[M,U])$, and iterating k steps exhausts every index-2^k invariant kernel.
- **U.Phi**: an irredundant generating marking of a finite two-group maps to a basis of its Frattini quotient; every binary coordinate assignment gives a character. All marked parabolics have the corresponding coordinate images.
- **U.Amalgam**: p-primary finite torsion is preserved by subgroups, finite p-group extensions and amalgamated free products. Proper embeddings in both factors make an amalgam infinite; finitely generated factors generate it.
- **U.Order**: finite normal factors and a finite quotient-index bound give the product upper bound; a marked finite quotient attaining that bound identifies the complete presentation.

U retains Burnside's basis theorem, lower-central graded Lie theory, finite nilpotency of finite two-groups and amalgam normal forms. Here we additionally retain the Gaussian-integer Euclidean algorithm, the precise cyclic-core theorem in §5 and the geometric assembly/extension theorems in §§4 and 8. The standard low-rank string-IP criteria, smooth square-quotient classification, universal geometric interpretation and exceptional infinite decomposition quoted by [GLP18] are stated where used. Exact GAP group and coset algorithms and NQ's universal nilpotent quotient algorithm are computational inputs. No finite classification catalogue supplies search completeness.

2. A canonical quotient and one finite classification problem

The application-specific bridge to U is a rigidity statement: a forest diagram forces independent edge commutators in degree two. Its proof uses actual IP in G; no intermediate quotient is presumed to retain IP.

Lemma A2.1 (descending lift). Let G be finite nilpotent and let $A \cap B = 1$. Put $D_j = \gamma_j(G)$, $A_j = A \cap D_j$, $B_j = B \cap D_j$. If the images of A_j, B_j generate D_j/D_{j+1} for every $j \geq m$, then $D_j = A_j B_j$ for every $j \geq m$.

Proof. Descend from a trivial term. Assuming $D_{j+1} = A_{j+1}B_{j+1}$, if $a \in A_j, b \in B_j$ have equal images in the next quotient, write $ab^{-1} = a'b'$ with $a' \in A_{j+1}, b' \in B_{j+1}$. Then $a'^{-1}a = b'b \in A \cap B$, so both images are trivial. Their images therefore form an internal direct sum in the abelian quotient. Counting gives $|D_j| = |A_j||B_j| = |A_jB_j|$, proving equality as sets. No prior normality or permutability of A, B is assumed. ■

Lemma A2.2 (graded support). Let a positively graded Lie algebra be generated in degree one by disjoint sets X, Y, with subalgebras L_X, L_Y. If $[x, y] \in L_Y$ for $x \in X, y \in Y$, then $L = L_X + L_Y$, degree by degree.

Proof. The derivation $\operatorname{ad} x$ preserves L_Y because it does so on its generators. Iterated brackets of those derivations also preserve L_Y, by Jacobi. Hence $[L_X, L_Y] \subseteq L_Y$, and $L_X + L_Y$ is a Lie subalgebra containing all generators. The argument holds in characteristic two. ■

Theorem A2.3 (forest quotient). Suppose G is a finite two-group generated by involutions r_i, with marked IP, and its noncommuting pairs form a forest T on d vertices and m edges. Then edge commutators are independent in $G'/\gamma_3(G)$, and $G/\gamma_3(G)$ is the marked group

$$(v, z)(w, t) = (v + w, z + t + \beta(v, w)), \quad v, w \in \mathbb{F}_2^d,\ z, t \in \mathbb{F}_2^m, \tag{A-F}$$

where, after orienting each edge, its component of β is $v_{\text{head}} w_{\text{tail}}$. Its order is 2^{d+m}.

Proof. IP gives irredundancy. U.Phi gives $G/G' = C_2^d$ and $G' \cap \langle r_i \rangle = 1$. The lower-central graded Lie ring is generated by these degree-one elements; all its graded components are $\mathbb{F}_2$-spaces by bilinearity. Edge commutators span degree two. If they satisfy a nonzero linear relation, root each forest component and choose a participating edge whose child endpoint has maximal depth in its component. Removing it separates the descendant set X from all other vertices Y, and no other participating edge is internal to X. The relation therefore writes the only nonzero cross bracket as a sum of brackets internal to Y. Lemma A2.2 shows every graded component is generated by pure X- and pure Y-brackets. Their representatives lie in $G_X \cap \gamma_j(G)$ and $G_Y \cap \gamma_j(G)$.

Actual IP gives $G_X \cap G_Y = 1$, so Lemma A2.1 gives $G' = (G' \cap G_X)(G' \cap G_Y)$. For the crossing endpoints $x \in X, y \in Y$, write $[r_x, r_y] = uv$ accordingly. Then $u \in G_X \cap G_{\{x\} \cup Y} = \langle r_x \rangle$, whence $u = 1$; next $[r_x, r_y] \in G_Y \cap G_{\{x,y\}} = \langle r_y \rangle$, so it too is one. This contradicts that the edge is noncommuting. Independence follows.

Bilinearity makes (A-F) associative; its coordinate markers are involutions with independent central edge commutators. It is generated by those markers, and its parabolic on I consists exactly of vertex coordinates supported on I and edge coordinates supported on edges internal to I. Thus it has IP. Collection in the corresponding class-two presentation gives at most 2^{d+m} elements, and this coordinate model attains the bound. Independence in $G'/\gamma_3(G)$ gives the same order for $G/\gamma_3(G)$; hence the marked map is an isomorphism. ■

Corollary A2.4 (F013 1.1–1.2). For every finite nondegenerate two-group string C-group of rank $d \geq 2$, $G/\gamma_3(G) \cong G(C_d)$, of order 2^{2d-1}; the associated polytope covers C_d.

Proof. Its diagram is a path, so $m = d - 1$. Conder's marked presentation has involutions, distant commutation, adjacent order four, and adjacent squares central. The model (A-F) realizes that presentation with the maximal collected order, identifying it with $G(C_d)$. The retained marked covering correspondence gives the geometric assertion. This also proves uniqueness of the minimal object in the declared finite nondegenerate class. ■

Notice that no intermediate quotient was assumed to retain IP. Lemma A2.1 is the step that prevents that circular assumption.

Corollary A2.5 (rank and excess classify the finite search). Let T be a specified forest on d vertices and m edges, and let k≥0. Set

$$F_T = \langle x_i \mid x_i^2 = 1,\ [x_i, x_j] = 1 \text{ for nonedges} \rangle, \quad U_{T,k} = F_T/\gamma_{k+3}(F_T), \quad K = \gamma_3(U_{T,k}).$$

The ordered IP markings with graph T and order 2^{d+m+k} are exactly the quotients $U_{T,k}/N$, where N is normal, contained in K, $[K : N] = 2^k$, and the images $(N(K \cap (U_{T,k})_J))/N$ have IP for every pair J,L. Generator-set equivalence is exactly the action of Aut(T) on these kernels.

Proof. Killing any set of generators preserves every defining relator of F_T. Collection in its class-two quotient gives at most d+m binary coordinates, and (A-F) realizes all of them, so this quotient is precisely the marked model Q_T. Every target has that same quotient by A2.3. These are exactly U.Nilpotent's hypotheses. Its all-parabolic condition and U.Ladder give the stated bijection and exhaustive search. An ordered isomorphism must commute with the generating map, hence has the same kernel. A permutation of the generator set preserves the noncommutation graph and extends to an automorphism of F_T and $U_{T,k}$; this proves the last claim. ■

This is used twice below: the order-256 seed has (d,k)=(3,3), while all nondegenerate order-1024 markings have (d,k)=(3,5),(4,3),(5,1). Their actual terminal counts require the complete computation in Appendix C. The parameter theorem proves exhaustion and termination, not those output values.

3. The affine square seed

Let

$$a(x, y) = (-x, y), \quad b(x, y) = (y, x), \quad c(x, y) = (x, 1 - y).$$

These generate $W = \mathbb{Z}^2 \rtimes D_8$, of type $\{4, 4\}$, with $ac = ca$. Its presentation is

$$W = \langle a, b, c \mid a^2 = b^2 = c^2 = (ab)^4 = (bc)^4 = (ac)^2 = 1 \rangle. \tag{A-S1}$$

Here is an algebraic normal-form proof. Put $d = bab$, $y_0 = cd$, and $x_0 = by_0b$. The relations give $[a, d] = 1$, $[a, y_0] = 1$, $dy_0d = y_0^{-1}$, $ax_0a = x_0^{-1}$, and conjugation by b swaps x_0, y_0. Substitution of $bc = x_0ab$ makes $(bc)^4 = 1$ equivalent to $[x_0, y_0] = 1$. Thus every word collects to $x_0^u y_0^v M$, $M \in \langle a, b \rangle = D_8$. In the affine model x_0, y_0 are the unit translations, so these forms are distinct. This proves (A-S1).

For $n \geq 5$, define the invariant lattice

$$\Lambda_n = \begin{cases} 2^e\mathbb{Z}^2, & n = 2e + 3, \\ 2^e\langle (1, 1), (1, -1) \rangle, & n = 2e + 4, \end{cases} \qquad e \geq 1,$$

and put $H_n = W/\Lambda_n$. Its order is 2^n; the full linear D_8 coordinate is retained, even if its action on the vertex quotient is not faithful. Its pair orders are $4, 4, 2$.

Write $T = abcb$, $U = cbab$, translations by $(-1, 0)$, $(0, 1)$. The extra relators for H_n can be taken as T^{2^e}, U^{2^e} in the square case, and $T^{2^e}U^{2^e}, T^{2^e}U^{-2^e}$ in the diagonal case. After deleting any marker, both T, U have order at most two: deleting a leaves conjugates of c, deleting c leaves conjugates of a, and deleting b leaves ac, ca. Therefore every lattice relator survives every deletion. The even Coxeter relators also survive. Thus H_n has Boolean deletions and IP by U.D.

Moreover,

$$\gamma_3(W) = 2\mathbb{Z}^2, \qquad \gamma_3(H_n) = 2\mathbb{Z}^2/\Lambda_n, \qquad |\gamma_3(H_n)| = 2^{n-5}. \tag{A-S2}$$

Indeed reduction modulo $2\mathbb{Z}^2$ makes the adjacent squares central; conversely $[[a,b],c]$ and $[[b,c],a]$ give the two doubled unit translations, up to signs. Thus H_5 is class two of order 32. The squared translations are not always independent cyclic subgroups: in H_6 they coincide and have order two. Formula (A-S2) uses a lattice index and does not make the false independence assertion.

Lemma A3.1 (square-map recognition). Every finite two-group string C-group of type $\{4,4\}$ and order 2^n is marked-isomorphic to H_n; necessarily $n \geq 5$.

Proof. The two dihedral parabolics have orders eight and intersect in order two, so their product already has 32 elements. The universal group (A-S1) maps onto the given group. IP gives irredundancy, so U.Phi gives elementary abelianization of rank three. Hence word parity is a homomorphism onto C_2, and the kernel of (A-S1) lies in the subgroup of orientation-preserving plane isometries. It contains no nonidentity torsion: a finite-order plane isometry fixes a point; the square tiling moves that point into a closed fundamental right triangle; its stabilizer is a subgroup of a vertex dihedral parabolic or an edge reflection group. Each embeds by the prescribed pair orders and IP. An orientation-preserving plane isometry without finite order is a translation. Thus the kernel is a D_8-invariant sublattice of $\mathbb{Z}^2$ of index 2^{n-3}. Identifying the lattice with an ideal of $\mathbb{Z}[i]$, its generator has norm a power of two. Unique factorization (from the Euclidean algorithm) makes it a unit times a power of $1+i$. Even powers give the square lattice and odd powers the diagonal lattice displayed above. There is exactly one such ideal at each index, proving the assertion. ■

The quotient $q_n : H_n \to H_5$ fixes the three named generators. Its kernel is $\gamma_3(H_n)$. Every proper marked parabolic has at most two generators, and its type is respectively 4,4 or2 both upstairs and downstairs. Each is the identical dihedral or order-four group; singleton and empty parabolics likewise embed. This verifies U.SK's full proper-support condition, rather than merely checking three involutions.

3.1. Complete square quotients and their decompositions

For $\delta \geq 0$, set

$$Q_\delta = \begin{cases} W/q\mathbb{Z}^2, & \delta \text{ even}, \quad q = 2^{(\delta+2)/2}, \\ W/q\langle (1,1),(1,-1)\rangle, & \delta \text{ odd}, \quad q = 2^{(\delta+1)/2}. \end{cases} \tag{A-S3}$$

Thus $Q_\delta = H_{\delta+5}$, and its order is $2^{\delta+5}$. It is a string C-group of type $\{4,4\}$, has all Boolean deletions, and its maximal-parabolic cosets have typed Helly by the retained regular-polytope correspondence.

The two words needed in the source are, in these affine coordinates,

$$[(ab)^2, c] = (0,-2;1), \qquad [(ab)^2,(bc)^2] = (-2,-2;1), \tag{A-S4}$$

where $(v;1)$ denotes translation by v. Indeed $(ab)^2$ is the half-turn $x \mapsto -x$, while $(bc)^2$ is the half-turn $x \mapsto (1,1) - x$; composing these with c gives the first formula, and composing the two half-turns twice gives the second. The normal closure in W of a translation consists of the lattice generated by its signed-permutation orbit. Therefore adjoining respectively

$$[(ab)^2, c]^{2^{\delta/2}} = 1 \quad \text{or} \quad [(ab)^2,(bc)^2]^{2^{(\delta-1)/2}} = 1 \tag{A-S5}$$

to (A-S1) gives exactly (A-S3), not just a quotient of the stated order.

Proposition A3.2 (the two square decompositions). The source presentations M_2 and M_1 have the decompositions displayed below. They hold in the same affine model for every integer $q \geq 1$, not only the powers of two in (A-S3). We use D_{2q} for the dihedral group of order $2q$, including $D_2 = C_2$ and $D_4 = C_2^2$. The source's presentations M_2 and M_1 adjoin respectively $(bcba)^q = 1$ and $(cba)^{2q} = 1$ to (A-S1).

Proof. Here bcba is translation by $(1,0)$, and $(cba)^2$ is translation by $(1,1)$. Their signed-permutation orbits therefore generate exactly the square and diagonal lattices in (A-S6)–(A-S7). This identifies those complete presentations with the indicated lattice quotients.

In the square case, $\langle a, c^b\rangle$ acts on the first coordinate and $\langle a^b, c\rangle$ on the second. Their translations have order q; each is D_{2q}, since its reflection has nonidentity linear part, also when q=1. They commute and intersect trivially. The reflection b interchanges the two factors and lies outside their product. They and b generate all three original generators. Hence

$$W/q\mathbb{Z}^2 \cong (D_{2q} \times D_{2q}) \rtimes C_2. \tag{A-S6}$$

In the diagonal case, put $A_0 = \langle b^a, b^c\rangle$ and $B_0 = \langle b, b^{ac}\rangle$. Their reflecting axes are respectively $x+y = 0, 1$ and $x-y = 0, -1$. The first factor has translations along $(1, 1)$, the second along $(1, -1)$, of order q in each case. Each reflection retains its nonidentity linear part, so the two factors have order 2q also at q=1. All cross-generators commute. Their linear parts show that an element of their intersection must be a translation; the independence of the two diagonal translations modulo the defining lattice then makes that intersection trivial. Thus $A_0B_0 \cong D_{2q} \times D_{2q}$.

Both a and c exchange the two factors by conjugation, so normalize their product. The two individual reflections a,c do not belong to the product by their linear parts. The only remaining possible nontrivial intersection with $\langle a, c\rangle$ is ac. Every translation part of an element of A_0B_0 with linear part -1 has coordinates of the same parity, whereas ac has translation part $(0, 1)$. The diagonal lattice preserves this parity condition, so ac is also outside the product. Finally b already lies in B_0, so the two factors and $\langle a, c\rangle$ generate the full group. Consequently

$$W/q\langle(1, 1), (1, -1)\rangle \cong (D_{2q} \times D_{2q}) \rtimes (C_2 \times C_2). \tag{A-S7}$$

These arguments specify the actions by the original affine generators and prove the decompositions, rather than only equality of orders. □

4. Prescribed local sections from supported kernels

Theorem A4.1 (edge inflation and exact orders). Let B have Boolean deletions and select pairs S={i,j} whose product orders q_S are even. For m_S divisible by q_S, attach the marked dihedral map $D_{2m_S} \to B_S = D_{2q_S}$. The resulting supported fiber group has IP, all deletions and

$$|H_J| = |B_J| \prod_{S\subseteq J} (m_S/q_S).$$

Every selected pair has product order m_S and every unselected pair retains its old order. If the base marking gives a regular hypertope, so does the constructed marking, with full type-preserving group H.

Proof. The dihedral quotient is injective on its proper parabolics, which are trivial or a single reflection. U.SK applies and its antichain clause identifies each selected local group. On an unselected pair every extra coordinate is a reflection or one; a base pair of odd order cannot have a single-reflection deletion. Thus the extra coordinates do not increase its base product order. For the geometric assertion the local dihedral markers are involutions, and the supports are pairs. Thus the corrected U.SK.Geometry applies to the regular base and proves the required flag property as well as IP. ■

For d≥3, n≥5, k_i≥2 and $\sum k_i \le n-1$, set $\delta = n-1-\sum k_i$ and $B = C_2^{d-3} \times H_{\delta+5}$, with the elementary factors first. Its initial d−3 consecutive pair orders are 2 and its final two are 4. Attach $m_i = 2^{k_i}$. The seed supplies all deletions, so A4.1 gives

$$|H| = 2^{d-3+\delta+5} \prod_{i\le d-3} 2^{k_i-1} \prod_{i>d-3} 2^{k_i-2} = 2^n.$$

Distant markers commute in every coordinate, so the marking is string. The geometric correspondence proves **[HFLa] Theorem 1.2**. Taking d=3 proves **[HFLb] Theorem 1.2**, including its stated n≥10 range, without another construction. These existence results do not imply the classification of §5.

Three-generator hypertopes

Fix integers $n \geq 10$, $s, t \geq 2$, $\ell \geq 1$, with $\delta = n-s-t-\ell \geq 0$. Let $P = P(n, s, t, \ell)$ have generators a,b,c and relations

$$a^2 = b^2 = c^2 = (ab)^{2^s} = (bc)^{2^t} = (ac)^{2^\ell} = 1,$$
$$[(ab)^4, c] = [a, (bc)^4] = [(ac)^2, b] = 1, \tag{A-H1}$$

together with the appropriate parity relation (A-S5). This is the exact defining presentation in [HFLh, Theorem 3.2], with the source markers renamed a,b,c.

Theorem A.H (three-generator hypertope presentation). P has order 2^n, its three pair-product orders are exactly $2^s, 2^t, 2^\ell$, and its maximal-parabolic coset system is a regular rank-three hypertope with full type-preserving automorphism group P. Moreover the three normal subgroups

$$K_{01} = \langle (ab)^4 \rangle, \quad K_{12} = \langle (bc)^4 \rangle, \quad K_{02} = \langle (ac)^2 \rangle$$

form an internal direct product

$$K \cong C_{2^{s-2}} \times C_{2^{t-2}} \times C_{2^{\ell-1}}, \qquad P/K \cong Q_\delta. \tag{A-H2}$$

The factors of order one are permitted. With the extension dot making no splitting assertion, the corrected structural conclusion is

$$P \cong K \,.\, \begin{cases} (D_{2^{(\delta+4)/2}} \times D_{2^{(\delta+4)/2}}) \rtimes C_2, & \delta \text{ even}, \\ (D_{2^{(\delta+3)/2}} \times D_{2^{(\delta+3)/2}}) \rtimes (C_2 \times C_2), & \delta \text{ odd}. \end{cases} \tag{A-H3}$$

Proof. Apply A4.1 to $Q_\delta = H_{\delta+5}$, with all three pair supports and target orders $2^s, 2^t, 2^\ell$. The base pair orders are $4, 4, 2$; the attached maps are therefore $D_{2^{s+1}} \to D_8$, $D_{2^{t+1}} \to D_8$ and $D_{2^{\ell+1}} \to D_4$. Section 3 supplies the base deletions and its regular string-polytope geometry. Each local marking consists of dihedral reflections, so A4.1's geometric clause applies. It constructs a regular hypertope with group H and the exact three pair orders, with

$$|H| = 2^{\delta+5}\, 2^{s-2}\, 2^{t-2}\, 2^{\ell-1} = 2^n.$$

The local orders and proper-parabolic injections used here are precisely those checked in A4.1; no independent second assembly proof is needed.

The marked tuple in H satisfies every relator of P. The pair-power relations have just been checked. For each commutator in (A-H1), the relevant power is trivial in the base and in every local factor other than its own support, while in that support the third marker is one. For either word in (A-S5), the base satisfies it by (A-S4)–(A-S5). In the $\{0, 1\}$ coordinate c is one; in the $\{1, 2\}$ coordinate a is one; in the $\{0, 2\}$ coordinate b is one. Thus each commutator word is trivial in every local coordinate. There is a marked epimorphism $P \to H$.

It remains to prove that this map has no kernel. In P each K_{ij} is normal: its two endpoints invert the pair-product power, and the third marker centralizes it by (A-H1). Their cyclic orders are at most 2^{s-2}, 2^{t-2} and $2^{\ell-1}$. Their normal product K is therefore finite of order at most $2^{s+t+\ell-5}$; neither commutation nor independence of the factors is being assumed for this bound. After quotienting by K, all remaining pair-power and commutator relations reduce exactly to (A-S1) and (A-S5). Section 3 identifies that quotient with Q_δ, of order $2^{\delta+5}$. Hence P is finite and

$$|P| \leq 2^{s+t+\ell-5} 2^{\delta+5} = 2^n = |H|.$$

The epimorphism is an isomorphism, by U.Order. This recognizes the bare source presentation and transfers the hypertope conclusion.

In H, the word $(ab)^4$ is trivial in the base and all coordinates except support $\{0,1\}$, where it generates exactly the kernel of $D_{2^{s+1}} \to D_8$. The corresponding statements hold for $(bc)^4$ and $(ac)^2$. Thus the displayed three subgroups are precisely the pure coordinate factors in U.SK. This proves the direct product and the exact quotient in (A-H2). Equations (A-S6)–(A-S7) now give (A-H3), including its action and parity. □

The bound $n \geq 10$ was not needed in the proof, but it is retained in the statement to match the declared source contract. No nonexistence assertion for $n < s + t + \ell$ follows from this construction.

Prescribed square sections

Theorem A4.2 (square kernels on a forest). Let T be a forest and choose distinct induced three-vertex paths S, with parameters $n_S \geq 5$. Attach $H_{n_S} \to H_5 = (Q_T)_S$ to the model (A-F). Then

$$|H_J| = 2^{|J|+|E(T[J])|+\sum_{S\subseteq J}(n_S-5)}, \qquad \gamma_3(H) = \prod_S \gamma_3(H_{n_S}), \qquad H/\gamma_3(H) = Q_T. \tag{A-P}$$

It has Boolean deletions and IP, and each selected triple is exactly H_{n_S}.

Proof. Deleting coordinates in (A-F) gives all base deletions and its parabolic order $2^{|J|+|E(T[J])|}$. Section3 verifies every proper-support injection and the local γ3 kernel. U.SK, including its antichain and lower-central clauses, gives every assertion. ■

For the d-vertex path, take all consecutive triples with parameters $n_3,\ldots,n_d$. Equation (A-P) gives the exact group order $2^{\sum n_j - 3(d-3)}$, all prescribed square sections and all FAP cuts. To prove flatness when d≥4, let L={0,…,d−2} and M={d−3,d−2,d−1}. Every support lies on one side. The vertex-and-edge counts in (A-F), together with IP, give $Q_T = (Q_T)_L(Q_T)_M$. U.SK therefore gives $H = H_L H_M$. The stabilizer of the base (d−4)-face contains H_M, while H_L is a facet stabilizer; taking inverses also gives $H = H_M H_L$. The coset face criterion proves (d−4,d−1)-flatness and hence flatness. The co-(−1)-face cut at d=3 is the tautological whole-polytope cut; the other two cuts are genuine seed deletions. This proves all clauses of **[CFHS25] Theorem 5.1**.

The universal presentation is a separate conclusion

Write G for the group H constructed in the path case above. The supported fiber group is not an arbitrary amalgam. To prove the source's universal-presentation claim, a further local normalization argument is needed.

Lemma A4.3 (four-marker centralization). Suppose a, b, c, d are involutions with $[a, c] = [a, d] = [b, d] = 1$ and $[(ab)^2, (cd)^2] = 1$. Then d commutes with $(cbab)^2$ and $(abcb)^2$.

Proof. Put $x = bab$, $E = (ab)^2 = ax$, $F = (cd)^2$. The element a commutes with F, so $[E, F] = 1$ implies $[x, F] = 1$. Also $[x, d] = 1$, hence $[x, Fd] = [x, cdc] = 1$. Conjugating by c gives $[cxc, d] = 1$. Both factors of $(cbab)^2 = (cxc)x$ commute with d. Finally $abcb = b(cbab)^{-1}b$, and b commutes with d, proving the second assertion. ■

First let P_0 have the string $[4, \ldots, 4]$ relations and every prescribed triple presentation from §3. All its defining relators are stable under every deletion assignment: the string Coxeter relators have even exponents, and the local lattice relators survive every deletion of their three markers by §3. Thus the assignments define endomorphisms of P_0, and U.D proves its IP without assuming finiteness. The constructed quotient G shows that each local triple embeds with its exact prescribed marking and that every generator is nontrivial. Hence P_0 itself is a string C-group of the required local type. Every marked string C-group of that local type satisfies its defining relators, so is its quotient. The standard marked

covering correspondence therefore constructs the universal regular polytope of that type directly. No separate assumption about a universal polytope's presentation or existence is needed.

Now let Q be the quotient of P_0 by

$$[(r_i r_{i+1})^2, (r_{i+2} r_{i+3})^2] = 1 \quad (0 \le i \le d-4). \tag{A-U}$$

The constructed group satisfies these relations. For (A-U), no tuple coordinate contains both disjoint edges; the edge missing one endpoint has square one in that coordinate. Hence $Q \twoheadrightarrow G$. Each triple embeds in Q, since its map onward to the embedded triple in G is injective. Its third lower-central subgroup K_i is generated by its two squared translations and has order $2^{n_{i+3}-5}$, by (A-S2). Lemma A4.3 and its reversed version show that the nearest outside markers centralize K_i, while the farther markers do so by string commutation. Thus every $K_i \lhd Q$. Their product K is finite of order at most $2^{\sum_j (n_j-5)}$. Modulo it all triples are class two; each edge commutator commutes with its endpoints and nearest outside markers, and with farther markers by string commutation. Thus Q/K is class two. Collecting markers and central edge commutators gives $|Q/K| \le 2^{2d-1}$. Therefore

$$|Q| \le 2^{\sum_j (n_j-5)+2d-1} = 2^{\sum_j n_j - 3(d-3)} = |G|.$$

U.Order, applied to the finite normal factors and the displayed quotient bound, identifies $Q = G$ as marked groups. Since P_0 is the universal type group just constructed, this is exactly its quotient by (A-U). The finite group presentation, the universal group and both intersection properties have all been proved using the same deletion calculus.

This proves **[CFHS25] Theorem 5.3**. No assertion about the order of the universal group P_0 itself is required.

Alternating assembly

We retain the following precise FAP case of Monson–Schulte's assembly theorem. If regular (d−1)-polytopes P,Q have a common facet K and FAP retractions $\Gamma(P) = N_P \rtimes \Gamma(K)$ and $\Gamma(Q) = N_Q \rtimes \Gamma(K)$, the tail-triangle group $(N_P \times N_Q) \rtimes \Gamma(K)$, with distinct final markers, gives an alternating semiregular d-polytope with these facets. Its full automorphism group is that group if P and Q are not isomorphic, and its extension by branch-swapping C2 when P=Q; in that case the polytope is regular. This is a retained geometric theorem, not a consequence of IP alone.

Apply A4.2 to the tree with common prefix and two final leaves, attaching the two lists $(n_3, \dots, n_{d-2}, n_{d-1})$ and $(n_3, \dots, n_{d-2}, m_{d-1})$. Write L,M for the facet label sets and C=L∩M. The two deletions map H into $H_L \times_{H_C} H_M$. They are jointly injective: every vertex coordinate, edge coordinate and supported kernel coordinate lies on one side and is detected there. Their image is the full fiber product, since (A-P) gives its order $|H_L||H_M|/|H_C|$. Split deletions identify it with the exact tail-triangle group in the retained theorem. For d=4 the common facet is a square; otherwise it has list $(n_3, \dots, n_{d-2})$. Its order is $2^{\sum_{j=3}^{d-2} n_j - 3(d-5)}$, including 8 for the empty list. The two facet kernels have orders $2^{n_{d-1}-3}$ and $2^{m_{d-1}-3}$. Thus the full automorphism order is

$$2^{\sum_{j=3}^{d-2} n_j + n_{d-1} + m_{d-1} - 3(d-3)}$$

when the final parameters differ, and twice that order when they agree. Unequal parameters give unequal facet orders and rule out an interchange; equal parameters give identical marked facets and their branch swap. These are exactly **[CFHS25] Theorem 6.1(a–b)**. No additional mixed co-(d−4)-section is prescribed.

5. Bounded cyclic cores and the eight high-type families

The supported constructions solve existence. Completeness at the high dihedral boundary needs a different local invariant, the cyclic core. Its bounded quotient is nevertheless recognized by the same square model, and its only finite seed is generated by the same U.Nilpotent/U.Ladder search as order 1024.

The retained **cyclic core theorem of Lucchini** says: if a cyclic proper subgroup M of a finite group H has core K, then $|M : K| < |H : M|$. Here K is the largest normal subgroup of H contained in M. We use the form recorded in Isaacs, *Finite Group Theory*, Theorem 2.20, also stated as L035 Proposition 2.9.

Theorem A5.1 (bounded cyclic quotient). Let $(G; a, b, c)$ be a rank-three string C-group of order 2^{t+s}, type $\{4, 2^t\}$, with $s \geq 3$ and $t \geq s - 2$. Put

$$r = bc, \quad m = 2^t, \quad d_0 = (ab)^2, \quad A = \langle r^{2^{s-2}} \rangle.$$

Then $A \lhd G$, and G/A is a string C-group of order 2^{2s-2} and type $\{4, 2^{s-2}\}$.

Proof. The even-word subgroup $H = \langle ab, bc \rangle$ has index at most two. IP and U.Phi give $G/G' = C_2^3$, where its image has dimension two, so the index is exactly two. For $M = \langle r \rangle$, the cyclic core theorem gives $|M : \mathrm{Core}_H M| < 2^{s-1}$, hence at most 2^{s-2}. Thus A is the subgroup of the cyclic core of its prescribed order, so is characteristic there and normal in H. The involution b inverts r; consequently $A \lhd G$. It meets $\langle a, b \rangle$ trivially by IP and lies in $\langle b, c \rangle$. Their quotient parabolics still intersect in $\langle bA \rangle$: equality of cosets gives membership in the original right parabolic, after multiplication by an element of A. The left parabolic embeds and the right has its stated dihedral order. This proves IP, independence, type and order in the quotient. ■

Since $\mathrm{Aut}(A)$ is abelian, every commutator centralizes A. In particular $[d_0, r^{2^{s-2}}] = 1$. If $A \neq 1$, its unique involution $z = r^{m/2}$ is central in G. The universal identities

$$r^a = d_0 r, \quad d_0^a = d_0, \quad r^c = r^{-1}, \quad d_0^c = d_0^r \tag{A-C1}$$

follow from the involution, string and order-four relations.

Hereafter $s \in \{3, 4, 5\}$, $n = t + s \geq 10$, so m is divisible by 32. The following eight models will realize every class. Let z be a distinguished central involution of the finite group D:

Branch	Group D	Right-conjugation automorphism T, fixing z
$s = 3$	$\langle d_0, z \rangle = C_2^2$	$T(d_0) = d_0 z^\epsilon$
$s = 4$	$\langle d_0, d_1, z \rangle = C_2^3$	$T(d_0) = d_1,\ T(d_1) = d_0 z^\epsilon$
$s = 5$	Four involutions d_0, d_1, d_2, z, z central, $[d_0, d_2] = 1$, $[d_0, d_1] = [d_1, d_2] = z^\beta$	$T(d_0) = d_1,\ T(d_1) = d_2,\ T(d_2) = d_0 d_1 d_2 z^\nu$

Here $\epsilon \in \{0, 1\}$, or $(\beta, \nu) \in \{0, 1\}^2$. For the last row an explicit group law on four binary coordinates is

$$(u, v, w, e)(u', v', w', e') = (u + u', v + v', w + w', e + e' + \beta(vu' + wv')). \tag{A-C2}$$

Its bilinear final term proves associativity and gives all sixteen normal forms $d_0^u d_1^v d_2^w z^e$. Thus the three D-orders are four, eight and sixteen. The displayed maps preserve their involution and commutator relations and have fourth power one. For example in the last row the two nontrivial commutator contributions cancel in $(d_0 d_1 d_2)^2$, and applying T four times to d_0 returns d_0; its orbit generates D together with z.

Adjoin a cyclic r of order m, with $x^r = T(x)$, and take the central quotient

$$N = (D \rtimes \langle r \rangle)/\langle z r^{-m/2} \rangle. \tag{A-C3}$$

The identified subgroup has order two and meets both D and $\langle r\rangle$ trivially, since $T^{m/2}=1$. Hence D embeds, r retains order m, and $|N|=|D|m/2$.

Define automorphisms $\mathcal{A},\mathcal{C}$ by

$$\mathcal{A}(r)=d_0r,\quad \mathcal{A}(d_0)=d_0,\qquad \mathcal{C}(r)=r^{-1},\quad \mathcal{C}(d_i)=T^{1-i}(d_0),\quad \mathcal{C}(z)=z.$$

In the first two rows $\mathcal{A}$ fixes D; in the third it fixes d_0,d_1,z and sends d_2 to d_2z^{β}. For an explicit verification, substituting the listed generator images into the defining involutions and commutators of D preserves each relation. On its generators the compatibility equalities are

$$\mathcal{A}(Tx)=T(\mathcal{A}(x)^{d_0}),\qquad \mathcal{C}(Tx)=T^{-1}\mathcal{C}(x).$$

They express precisely preservation of $x^r=T(x)$. Collection gives $(d_0r)^4=r^4$ in the first two rows and r^4z^{ν} in the last. Since eight divides $m/2$, this preserves $r^m=1$ and $r^{m/2}=z$. Finally $\mathcal{A}^2=\mathcal{C}^2=1$ and $\mathcal{A}\mathcal{C}=\mathcal{C}\mathcal{A}$ on r,d_0, which generate N. For instance both composites send r to $r^{-1}d_0$. These checks prove the automorphism assertions for every admissible m, not only for a computational sample.

Set $G=N\rtimes\langle a,c\rangle$, where a,c are commuting involutions inducing these automorphisms, and put $b=rc$. Then a,b,c are involutions, $[a,c]=1$, $bc=r$, and $(ab)^2=d_0$. They generate G, of order 2^sm, with pair orders $4,m,2$. In G/N, the images of a,b are independent whereas the right parabolic has image $\langle cN\rangle$. Hence the only candidates for their intersection are $1,b,d_0,d_0b$. The normal form (A-C3) gives $d_0\notin\langle r\rangle$, leaving exactly $\langle b\rangle$. The three generators are independent (already $a\notin\langle b,c\rangle$, and the pair orders exclude the others), so this proves rank-three IP. Direct substitution yields

$$\begin{array}{ll} s=3: & [d_0,c]=z^{\epsilon},\\ s=4: & [d_0,r^2]=z^{\epsilon},\\ s=5: & [a,r^2]^2=z^{\beta},\qquad [a,r^4]=z^{\nu}. \end{array}\tag{A-C4}$$

They are invariant words of the ordered marking and distinguish the two, two and four models within their branches.

Completeness and bare presentation recognition

The eight source presentations have common relations

$$a^2=b^2=c^2=(ac)^2=(ab)^4=r^m=1,\qquad r=bc,$$

and the following additional relations, with $z=r^{m/2}$:

Source name	m	Extra relations
G_1,G_2	2^{n-3}	$[d_0,c]=z^{\epsilon}$, $\epsilon=0,1$ respectively
G_3,G_4	2^{n-4}	$[d_0,r^2]=z^{\epsilon}$, $\epsilon=0,1$ respectively
G_5,G_6,G_7,G_8	2^{n-5}	$[a,r^2]^2=z^{\beta}$, $[a,r^4]=z^{\nu}$, with $(\beta,\nu)=$ $(0,0),(1,0),(0,1),(1,1)$

Theorem A5.2 (high-type completeness). Every group in the three classes of L035 Theorem 1.3 is exactly one of the marked models of §5; each displayed source presentation defines that model.

Proof, first branch. For $s=3$, Theorem A5.1 gives a quotient of type $\{4,2\}$ and order sixteen, necessarily $D_8\times C_2$. Thus $h=[d_0,r]\in A=\langle r^2\rangle$. Since it commutes with r, and d_0 centralizes A,

$h^2 = [d_0, r^2] = 1$. Hence $h = 1$ or z, and (A-C1) makes this exactly $[d_0, c] = z^\epsilon$. The conjugates of d_0 obey $d_1 = d_0 z^\epsilon$; (A-C1) forces all the first model's actions and normal forms. The resulting surjection from that model onto the actual group is between groups of the same order, hence an isomorphism.

Second branch. For $s = 4$, the quotient of order 64 and type $\{4, 4\}$ is H_6, by Lemma A3.1. In that affine quotient $[d_0, r^2] = 1$, so upstairs $h = [d_0, r^2] \in \langle r^4 \rangle$. Again $h^2 = [d_0, r^4] = 1$, giving $d_2 = d_0 z^\epsilon$, where $d_i = d_0^{r^i}$. Conjugation by a fixes d_0, d_1. Applying it to the equation for d_2, with z central, gives $d_2^a = (d_1^{d_0})^r = d_2$, so $[d_0, d_1] = 1$. All second-model relations follow, and the same order comparison proves uniqueness.

The bounded seed for the third branch. The same universal-kernel theorem used for the order-1024 classification gives the following finite fact: every order-256 marked string C-group of type $\{4, 8\}$, with $r = bc, d_0 = (ab)^2$, satisfies exactly one row.

Row	Relation pattern	Ordered marked classes
I	$[a, r^2]^2 = 1,\ [a, r^4] = 1$	1
II	$[a, r^2]^2 = r^4,\ [a, r^4] = 1$	1
III	$[d_0, r^2] = r^4$	1
IV	$[(r^2)^a, r] = r^4$	1

Apply U.Nilpotent and U.Ladder, as specialized in A2.5 with d=3 and k=3. The universal class-five group has order 2^{18} and third lower-central subgroup of order 2^{13}. Its invariant index-two ladder has 7, 23 and 61 kernels. Testing all 64 subset pairs for each final candidate gives 37 ordered IP markings, four with the indicated orientation of type {4,8}. The census program in Appendix C evaluates every displayed word on all four and verifies that each belongs to exactly one row, one marking per row. Section 7 states the complete output and the shared finite-search proof. Thus this bounded input is now conditional on exact GAP/NQ algorithms, with no abstract-group catalogue. No SmallGroups identifier or catalogue multiplicity enters these relation implications.

Third-branch lifts. For $s = 5$, Theorem A5.1 gives $A = \langle r^8 \rangle$ and the table applies to G/A. In row III we would have $[d_0, r^2] = r^e$, $e \equiv 4 \pmod 8$. It commutes with r, so $[d_0, r^8] = r^{4e} = 1$, impossible for $m \geq 32$, since $v_2(4e) = 4$. In row IV put $x = (r^2)^a$. The relation would give $[x, r] = r^e$, $e \equiv 4 \pmod 8$, hence $r^x = r^{1-e}$. The unit $1 - e \equiv 5 \pmod 8$ has order $m/4$ modulo m: successive squaring gives $v_2((1-e)^{2^j} - 1) = j + 2$. But $x^{m/8} = (r^{m/4})^a \in A$, by normality, so it centralizes r, contradicting that order. Thus III and IV cannot lift.

In I and II, $[a, r^4] \in A$, so a maps r^4 to an odd power of itself and $M = \langle r^4 \rangle \lhd G$. Quotienting gives a marked $\{4, 4\}$ group of order 128, hence $H_7 = W/4\mathbb{Z}^2$. Let $N = \langle d_0, r \rangle$. Equations (A-C1) make it normal, with index four: modulo N, a, c commute, and their independence follows in the rank-three Frattini quotient, where N has the one-dimensional image spanned by r. The cyclic normal subgroup M is central in N, because d_0 is a commutator and r centralizes its own powers.

In H_7, the conjugates d_i, $0 \leq i \leq 3$, have linear part $-I$ and translation parts $(0,0), (0,2), (2,2), (2,0)$, up to the choice of coordinate directions. They generate C_2^3, with $d_3 = d_0 d_1 d_2$. Upstairs, every $[d_i, d_j]$ is in central M and has order at most two. Put $[d_0, d_1] = z^\beta$; conjugating by r gives $[d_1, d_2] = z^\beta$. Write $d_3 = d_0 d_1 d_2 u$ with $u \in M$. Conjugation invariance gives

$$[d_1, d_2] = [d_2, d_3] = [d_2, d_0][d_2, d_1],$$

so $[d_0, d_2] = 1$. Consequently $(d_0 d_1 d_2)^2 = 1$; because d_3 is an involution, $u^2 = 1$, giving $u = z^\nu$. One further conjugation gives $d_4 = d_0$. These are precisely the final row of the table in §5, and (A-C1) forces the actions of the commuting complement $\langle a, c \rangle$. The model maps onto G and has its order $32m$, so is isomorphic. This proves completeness of the eight models.

It remains to recognize the **bare presentations**, where the target order cannot yet be assumed. Let P be one of them and $z = r^{m/2}$, initially of order at most two. For G_1, G_2, if $\epsilon = 1$, a centralizes $z = [d_0, c]$ since it centralizes d_0, c; b, c centralize it since they invert r. If $\epsilon = 0$, d_0 is central; (A-C1) and the evenness of $m/2$ then make z central too. Collection gives at most $8m$ normal forms, attained by the first model.

For G_3, G_4, $d_2 = d_0 z^\epsilon$. If $\epsilon = 0$, applying a as in the second-branch argument gives $[d_0, d_1] = 1$, after which the model relations and centrality of z follow. If $\epsilon = 1$, $[d_0, r^2] = z$ implies that conjugation by d_0 inverts z; as $z^2 = 1$, it centralizes z. Thus z commutes with r and all d_i. Write $j = [d_0, d_1]$. Applying a to the equation for d_2, and using (A-C1), gives $z^a = jz$. Since a fixes d_0, d_1, it fixes j, and $a^2 = 1$ gives $j^2 = 1$. Hence d_0, d_1 centralize j. Also $j^r = [d_1, d_2] = [d_1, d_0 z] = j^{-1} = j$, and c exchanges d_0, d_1, so j is central. Collection yields $(d_0 r)^4 = jr^4$. Since $m/8$ is even, $z^a = (d_0 r)^{m/2} = z$, so $j = 1$. The second model relations now give at most $16m$ forms, all attained by that model.

For G_5–G_8, put $w = [a, r^2]$. If $\beta = 1$, conjugation by a inverts w, and so fixes $z = w^2$. If $\nu = 1$, it inverts $[a, r^4] = z$, again fixing z. If both are zero, a centralizes r^4 and therefore z. Thus z is central in every case, and $[a, r^4] = z^\nu$ makes $M = \langle r^4 \rangle$ normal of order at most $m/4$. Modulo M, the presentation is W with the extra relation $[a, r^2]^2 = 1$. In the affine model $[a, r^2]$ is a doubled unit translation, so its square and conjugates kill precisely $4\mathbb{Z}^2$. Thus the quotient has order 128, and $|P| \le 128(m/4) = 32m$. The constructed model attains this bound and satisfies every relator, so the order comparison principle identifies P with it. This completes presentation recognition and the theorem. ■

The preceding theorem proves the full marked interpretation of L035 Theorem 1.3; the listed word invariants distinguish all the declared families. It does not classify arbitrary two-groups of order 2^n, other types, or the source's unnumbered exceptional $n < 10$ discussion.

6. Central descent and complete binary lifts

This branch concerns the principal nondegenerate classification in Gomi–Loyola–De Las Peñas [GLP18], with the marking retained. Its equivalence sends the distinguished generator set onto the other set. For a connected string diagram this permits precisely the two orders related by reversal. It does not identify arbitrary markings of an abstractly isomorphic group. The rank-two case is the unique dihedral marking of order 1024. The three higher-rank totals below are 125, 105 and 8; all stated types, facets and universal orders are part of the conclusion.

The specifically retained prior results in this branch are the standard string intersection criteria quoted as [GLP18] Lemmas 2.2–2.4, the universal geometric interpretation quoted as its Theorem 4.2, the finite smooth square-quotient classification quoted as its Lemma 5.1, and the exceptional infinite universal decomposition quoted as its Theorem 5.3. Their relevant precise forms are stated below. Exact finite-group and complete coset-enumeration algorithms in GAP, and NQ's universal nilpotent quotient algorithm, are computational inputs. The Atlas supplies definitions of named marked presentations, all included in the defining-data appendix; neither its completeness nor a SmallGroups catalogue is assumed.

6.1. Preliminary adapters and exact quotient intersections

Proposition N1 (endpoint cosets). A finite string C-group of rank d and adjacent orders $m_1,\dots,m_{(d-1)}$ has order at least $2\prod m_i$. In particular, a nondegenerate string two-group of order 1024 has rank at most five.

Proof. Put $a=r_0$, $b=r_1$ and $H=\langle r_1,\dots,r_{(d-1)}\rangle$. IP gives $H\cap\langle a,b\rangle=\langle b\rangle$. If $H(ab)^i=H(ab)^j$, then $(ab)^{i-j}$ lies in the intersection of the rotation subgroup with $\langle b\rangle$, which is trivial. The m_1 such cosets, with $0\le i<m_1$, are therefore distinct. Iterating gives the bound. Each m_i is at least four in the stated two-group case, so $2^{2d-1}\le 1024$, forcing $d\le 5$. ■

The smooth Coxeter quotient assertion in source Theorem 2.7 is the homomorphism from the string Coxeter presentation sending each distinguished generator to its marked involution; the specified adjacent orders make it smooth. A disconnected string diagram yields a direct product of its component subgroups: they commute and IP makes each meet the subgroup on the other components trivially. This gives the source's preliminary reduction; a separate census of every degenerate product is not added to the principal contract. Source Lemma 2.8 is the coordinate-character consequence of Burnside's basis theorem: an irredundant generating list of a finite two-group maps to a basis of $G/\Phi(G)$, and every binary assignment extends to a homomorphism to C_2. No infinite-group assertion is made here.

For rank three write $L=\langle a,b\rangle$ and $R=\langle b,c\rangle$. The retained rank-three criterion says that these two rank-two marked string C-groups, together with $[a,c]=1$ and $L\cap R=\langle b\rangle$, give full IP. Precisely, for d=4 or 5, the versions quoted in source Lemmas 2.3–2.4 require the prefix $H=\langle r_0,\ldots,r_{(d-2)}\rangle$ to have IP and each equality $H\cap\langle r_j,\ldots,r_{(d-1)}\rangle=\langle r_j,\ldots,r_{(d-2)}\rangle$ for $j=1,\ldots,d-1$, with the final right side trivial. These are respectively all four and all five source hypotheses. All finite certificates below instead test every subset pair directly.

If $N\lhd G$ and $N\cap RL=1$, U.Q says the quotient facets have exactly their old intersection and both embed; their product orders are unchanged. This is source Lemma 3.1. If $N\le L$ and the displayed images of a,b still give a rank-two marking on L/N, then $N\cap R\le\langle b\rangle$ and $b\notin N$, so R embeds. The last clause of U.Q gives the same facet intersection and the rank-three criterion gives IP. This is source Lemma 3.2, including its symmetric case. An unrelated marking on the abstract quotient L/N would not suffice.

6.2. A safe central quotient and the three complete lift patterns

Assume throughout this subsection that G is a finite marked string two-group $\langle a,b,c\rangle$, with $m=|ab|\ge4$ and $n=|bc|\ge4$. Put $x=(ab)^{m/2}$ and $y=(bc)^{n/2}$.

Lemma N2 (central elements in an end product).

$$Z(G)\cap RL\subseteq\{1,x,y,yx,ybx\}. \tag{N1}$$

If yx is central, x and y are separately central. The element ybx is never central.

Proof. Every element of RL has a unique expression $h(ab)^j$ with $h\in R$ and $0\le j<m$. Equality of two expressions would put a rotation in $L\cap R=\langle b\rangle$. Conjugation by b negates j, so centrality implies j=0 or m/2. The former case gives an element of $Z(R)=\{1,y\}$. In the latter case write $z=hx$; since x commutes with b, h lies in $C_R(b)=\{1,b,y,yb\}$. The choice h=b fails because the reflection bx does not commute with a when $m\ge4$. This proves (N1).

If yx commutes with a, then y does, because x already does; commuting with c similarly makes x commute with c. Thus both half-rotations are central. Finally $x,y\in\Phi(G)$, since they are even powers of rotations. If z=ybx were central, then a,z,c would have the same Frattini basis as a,b,c and would generate G. A central z is an involution here: $z\in C_G(x)$ implies y commutes with x, and y,b,x are commuting involutions. Thus a,z,c are three commuting involutions, contradicting $|ab|\ge4$. ■

Theorem N3 (central descent with IP). G has a central order-two subgroup N such that the displayed images of a,b,c give a rank-three string C-group on G/N. Successive such descents, allowing degenerate intermediate markings, stop at C_2^3.

Proof. If x is central, take $N=\langle x\rangle$. The marked dihedral quotient L/N has rotation order $m/2\ge2$ and R embeds because $N\cap R=1$. The $N\le L$ clause of U.Q proves the quotient intersection. The same reasoning applies to central y. If neither half-rotation is central, choose any central involution z, which exists in every nontrivial finite two-group. Lemma N2 puts z outside RL; now the injective-ends clause of U.Q proves IP and preserves both end groups. A degenerate rank-three marking is C_2 times a dihedral marking.

Unless it is C_2^3, the central rotation involution of its nonabelian dihedral factor can be factored out while preserving rank three. Orders strictly decrease by two, so this process terminates exactly at C_2^3. ■

This proves source Corollary 3.3, Lemmas 3.4–3.5 and Theorem 3.6. The bottom boundary is necessary in source Corollary 3.7: a rank-three order-eight group cannot have a rank-three order-four quotient. Its literal unrestricted wording is not asserted.

Theorem N4 (three lift patterns with complete presentations). Let Q be a finite rank-three marked string C-group with a complete presentation in its involutions a_0,b_0,c_0, including Coxeter relations and additional relators $R_1,\ldots,R_t$. Let $G=\langle a,b,c\rangle$ be a central extension with kernel $\langle z\rangle$ of order two, mapping those involutions respectively to a_0,b_0,c_0. Suppose the lifts are involutions, $[a,c]=1$ and G has IP. Exactly three order patterns are possible: the left adjacent order doubles, the right adjacent order doubles, or neither doubles. In each case a complete presentation is obtained by adjoining central z with $z^2=1$, each defining Coxeter word of Q set equal to its actual value 1 or z (in particular $(ab)^p=z$ when its quotient order p doubles), and equations $R_i=z^{\epsilon_i}$, $\varepsilon_i\in\{0,1\}$. At least one relation expresses z as a word in a,b,c, so z can be eliminated.

Proof. Each adjacent order is unchanged or doubled. If both doubled, z would be the halfway rotation in both L and R and hence in their intersection $\langle b\rangle$, impossible for a nontrivial dihedral rotation. Every defining word of Q lifts to 1 or z. The indicated presentation P maps onto G and has $P/\langle z\rangle=Q$; centrality and $z^2=1$ imply $|P|\le 2|Q|=|G|$, proving that map an isomorphism. When an adjacent order doubles, its half-power is z. If neither does and every additional relator lifted to 1, the generating tuple would define a surjection $Q\to G$, contradicting the orders. ■

This is source Theorem 3.8 and Table 1. It classifies successful extensions; it does not assert that every binary lift choice has the required order or IP. Those properties and equality of ordered markings must still be checked.

7. Exhaustive bounded classification and named objects

Apply A2.5 (U.Nilpotent/U.Ladder) to the path on d vertices with $k=\log_2|G|-(2d-1)$. The census program of Appendix C is a single parameterized program for the four cases in the following table. NQ computes the universal nilpotent quotient of class k+2. The original generators are transported through its epimorphism and pc isomorphism; they are never replaced by an arbitrary generating list. Every normal kernel in the invariant ladder is generated, with only equal subgroups deduplicated. The binary canonical subgroup key is used only after all ambient relative pc orders have been checked to be two.

d	k	Order of U	Order of $\gamma_3(U)$	Kernels	IP	Fixed	Classes	Meets
3	3	2^{18}	2^{13}	61	37	3	20	3904
3	5	2^{38}	2^{33}	401	235	15	125	25664
4	3	2^{34}	2^{27}	1483	199	11	105	379648
5	1	2^{16}	2^{7}	127	15	1	8	130048

Here Kernels counts terminal normal subgroups, IP counts ordered markings satisfying every intersection test, Fixed counts reversal-fixed markings, and Meets counts actual subgroup-intersection checks. The ladder sizes are respectively (7,23,61), (7,23,61,147,401), (31,267,1483) and (127). These are computed outputs, never candidate filters. All 539264 subset-pair intersections are tested, including every candidate rejected by the kernel-profile criterion, and the two IP decisions agree for every candidate. Burnside's orbit formula gives (ordered+fixed)/2 reversal classes. The four ordered type-{4,8} markings in the first

case satisfy the four distinct word rows in §5, one each; this is the same finite input used in the eight-family completeness proof. The last three cases prove the classification at order 1024, with the rank bound in Proposition N1 proving that no further nondegenerate rank is omitted.

For rank three the twelve type counts, up to reversal, are:

Type	Count	Type	Count
{4,4}	1	{8,8}	22
{4,8}	8	{8,16}	40
{4,16}	14	{8,32}	8
{4,32}	4	{8,64}	4
{4,64}	2	{16,16}	10
{4,128}	2	{16,32}	10

These are exactly source Table 5. The complete presentations defining all named facets, and the full 107 rows of Table 6, appear as literal mathematical data in the complete defining-data artifact of Appendix C. For every one of the 199 rank-four and 15 rank-five ordered markings, the presentation checker of Appendix C reconstructs the quotient from its exact new kernel key and proves unique generator-preserving isomorphisms from both named end groups: 428 exact facet identifications. It computes dual names from reversed generators, rather than assuming that equal-looking types or letter suffixes are self-dual. Under reversal a facet pair (A,B) becomes (dual(B),dual(A)). Counts are compared after this operation for each of the 99 rank-four and eight rank-five rows individually.

The four printed type-{4,16,4} rows in Table 6 have right-hand type {4,16}; their ordered vertex-figure markings necessarily have type {16,4}. The data use precisely these duals, preserving every letter suffix and count. This qualification was checked on printed page 16; it is not an inferred change to class counts.

For each of the 106 finite rows, the complete universal presentation is the union of the two complete facet presentations, shifted to overlapping generator lists, together with the distant-end commutation relation. Completed Todd–Coxeter enumeration at the trivial subgroup gives 101 orders of 1024 and five orders of 2048. The finite checker also verifies actual IP and both exact facet embeddings. The remaining row is the explicitly retained infinite universal of §7.4; it is never sent to an unbounded finite-order test.

7.2. Every printed rank-three presentation and its corrections

The complete finite-presentation protocol in the presentation checker of Appendix C takes the bare displayed relators, performs complete trivial-subgroup Todd–Coxeter enumeration, and checks that the resulting permutation action has the table's full order. It evaluates every relator, proves generation by the specified marked involutions, and tests all 4^d marked-subgroup intersections. Thus its order bounds concern the raw presented groups, not just finite images or nilpotent truncations.

Write C(p,q) for the rank-three Coxeter relations of type {p,q}. In source Table 2, adjoin to C(4,8) the following pairs, set equal to 1:

Name	R_1	R_2
512a	$(abcb)^4$	$(ca(bc)^3ab)^2$
512b	$(cab)^8$	$(a(bc)^3b)^2$
512c	$(c(abcb)^2)^2$	$(ca(bc)^3ab)^2$
512d	$(a(bc)^3b)^2$	$(cab)^5c(ba)^2(cb)^2ab$

All four raw groups have order 512 and are pairwise inequivalent as ordered markings; their complete primary Atlas presentations are identified exactly. Class I of source Example 3.9 would require an order-

512 quotient of C(2,8), whose order is 32, so contributes none. Class II uses C(4,4) with $(abcb)^8=1$, of order 512; its two lifts have central $\tau=(bc)^4$ and $(abcb)^8=1$ or τ. They define the two distinct targets called 1024a and 1024b. For each Table 2 row, the three Class III choices $(R_1,R_2)=(\tau,1),(1,\tau),(\tau,\tau)$, with central $\tau^2=1$, yield exactly:

Base	$(\tau,1)$	$(1,\tau)$	(τ,τ)
512a	1024c	1024d	1024e
512b	1024a	1024c	1024f
512c	1024c	1024g	1024h
512d	512d	1024a	512d

Quotienting by the actual τ subgroup recovers the named base marking in every case. In the two order-512 outcomes $\tau=1$. All eight order-1024 targets are pairwise inequivalent as ordered markings and match the eight type-{4,8} quotients in the unfiltered census bijectively, proving completeness independent of any source table assertion.

Separately evaluating every literal Table 4 row gives the ordered-label map

$$(a,b,c,d,e,f,g,h)_{\text{printed}} \longmapsto (a,b,c,d,c,a,g,c)_{\text{Table 3}}. \tag{N2}$$

To obtain its intended eight targets, the second displayed additional relation in each of printed rows e,f,h must have right-hand side τ instead of 1. These are exactly the already checked (τ,τ) lifts of Table 2 rows a,b,c. No other row or count changes. Printed page 10 was inspected visually; the diagnosis is about the literal source, without claiming priority or unique editorial intent for the repair.

7.3. Universal presentations and all finite examples

For source Definition 4.1 use the combined facet presentation described in §7. The retained geometric universal principle, source Theorem 4.2, states that a nonempty class of regular polytopes with those specified facet and vertex-figure groups has that marked universal string C-group and every member is its smooth quotient. At the group-presentation level the epimorphism is immediate from the defining words; the abstract-polytope interpretation remains an explicit background input.

The bare presentations in Examples 4.3–4.5 give:

Object	Raw order	Type	Ordered end labels
Example 4.3 universal	512	{4,4,4}	{4,4}*32, {4,4}*128
Example 4.4 universal	1024	{4,4,4}	{4,4}*32, {4,4}*256
Example 4.5 universal	2048	{4,4,8}	{4,4}*64, {4,8}*128a
Example 4.5 first quotient	1024	{4,4,8}	same two labels
Example 4.5 second quotient	1024	{4,4,8}	same two labels
Example 4.5 third quotient	1024	{4,4,4}	not in the prescribed smooth class

The first two universal presentations add $(abcb)^2=(bcdc)^4=1$ and $(abcb)^2=(dbc)^8=1$ respectively to the type-{4,4,4} Coxeter relations. The third adds

$$(cab)^4 = 1, \qquad dbcd(cb)^2(dc)^2bc = 1 \tag{N3}$$

to the type-{4,4,8} relations. The complete named facet words in the complete defining-data artifact of Appendix C differ, where necessary, only by cyclic conjugation from these displayed source words. Their exact generator-preserving isomorphisms and the full raw order/IP tests verify each row above.

For the order-2048 group the three printed central-involution words are

$$bcdcbab(dc)^2dba, \qquad (bcdcba)^2, \qquad (cd)^4. \tag{N4}$$

The checker independently enumerates all central involutions and obtains exactly these three distinct elements. Adding each as a raw relator gives the three quotient rows above. The first two are inequivalent marked groups, with unchanged type and end markings. Every normal order-two subgroup is central, so these three exhaust all order-1024 quotients of this universal, and exactly two belong to the prescribed smooth class. Example 4.3 has no quotient of order 1024; Example 4.4 has only itself. These statements concern these exact orders and do not extrapolate centrality to larger kernels.

7.4. Square-lattice centers and the exceptional infinite row

Retain the finite smooth square-quotient classification quoted in source Lemma 5.1: a finite smooth quotient of [4,4] lies in one of the two families obtained by adding $(abcb)^s$=1, of order 8s², or $(abc)^{2s}$=1, of order 16s², for a positive integer s. The former has |abcb|=s and |abc|=2s; the latter has |abcb|=2s and |abc|=2s. The square and diagonal decompositions are now supplied by A3.2, including s=1. In the marking used here they are

$$(\langle a, c^b\rangle \times \langle c, a^b\rangle) \rtimes \langle b\rangle, \qquad (\langle b, b^{ac}\rangle \times \langle b^a, b^c\rangle) \rtimes \langle a, c\rangle,$$

respectively. Each factor has order 2s and the complements have orders two and four. To check the marking in the second expression, its two pairs of reflection axes are x−y=0,1 and x+y=1,0; their products translate by (−1,1) and (1,1). The perpendicular factors, independent translations and complement parity check are exactly those in A3.2. The source instead prints the pairs $\langle b, b^a\rangle$, $\langle b^c, b^{ac}\rangle$. That pairing is erroneous for general s: $bb^a = (ba)^2$ has order two, so its first factor always has order four. The classification is retained; this structural correction and its proof are not assumed from it. Only the finite scope is used; the infinite Coxeter group itself is not asserted to admit such an s.

Use the common affine model of §3 after reversing its endpoints and conjugating all three generators by b. Since the lattices are invariant under the coordinate swap, the same square and diagonal quotients have coordinates

$$a(x, y) = (1 - x, y), \quad b(x, y) = (y, x), \quad c(x, y) = (x, -y). \tag{N5}$$

Its group is W=$\mathbb{Z}^2 \rtimes D_8$; t=abcb translates by (1,0), (abc)² translates by (1,1), (bc)² is (0,−I), and (ab)² is ((1,1),−I). The two finite families are W/s$\mathbb{Z}^2$ and W/L_s, where L_s=⟨(s,s),(s,−s)⟩. The §3 affine normal forms establish these identifications; equivalently the displayed source relations give surjections onto these affine images of the stated orders, hence isomorphisms.

Proposition N5 (all source center formulas). For s=2^m, in W/s$\mathbb{Z}^2$ the center is ⟨(ab)²,(bc)²⟩ when m=1 and $\langle (abc)^s\rangle$ when m≥2. In W/L_s it is ⟨ac,(ab)²,(bc)²⟩ when m=0 and $\langle (abcb)^s\rangle$ when m≥1.

Proof. In a central affine pair (v,M), commutation with all translations forces M to act trivially on T=$\mathbb{Z}^2$/L. If the signed-permutation action on T is faithful, M=I and the center is T^{D_8}.

For L=s$\mathbb{Z}^2$, s≥4, faithfulness is immediate on the two coordinate vectors. Invariance gives 2x=2y=0 and x=y modulo s, leaving precisely 0 and (s/2,s/2). For s=2, the action kernel is the diagonal sign subgroup; to commute also with the linear D_8 subgroup, M must lie in {I,−I} and v in {0,(1,1)}. This gives the first exceptional center. No first-family formula at m=0 is claimed in the source or here.

For L=L_s with s≥2, a nontrivial diagonal sign change cannot act trivially, since (2,0) and (0,2) do not lie in L_s. A signed swap cannot act trivially because its difference on a coordinate vector has a coordinate ±1, not divisible by s. A fixed translation satisfies (2x,0),(0,2y)∈L_s, hence s divides both x,y; modulo L_s its class is 0 or (s,0), both swap-invariant. For s=1 the lattice consists of vectors with even coordinate sum, so T=C_2 with trivial D_8 action and the group is $C_2\times D_8$. Here ac=(t,−I), $(bc)^2=(0,-I)$, and $(ab)^2=(bc)^2$ in the quotient, giving exactly the claimed generating set. Its three displayed generators are not independent. ■

For the sole infinite row of Table 6 retain the exact source Theorem 5.3, attributed there to McMullen–Schulte: the universal type-{4,4,4} group with extra relations $(abcb)^4=(bcd)^4=1$ has

$$U = (W_1 \times W_2) \rtimes \langle b, d\rangle, \quad \langle b, d\rangle \cong C_2^2, \quad W_1 = \langle c^{bd}, a, c^b\rangle \cong [4, 4], \quad W_2 = \langle c, a^b, c^d\rangle = W_1^b. \tag{N6}$$

In particular it is infinite. This theorem, including its marked decomposition, is retained and is not a result of finite coset enumeration.

Proposition N6 (the exceptional kernel location). If π:U→G is a marked epimorphism to a finite string two-group with IP, then $\ker\pi\le W_1\times W_2$ and

$$G \cong ((W_1 \times W_2)/\ker\pi) \rtimes \langle\pi(b), \pi(d)\rangle. \tag{N7}$$

Proof. IP makes the four marked generators irredundant. Their images form a basis of G/Φ(G). The b- and d-coordinate characters, composed with π, agree on every generator with the projection $U\to C_2^2$ in (N6): the displayed W_i generators have even b- and d-parities. Thus every element of kerπ lies in $W_1\times W_2$. The image of ⟨b,d⟩ still has order four by these same coordinate characters and meets the image of $W_1\times W_2$ trivially, proving the split quotient formula. ■

This proves source Lemma 5.4 without assuming quotients preserve IP. At order 1024 the first factor in (N7) has order 256. Finally the two source Theorem 5.5 presentations share the type-{4,4,4} Coxeter relators and $(abcb)^4=(bcd)^4=1$. The first adds $(dcbabdc)^2=1$. The second adds

$$(bdcdba)^2(cbab)^2 = 1, \qquad (abcb)^2(babdcd)^2 = 1. \tag{N8}$$

the presentation checker of Appendix C proves directly that these two raw groups have order 1024 and full IP, and identifies them with the two and only two census markings whose ordered ends are {4,4}*128 and {4,4}*64. Unequal end orders rule out their becoming equal under reversal. This proves the exact exceptional classification without repeating the source's separate case analysis or presuming its presentations finite.

The finite execution in the presentation checker of Appendix C totals 214 complete presentations and 43840 actual IP meets: 72 defining Atlas models, 34 source example/literal-table presentations, the two exceptional presentations, and 106 finite universal presentations. Its distinct logical role from the census program of Appendix C is preserved: the universal-kernel proof gives exhaustion, while complete raw coset tables and exact marked isomorphisms give presentation and name recognition. All code and mathematical data are included as complete external appendices with the bindings in Appendix C. These are

exact finite computations conditional on the stated algorithms, not Lean proofs or experimental extrapolations. The concluding source suggestion about geometric realizations, a separate classification of all degenerate products, and historical novelty of the corrections are outside the contract.

8. Free extensions and two-primary torsion

For the infinite statements we retain the following previously established **free-extension identifications**, including their geometric existence, universality and symmetry conclusions. All input ranks here satisfy the source's standing convention $d \geq 3$:

1. If K is regular with facet F, its free extension is regular, of rank one higher, with group $\Gamma(K) *_{\Gamma(F)} (\Gamma(F) \times C_2)$.
2. If K is oriented chiral with directly regular facets F, its free extension is oriented chiral, of rank one higher, with group $\Gamma(K) *_{\Gamma^+(F)} \Gamma(F)$, where $\Gamma(F) = \Gamma^+(F) \rtimes C_2$.
3. If regular d-polytopes P, Q have isomorphic facets K, their universal alternating semiregular $(d+1)$-polytope has group $\Gamma(P) *_{\Gamma(K)} \Gamma(Q)$, with an additional branch-swapping C_2 precisely if $P \cong Q$.

The first is the standard regular free-extension construction; the second is Schulte–Weiss, *Free extensions of chiral polytopes* (1995); the third is the universal alternating construction of Monson–Schulte (2012, 2021). The embeddings are the indicated face or facet-rotation subgroups, not an assumed embedding of a full facet group in a chiral automorphism group.

In each of the three free-extension identifications the common subgroup is proper in both factors, as the face/flag action or the new reflection shows. In the second factor of the first two constructions its index is two. Every factor has two-primary torsion: in the chiral case the **rotation subgroup** of the facet embeds in $\Gamma(K)$, and its extension by C_2 suffices to prove the property for $\Gamma(F)$. All factors are finitely generated by their finite marked sets. U.Amalgam and, when needed, its finite-extension clause prove exactly **F016 Theorems 7.1, 7.2 and 7.3**, including infinite input groups. No assertion that all elements have finite order is made.

9. Source contracts, corrections and attribution

The F013 common-quotient theorem is recovered by a different kernel. Its specified auxiliary subgroup is false in general. We include the proof of the correction so that no false historical claim is counted as a corollary of this manuscript.

In the path model (A-F), name the central edge coordinates $c_1, \dots, c_{d-1}$, and put

$$\sigma_i = r_{i-1} r_i, \quad A_i = \langle \sigma_1, \dots, \sigma_{i-1}, \sigma_i^2, \sigma_{i+1}, \dots, \sigma_{d-1} \rangle, \quad C = \bigcap_i \Phi(A_i). \tag{A-X1}$$

These are exactly the source's equations (7)–(9), and $\sigma_i^2 = c_i$. The vertex images of the rotations are independent and span the even-weight subspace. Each A_i contains all central coordinates, with vertex-image dimension $d-2$. Thus $|A_i| = 2^{2d-3}$. Squaring a collected class-two word shows its square subgroup is generated by the squares and pair commutators of the generators; Burnside identifies it with $\Phi(A_i)$. The endpoint computations give

$$\Phi(A_1) = \langle c_2, \dots, c_{d-1} \rangle, \qquad \Phi(A_{d-1}) = \langle c_1, \dots, c_{d-2} \rangle.$$

All listed coordinates are squares; no missing endpoint coordinate can arise because every remaining rotation has the missing endpoint vertex coordinate zero. For an internal i, all coordinates except c_i are squares, and $[\sigma_{i-1}, \sigma_{i+1}] = c_i$. Hence $\Phi(A_i) = \langle c_1, \dots, c_{d-1} \rangle$ internally and

$$C = \langle c_2, \dots, c_{d-2} \rangle, \qquad |C| = 2^{d-3} \quad (d \geq 3). \tag{A-X2}$$

For $d = 2$, separately, $C = 1$. For every $d \geq 4$, this model has order 2^{2d-1} but $C \neq 1 = \gamma_3(G)$. It contradicts F013 v1 Theorem 4.2(c)'s kernel order, (d)'s quotient order and prescribed identification, and (e)'s equality with $\gamma_3(G)$. The actual quotient order is 2^{d+2}; it still has IP, with interior adjacent orders two. Thus this counterexample does not refute the separate main common-quotient existence conclusion, nor every clause of 4.2(d).

Two F016 proof repairs also remain explicit. In a diagonal square-map quotient the perpendicular squared translation subgroups can coincide, so their orders cannot be multiplied as in its page-11 argument; (A-S2) uses the correct lattice index. In its chiral extension proof the full facet automorphism group need not embed in the chiral group; §8 only uses its rotation subgroup and a finite index-two extension.

The [HFLh] structural display reverses the square and diagonal parity cases on both pp. 7 and 8 of the retained author version. Its stated dihedral exponents are nonintegral in those respective parities. The affine quotient calculation in §3 gives the corrected assignments in (A-H3), agreeing with the source's preceding calculation. The second cyclic subgroup's displayed definition also repeats the first pair word; its correct generator is $(bc)^4$, as used in Theorem A.H. These literal erroneous displays are not asserted as true. Proposition A3.2 rederives both decompositions of [HFLh, Lemma 3.1], already attributed there to McMullen–Schulte, for every integer $q \geq 2$. Theorem A.H recovers its complete principal Theorem 3.2 with these stated corrections. It does not prove Conjecture 3.3 or infer nonexistence from the construction.

The diagonal factor pairing quoted in [GLP18, Lemma 5.1(ii)] is also corrected in §7.4. Its printed first factor has order four independently of the lattice parameter; A3.2 supplies the correct two factors and complement for every positive parameter. The affine center proofs and finite classification certificates do not use the erroneous pairing. The source's two-family classification remains a separate retained input.

The exact intake is the six specified editions in the references. The following is the principal-result map; the companion ledger expands every normalized statement, multipart clause and all 33 [GLP18] numbered/table records.

Source	Target contract	Derivation and actual U call
[HFLa], Theorem 1.2	Every admissible rank, order and sequence of 2-power adjacent orders	A4.1; square seed; U.SK with two-generator proper-parabolic checks
[HFLb], Theorem 1.2	Rank-three existence in the stated n≥10 range	d=3 specialization of A4.1
[HFLb], Theorem 1.3	All 2+2+4 high-type families; completeness, bare orders/IP and marking distinction	§5; A3.1; cyclic-core input; A2.5/U.Nilpotent/U.Ladder seed and exact relation checks
[HFLh], Theorem 3.2 and auxiliary Lemma 3.1	Exact three-generator presentation, order, pair orders, regular geometry, cyclic kernel product and corrected quotient structure	§3 affine words/decompositions; A4.1 with U.SK.Geometry; A.H sharp presentation bound via U.Order
[HS26], Theorems 1.1–1.2	Canonical marked quotient and geometric cover for every finite nondegenerate rank	A2.1–A2.4; U.Phi; retained cover correspondence
[CFHS25], Theorem 5.1(a–d)	Exact prescribed sections/order, flatness and all required FAP cuts	A4.2 path binding; U.SK antichain, γ3, separator and deletion clauses
[CFHS25], Theorem 5.3	Exact quotient of the universal local presentation	A4.3 normalization, actual local embeddings and finite order bound

Source	Target contract	Derivation and actual U call
[CFHS25], Theorem 6.1(a–b)	Alternating semiregular assembly and exact full automorphism order	A4.2 fork binding; two-deletion fiber isomorphism; retained assembly theorem
[CFHS25], Theorems 7.1–7.3	Regular, chiral and alternating free extensions, including infinite input	§8; all geometric embeddings/properness checked; U.Amalgam
[GLP18], §§2–3/Table 1	Marked quotient criteria, central descent and all binary lift patterns	§6; U.Q/U.Phi; explicit center-in-product argument
[GLP18], Example 3.9/Tables 2–4	Every stated raw presentation and intended named target; diagnosed literal discrepancies	§7.2; complete coset tables and generator-preserving isomorphisms
[GLP18], Examples 4.3–4.5	Exact universals and their order 1024 smooth quotients	§7.3; complete raw presentation orders/IP and all central involutions
[GLP18], Lemmas 5.1–5.4/Theorem 5.5	Lattice centers, retained infinite decomposition, kernel location and exactly two exceptional targets	§7.4; A3.1 affine model; U.Phi; exact census and presentations
[GLP18], Tables 5–6	All 125 rank 3,105 rank 4 and 8 rank 5 reversal classes; all 107 facet rows and universal orders	A2.5/U.Nilpotent/U.Ladder plus all-parabolic census and exact named-object recognition

The [HS26] auxiliary kernel assertions are corrected above; its two principal existence conclusions remain recovered. The [CFHS25] diagonal-translation and chiral-facet proof steps use the corrected arguments stated here. In [GLP18], Corollary 3.7 stops at order8, Table 4's literal rows remain distinguished from their repaired targets, and reversed facet markings in Table 6 are explicit. The retained inputs are part of the contract, not silently counted as newly proved results. No open question, historical novelty claim, unnumbered small-n classification or separate census of degenerate products is included. The power-polytope recollections in [CFHS25] are not extra source targets.

This scientific organization and the supported-kernel applications are repository work, with the support-inclusion simplification credited to independent reviewer 122. The source authors receive credit for the original results and constructions. The classical ingredients of U and the new assembly carry no historical-priority claim. Removing one declared U dependency only invalidates the recorded derivation; it does not show that the conclusion has no other proof.

Appendix C. Complete computational proof interface

There are four complete external artifacts, all part of this volume's proof cost: `phase78_n009_integration_census.g`, `phase78_n009_integration_finite.g`, `phase78_n009_integration_data.g` and `phase78_n009_integration_run.py`. Their legacy filenames identify established reusable artifacts; no surrounding manuscript is imported by those filenames. The first three contain every executed mathematical instruction and all 72 named Atlas presentations,107 Table 6 rows and source relators. The fourth is the complete no-clobber orchestration and output checker. Source editions and artifacts are hash-bound in the companion ledger with their retained verified executions.

The exact algorithms assumed are GAP's finite permutation/pc group arithmetic, subgroup intersection and complete Todd–Coxeter enumeration, and NQ's universal nilpotent quotient algorithm. A completed coset table at the trivial subgroup establishes the raw presentation order; a finite image alone would not

suffice. No source count is used to prune candidate generation, and no saved certificate, private primary file, permutation-array atlas or SmallGroups catalogue is read.

For every (d,k), the census constructs $U_{T,k}$ from its original marked nilpotent quotient map, transports those same generators, and enumerates every U-normal index-two hyperplane preimage k times. Only equal subgroups are deduplicated. Its binary subgroup key is used only after checking that every ambient relative pc order is two. A2.5 supplies completeness. Every terminal candidate is subjected both to the kernel-profile test and to all 4^d actual marked subgroup meets, including rejected candidates. Reversal is computed on the markings; the observed fixed count and the two-element orbit formula give the generator-set classes. The order 256 seed evaluates the four distinct word patterns on each oriented type-{4,8} marking, so its use in §5 is an explicit finite implication with no hidden classification assumption.

The presentation checker constructs all raw groups from complete relators, finishes the trivial-subgroup coset table, evaluates all defining relations, checks generation, exact involution/product orders and all subset intersections. It verifies each facet name by an actual generator-preserving isomorphism. Dual names are computed using reversed generators. It enumerates all central involutions in the order 2048 example and checks every quotient. For each finite Table 6 row it combines both complete facet presentations and distant-end commutation, proves the raw universal order and verifies both facet embeddings. The one infinite row invokes the precisely stated retained decomposition; it is never sent to an unbounded finite-order computation.

The complete run requires 539264 census meets,214 raw presentations and 43840 additional meets,428 exact facet identifications and all 107 Table 6 rows. These are expected results checked after exhaustive generation, not assumptions proving it. Successful replay establishes the finite conditional obligations, not a Lean formalization or a substitute for the parameter proofs.

From the public repository root, run:

```
python3 -B -O research/case_study/phase78_n009_integration_run.p
y --output-dir /private/tmp/NEW-two-group-certificate
```

The output path must not exist. The artifact manifest records source/program hashes, installed versions, complete logs and all generated mathematical data. The application ledger charges the four source files in full, plus the complete shared U body and all written hypotheses in A. Rendered mathematical pages, certificate bytes and any cumulative multi-volume accounting will be reported separately after complete review; the supported-kernel acquisition cost is not waived because it has several users.

References and source-version boundary

[U] *Marked groups, kernels and intersections*, companion manuscript, version 3.

[HFLh] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-hypertopes with* 2^n *chambers*, arXiv:1901.09034v1, 23 January 2019. The retained author PDF has eleven pages, SHA-256 `9d15089679be3828ff9291b4926ef3e3bd1a602449c67ce76ea17c7d9d1a6bfd`. The structural parity and repeated-pair-word corrections are given in §9; no journal-version collation is inferred.

[HFLa] D.-D. Hou, Y.-Q. Feng and D. Leemans, *On regular polytopes of 2-power order*, revised author PDF dated 8 November 2019, journal 2020, DOI 10.1007/s00454-019-00119-5. Exact retained primary file: `sources/catalogue_L032.pdf`.

[HFLb] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-polytopes of order* 2^n, arXiv:1802.08947v2, 22 January 2019; journal catalogue title *2-groups behaving as automorphism groups of regular 3-polytopes*. Exact retained primary file: `sources/catalogue_L035.pdf`.

[HS26] D.-D. Hou and E. Schulte, *String C-groups of 2-power order project onto a common string C-group*, arXiv:2605.20691v1, submitted 20 May 2026, PDF dated 21 May. Exact retained primary file: `sources/catalogue_F013.pdf`.

[CFHS25] G. Cunningham, Y.-Q. Feng, D.-D. Hou and E. Schulte, *Symmetric polytopes whose automorphism groups are 2-groups*, arXiv:2512.15511v1, submitted 17 December 2025, PDF dated 18 December. Exact retained primary file: `sources/symmetric_2groups2025.pdf`.

[GLP18] Y. Gomi, M. L. Loyola and M. L. A. N. De Las Peñas, *String C-groups of order 1024*, Contributions to Discrete Mathematics **13**(1) (2018), 1–22, DOI 10.55016/ojs/cdm.v13i1.62475. Primary journal PDF, SHA256 `0618a57cb5ff35c13587877818a2e77bce442cf2c561f204b6d5e5e095fa2fda`.

Its quoted standard intersection criteria are attributed to M. Conder and D. Oliveros, *The intersection condition for regular polytopes*, JCTA **120** (2013), 1291–1304. The universal and finite square-quotient inputs are quoted from McMullen–Schulte, *Abstract Regular Polytopes* (2002), the latter on pp. 364–365, 368. Its exceptional infinite decomposition is quoted from McMullen–Schulte, *Regular polytopes from twisted Coxeter groups and unitary reflexion groups*, Advances in Mathematics **82** (1990), 35–87. Those earlier proofs are retained in the precise quoted forms; no claim of fresh acquisition or rederivation is made.

The named-object data are credited to M. I. Hartley's Atlas of Small Regular Polytopes; the complete defining-data artifact records the exact primary URL and snapshot hash for every defining presentation. The URL may contain a catalogue identifier, but replay neither calls SmallGroups nor assumes a catalogue is complete. The entire new finite search and recognition were run with GAP 4.15.1 and NQ 2.5.11.

The retained cyclic-core result is Lucchini's theorem in the form recorded by Isaacs, *Finite Group Theory*, Theorem 2.20. The retained alternating assembly is Monson–Schulte, *The assembly problem for alternating semiregular polytopes* (2020), Theorem 5.3 in its FAP case. The chiral free-extension theorem is Schulte–Weiss, *Free extensions of chiral polytopes* (1995); the universal alternating construction is Monson–Schulte (2012,2021), in the exact forms stated in §8. These prior proofs are retained rather than included as additional compressed sources. The regular-polytope and marked-cover correspondences are the standard McMullen–Schulte framework, with the scopes stated in §1.

Complete computational proof appendix

The following four programs are reproduced in full and charged once. Their exact source files accompany the manuscript. The runner executes the three GAP inputs; GAP and its NQ package remain named computational dependencies.

phase78_n009_integration_census.g

```
# Complete path-marked two-group census, parameterized by rank a
nd excess.
# Theorems Q5.5/Q5.6 prove candidate completeness and profile re
cognition.
# Adapted from the reviewed phase21/22 ladder; original markings
 are preserved.
# Every candidate, including profile rejections, receives all 4^
d actual meets.
# Four parameter cases share this program; no source count or na
me filters it.
# The order-256 case additionally checks the four exact L035 see
d word patterns.
# Use a fresh working directory. Existing mathematical output pa
ths are refused.
if LoadPackage("nq")=fail then Error("NQ required");fi;
```

```
SizeScreen([240,40]);;
Run78:=function(d,excess78,prefix78)
 local F,f,rel,i,j,Q,epi,iso,G,r,lc,K,pc,key,levels,M,C,pi,V,
  maxes,L,N,known,nxt,k,edges,maxdim,dim,at,start,allChecks,ss,
  pars,good,accepted,hist,typ,kernelType,rev,selfdual,dualclasse
s,
  acceptedKernels,totalCandidates,gi,gj,pos,typcounts,typekey,
  inds,ambientRows,pcweights,ks,ps,kg,qp,kpars,kchecks,profileGo
od,
  rejectedProfiles,acceptedSelfTypes,selfTypeKey,facetOrders,fac
ets,
  acceptedDetails,checkpointName,a78,b78,c78,r78,d078,z78,patter
ns78,seed78;
 if not (d>=3 and excess78>=1) then Error("Rank/excess input");f
i;
 for k in [1..excess78] do
  if IsExistingFile(Concatenation(prefix78,".level",String(k),".
json")) then Error("Existing checkpoint");fi;
 od;
 if IsExistingFile(Concatenation(prefix78,".accepted.json")) or
IsExistingFile(Concatenation(prefix78,".seed.json")) then Error(
"Existing output");fi;
 seed78:=[];
 F:=FreeGroup(d);f:=GeneratorsOfGroup(F);rel:=List(f,x->x^2);
 for i in [1..d] do for j in [i+1..d] do
  if j-i>1 then Add(rel,Comm(f[i],f[j]));fi;
 od;od;
 Q:=F/rel;epi:=NqEpimorphismNilpotentQuotient(Q,excess78+2);
 iso:=IsomorphismPcGroup(Image(epi));G:=Image(iso);
 r:=List(GeneratorsOfGroup(Q),x->Image(iso,Image(epi,x)));
 lc:=LowerCentralSeries(G);K:=lc[3];pc:=Pcgs(G);pcweights:=List(
[1..Length(pc)],t->2^(t-1));
 if not ForAll(RelativeOrders(pc),n->n=2) then Error("Binary can
onical-pcgs key requires relative orders two");fi;
 if Index(G,K)<>2^(2*d-1) then Error("Universal canonical quotie
nt order");fi;
 key:=function(H)
  return List(CanonicalPcgs(InducedPcgsByGenerators(pc,Generator
sOfGroup(H))),
   x->ExponentsOfPcElement(pc,x)*pcweights);
 end;
 Print("VERSION GAP=",GAPInfo.Version," NQ=",PackageInfo("nq")[1
].Version,"\n");
 Print("UNIVERSAL rank=",d," excess=",excess78," order=",Size(G)
," gamma3=",Size(K)," abelianization_type=",AbelianInvariants(K)
,"\n");
 levels:=[K];start:=Runtime();
 for k in [1..excess78] do
  known:=NewDictionary([],true);nxt:=[];edges:=0;maxdim:=0;at:=0
;
  for M in levels do
   C:=ClosureGroup(CommutatorSubgroup(M,G),FrattiniSubgroup(M));
```

```
   pi:=NaturalHomomorphismByNormalSubgroup(M,C);V:=Image(pi);
   if not IsElementaryAbelian(V) then Error("Coinvariant square
quotient");fi;
   dim:=LogInt(Size(V),2);maxdim:=Maximum(maxdim,dim);
   maxes:=MaximalSubgroups(V);
   if Length(maxes)<>2^dim-1 then Error("All hyperplanes");fi;
   for L in maxes do
    N:=PreImage(pi,L);edges:=edges+1;
    inds:=key(N);
    if LookupDictionary(known,inds)=fail then
     if Index(K,N)<>2^k or not IsNormal(G,N) then Error("Invaria
nt ladder kernel");fi;
     AddDictionary(known,inds,Length(nxt)+1);Add(nxt,N);
    fi;
   od;
   at:=at+1;
   if at mod 100=0 then Print("PROGRESS depth=",k," parents=",at
,"/",Length(levels)," unique=",Length(nxt)," elapsed_ms=",Runtim
e()-start,"\n");fi;
  od;
  Print("LEVEL index=",2^k," parents=",Length(levels)," hyperpla
ne_edges=",edges," unique=",Length(nxt)," max_coinvariant_dimens
ion=",maxdim," elapsed_ms=",Runtime()-start,"\n");
  levels:=nxt;
  if true then
   checkpointName:=Concatenation(prefix78,".level",String(k),".j
son");
   PrintTo(checkpointName,List(levels,key),"\n");
  fi;
 od;
 # Exact actual-IP checks for every final candidate, including e
very meet.
 ss:=Combinations([1..d]);allChecks:=0;accepted:=0;selfdual:=0;
 ks:=List(ss,S->Intersection(K,Subgroup(G,r{S})));
 kchecks:=0;rejectedProfiles:=0;acceptedSelfTypes:=[];acceptedDe
tails:=[];
 acceptedKernels:=[];hist:=[];typcounts:=[];totalCandidates:=Len
gth(levels);
 rev:=GroupHomomorphismByImages(G,G,r,Reversed(r));
 if rev=fail or not IsBijective(rev) then Error("Marked reversal
 automorphism");fi;
 at:=0;
 for N in levels do
  # The kernel quotient has order 2^excess78 and need not be abe
lian.
  qp:=NaturalHomomorphismByNormalSubgroup(K,N);kg:=Image(qp);
  if Size(kg)<>2^excess78 then Error("Small kernel quotient");fi
;
  kpars:=List(ks,H->Image(qp,H));profileGood:=true;
  for i in [1..Length(ss)] do
   for j in [i+1..Length(ss)] do
    pos:=Position(ss,Intersection(ss[i],ss[j]));
```

```
    kchecks:=kchecks+1;
    if Intersection(kpars[i],kpars[j])<>kpars[pos] then profileG
ood:=false;break;fi;
   od;
   if not profileGood then break;fi;
  od;
  if not profileGood then
   rejectedProfiles:=rejectedProfiles+1;
   if at mod 100=0 then Print("PROFILE_PROGRESS candidates=",at,
"/",totalCandidates," accepted=",accepted," elapsed_ms=",Runtime
()-start,"\n");fi;
  fi;
  pi:=NaturalHomomorphismByNormalSubgroup(G,N);Q:=Image(pi);
  f:=List(r,x->Image(pi,x));
  if Size(Q)<>2^(2*d-1+excess78) or Subgroup(Q,f)<>Q then Error(
"Candidate order/marking");fi;
  pars:=List(ss,S->Subgroup(Q,f{S}));good:=true;
  for i in [1..Length(ss)] do for j in [1..Length(ss)] do
   pos:=Position(ss,Intersection(ss[i],ss[j]));
   if Intersection(pars[i],pars[j])<>pars[pos] then good:=false;
fi;
   allChecks:=allChecks+1;
  od;od;
  if good<>profileGood then Error("Kernel-profile and actual-IP
criteria disagree");fi;
  if good then
   accepted:=accepted+1;Add(acceptedKernels,key(N));
   if Image(rev,N)=N then selfdual:=selfdual+1;fi;
   typ:=List([1..d-1],i->Order(f[i]*f[i+1]));
   if Minimum(typ)<4 then Error("Degenerate marking escaped cano
nical quotient");fi;
   kernelType:=AbelianInvariants(Image(pi,K)/DerivedSubgroup(Ima
ge(pi,K)));
   Add(hist,[Length(LowerCentralSeries(Q))-1,kernelType]);
   typekey:=Minimum(typ,Reversed(typ));Add(typcounts,typekey);
   if Image(rev,N)=N then Add(acceptedSelfTypes,typekey);fi;
   facetOrders:=[Size(Subgroup(Q,f{[1..d-1]})),Size(Subgroup(Q,f
{[2..d]}))];
   Add(acceptedDetails,[key(N),typ,Length(LowerCentralSeries(Q))
-1,kernelType,Image(rev,N)=N,facetOrders]);
   if d=3 and excess78=3 and typ=[4,8] then
    a78:=f[1];b78:=f[2];c78:=f[3];r78:=b78*c78;d078:=(a78*b78)^2
;z78:=r78^4;
    patterns78:=[];
    if Comm(a78,r78^2)^2=One(Q) and Comm(a78,r78^4)=One(Q) then
Add(patterns78,1);fi;
    if Comm(a78,r78^2)^2=z78 and Comm(a78,r78^4)=One(Q) then Add
(patterns78,2);fi;
    if Comm(d078,r78^2)=z78 then Add(patterns78,3);fi;
    if Comm((r78^2)^a78,r78)=z78 then Add(patterns78,4);fi;
    if Length(patterns78)<>1 then Error("Unclassified or overlap
ping order-256 word pattern");fi;
```

```
    Add(seed78,[key(N),patterns78[1]]);
   fi;
  fi;
  at:=at+1;
  if at mod 100=0 then Print("IP_PROGRESS candidates=",at,"/",to
talCandidates," accepted=",accepted," elapsed_ms=",Runtime()-sta
rt,"\n");fi;
 od;
 dualclasses:=(accepted+selfdual)/2;
 Print("RESULT rank=",d," excess=",excess78," candidates=",total
Candidates," ordered=",accepted," selfdual=",selfdual," up_to_re
versal=",dualclasses," actual_meets=",allChecks," elapsed_ms=",R
untime()-start,"\n");
 Print("CLASS_KERNEL_ABELIANIZATION_HISTOGRAM ",Collected(hist),
"\n");
 Print("ORDERED_CANONICAL_TYPE_HISTOGRAM ",Collected(typcounts),
"\n");
 Print("SELFDUAL_TYPE_HISTOGRAM ",Collected(acceptedSelfTypes),"
\n");
 Print("KERNEL_PROFILE_CHECKS comparisons=",kchecks," rejected="
,rejectedProfiles," directly_verified=",accepted,"\n");
 if true then
  PrintTo(Concatenation(prefix78,".accepted.json"),acceptedDetai
ls,"\n");
 fi;
 PrintTo(Concatenation(prefix78,".seed.json"),seed78,"\n");
 if Length(seed78)>0 then Print("SEED_WORD_PATTERNS ",Collected(
List(seed78,x->x[2])),"\n");fi;
 Print("PASS: complete invariant-kernel ladder and all-candidate
 actual-IP cross-check; counts are output, never candidate filte
rs.\n");
 return rec(rank:=d,ordered:=accepted,selfdual:=selfdual,dualcla
sses:=dualclasses,kernel_keys:=acceptedKernels);
end;;
if not IsBound(cases78) then cases78:=[[3,3],[3,5],[4,3],[5,1]];
fi;
results78:=[];
for case78 in cases78 do
 Add(results78,Run78(case78[1],case78[2],Concatenation("phase78_
d",String(case78[1]),"_k",String(case78[2]))));
od;
QUIT;
```

phase78_n009_integration_finite.g

```
# Source-bound raw presentations, all N009 finite table labels a
nd universal orders.
# All orders below come from COMPLETE Todd--Coxeter coset tables
, not NQ bounds.
SizeScreen([240,40]);;
report24:=[];;meet24:=0;;tc24:=0;;
Cox24:=function(f,typ)
```

```
 local n,rels,i,j;n:=Length(typ)+1;rels:=List(f{[1..n]},x->x^2);
 for i in [1..n] do for j in [i+1..n] do
  if j=i+1 then Add(rels,(f[i]*f[j])^typ[i]);else Add(rels,Comm(
f[i],f[j]));fi;
 od;od;return rels;
end;;
TableWords24:=function(f)
 local a,b,c;a:=f[1];b:=f[2];c:=f[3];
 return [[(a*b*c*b)^4,(c*a*(b*c)^3*a*b)^2],
 [(c*a*b)^8,(a*(b*c)^3*b)^2],
 [(c*(a*b*c*b)^2)^2,(c*a*(b*c)^3*a*b)^2],
 [(a*(b*c)^3*b)^2,(c*a*b)^5*c*(b*a)^2*(c*b)^2*a*b]];
end;;
Build24:=function(name,F,rels,rank)
 local fg,table,n,gs,G,mark,sets,pars,i,j,typ;
 fg:=GeneratorsOfGroup(F);table:=CosetTableFromGensAndRels(fg,re
ls,[]);n:=Length(table[1]);
 gs:=List([1..Length(fg)],i->PermList(table[2*i-1]));G:=Group(gs
);mark:=gs{[1..rank]};
 if Size(G)<>n or Subgroup(G,mark)<>G then Error("Complete regul
ar coset action/order/generation",name);fi;
 if not ForAll(rels,w->MappedWord(w,fg,gs)=One(G)) then Error("R
aw relators",name);fi;
 sets:=Combinations([1..rank]);pars:=List(sets,S->Subgroup(G,mar
k{S}));
 for i in [1..Length(sets)] do for j in [1..Length(sets)] do
  if Intersection(pars[i],pars[j])<>pars[Position(sets,Intersect
ion(sets[i],sets[j]))] then Error("Actual IP",name);fi;
  meet24:=meet24+1;
 od;od;
 typ:=List([1..rank-1],i->Order(mark[i]*mark[i+1]));
 Add(report24,[name,n,typ]);tc24:=tc24+1;
 Print("PRESENTATION ",name," order=",n," type=",typ," IP=PASS\n
");
 return rec(name:=name,group:=G,mark:=mark,allgens:=gs,order:=n,
F:=F,rels:=rels);
end;;
Same24:=function(A,B)
 local h;h:=GroupHomomorphismByImages(A.group,B.group,A.mark,B.m
ark);return h<>fail and IsBijective(h);
end;;

f:=[];;model:=fail;;pair78:=[];;
Read("phase78_n009_integration_data.g");;
for A in atlas78 do
 A.words:=List(A.words,w->List(w,c->Int([c])));
 F:=FreeGroup(Length(A.typ)+1);f:=GeneratorsOfGroup(F);
 rel:=List(A.words,w->Product(List(w,i->f[i])));
 model:=Build24(Concatenation("Atlas_",A.label),F,rel,Length(f))
;
 if model.order<>A.order or List([1..Length(A.typ)],i->Order(mod
el.mark[i]*model.mark[i+1]))<>A.typ then Error("Complete Atlas o
```

```
rder/type",A.label);fi;
 A.group:=model.group;A.gens:=model.mark;
od;
atlas22:=Filtered(atlas78,A->Length(A.typ)=2 and A.order<512);;
atlas4_22:=Filtered(atlas78,A->Length(A.typ)=3);;
atlas24:=Filtered(atlas78,A->Length(A.typ)=2 and A.order=512);;
if [Length(atlas22),Length(atlas4_22),Length(atlas24)]<>[56,11,5
] then Error("Library partition");fi;
Identify78:=function(H,gs,library)
 local typ,matches,A,hom;
 typ:=List([1..Length(gs)-1],i->Order(gs[i]*gs[i+1]));matches:=[
];
 for A in library do
  if A.order=Size(H) and A.typ=typ then
   hom:=GroupHomomorphismByImages(A.group,H,A.gens,gs);
   if hom<>fail and IsBijective(hom) then Add(matches,A.label);f
i;
  fi;
 od;
 if Length(matches)<>1 then Error("Unique ordered marked label",
matches,typ,Size(H));fi;
 return matches[1];
end;;
duals78:=List(Concatenation(atlas22,atlas4_22),A->[A.label,Ident
ify78(A.group,Reversed(A.gens),Concatenation(atlas22,atlas4_22))
]);;
Dual78:=function(label) return First(duals78,x->x[1]=label)[2];e
nd;;
Canon78:=function(pair) return Minimum(pair,[Dual78(pair[2]),Dua
l78(pair[1])]);end;;
F:=FreeGroup(3);f:=GeneratorsOfGroup(F);words:=TableWords24(f);;
base24:=List([1..4],i->Build24(Concatenation("Table2_512",[CharI
nt(96+i)]),F,Concatenation(Cox24(f,[4,8]),words[i]),3));;
if not ForAll(base24,A->A.order=512) then Error("Table2 orders")
;fi;
for i in [1..4] do for j in [i+1..4] do if Same24(base24[i],base
24[j]) then Error("Distinct Table2 markings");fi;od;od;
low24:=Build24("Example3.9_classI_base",F,Cox24(f,[2,8]),3);;
if low24.order<>32 then Error("ClassI boundary");fi;
q44:=Build24("Example3.9_classII_base",F,Concatenation(Cox24(f,[
4,4]),[(f[1]*f[2]*f[3]*f[2])^8]),3);;
if q44.order<>512 then Error("ClassII base");fi;
F4:=FreeGroup(4);f4:=GeneratorsOfGroup(F4);t:=f4[4];a:=f4[1];b:=
f4[2];c:=f4[3];;
central24:=Concatenation([t^2],List(f4{[1..3]},x->Comm(t,x)));;
classII24:=[];;
for eps in [0,1] do
 A:=Build24(Concatenation("Example3.9_classII_",String(eps)),F4,
Concatenation(Cox24(f4,[4,8]),central24,[t^-1*(b*c)^4,(a*b*c*b)^
8*t^-eps]),3);;
 if A.order<>1024 or Order(A.allgens[4])<>2 then Error("ClassII
lift");fi;Add(classII24,A);
```

```
od;
words4:=TableWords24(f4);patterns24:=[[1,0],[0,1],[1,1]];lifts24
:=[];;pi:=fail;;
for i in [1..4] do
 row:=[];
 for eps in patterns24 do
  A:=Build24(Concatenation("Table3_",String(i),"_",String(eps)),
F4,Concatenation(Cox24(f4,[4,8]),central24,[words4[i][1]*t^-eps[
1],words4[i][2]*t^-eps[2]]),3);;
  Add(row,A);
  N:=Subgroup(A.group,[A.allgens[4]]);pi:=NaturalHomomorphismByN
ormalSubgroup(A.group,N);Q:=Image(pi);;
  qm:=List(A.mark,x->Image(pi,x));h:=GroupHomomorphismByImages(b
ase24[i].group,Q,base24[i].mark,qm);;
  if h=fail or not IsBijective(h) then Error("Lift quotient is e
xact Table2 marking");fi;
 od;Add(lifts24,row);
od;
# Names a,b from Example3.9 ClassII; c,d,e,f,g,h from the explic
it Table3 assignments.
refs24:=[classII24[1],classII24[2],lifts24[1][1],lifts24[1][2],l
ifts24[1][3],lifts24[2][3],lifts24[3][2],lifts24[3][3]];;
if not ForAll(refs24,A->A.order=1024) then Error("Eight target o
rders");fi;
for i in [1..8] do for j in [i+1..8] do if Same24(refs24[i],refs
24[j]) then Error("Eight targets not distinct",i,j);fi;od;od;
expected24:=[[3,4,5],[1,3,6],[3,7,8],[9,1,9]];;table3matches24:=
[];;
for i in [1..4] do for j in [1..3] do
 k:=expected24[i][j];if k=9 then B:=base24[4];else B:=refs24[k];
fi;
 if not Same24(lifts24[i][j],B) then Error("Exact Table3 assignm
ent",i,j,k);fi;
 Add(table3matches24,[i,patterns24[j],k,lifts24[i][j].order,Orde
r(lifts24[i][j].allgens[4])]);
od;od;
# Literal Table4, preserving each printed e (identity) even when
 Table3 disagrees.
literal24:=[];;
for i in [1..8] do
 if i<=2 then rel:=Concatenation(Cox24(f4,[4,8]),central24,[t^-1
*(b*c)^4,(a*b*c*b)^8*t^-(i-1)]);
 elif i in [3,5] then rel:=Concatenation(Cox24(f4,[4,8]),central
24,[words4[1][1]*t^-1,words4[1][2]]);
 elif i=4 then rel:=Concatenation(Cox24(f4,[4,8]),central24,[wor
ds4[1][1],words4[1][2]*t^-1]);
 elif i=6 then rel:=Concatenation(Cox24(f4,[4,8]),central24,[wor
ds4[2][1]*t^-1,words4[2][2]]);
 elif i=7 then rel:=Concatenation(Cox24(f4,[4,8]),central24,[wor
ds4[3][1],words4[3][2]*t^-1]);
 else rel:=Concatenation(Cox24(f4,[4,8]),central24,[words4[3][1]
*t^-1,words4[3][2]]);fi;
```

```
 A:=Build24(Concatenation("Table4_literal_",String(i)),F4,rel,3)
;;
 matches:=Filtered([1..8],j->Same24(A,refs24[j]));if Length(matc
hes)<>1 then Error("Literal Table4 identification");fi;
 Add(literal24,[i,matches[1]]);
od;
if literal24<>[[1,1],[2,2],[3,3],[4,4],[5,3],[6,1],[7,7],[8,3]]
then Error("Unexpected literal Table4 mapping",literal24);fi;
Print("TABLE3_MATCHES ",table3matches24,"\n");Print("TABLE4_LITE
RAL_MATCHES ",literal24,"\n");
# Examples4.3--4.5, using exactly their printed raw relators.
a:=f4[1];b:=f4[2];c:=f4[3];d:=f4[4];;
ex43:=Build24("Example4.3",F4,Concatenation(Cox24(f4,[4,4,4]),[(
a*b*c*b)^2,(b*c*d*c)^4]),4);;
ex44:=Build24("Example4.4",F4,Concatenation(Cox24(f4,[4,4,4]),[(
a*b*c*b)^2,(d*b*c)^8]),4);;
ex45rel:=Concatenation(Cox24(f4,[4,4,8]),[(c*a*b)^4,d*b*c*d*(c*b
)^2*(d*c)^2*b*c]);;
ex45:=Build24("Example4.5_universal",F4,ex45rel,4);;
if [ex43.order,ex44.order,ex45.order]<>[512,1024,2048] then Erro
r("Example raw orders");fi;
tauwords24:=[b*c*d*c*b*a*b*(d*c)^2*d*b*a,(b*c*d*c*b*a)^2,(c*d)^4
];;
tauimages24:=List(tauwords24,w->MappedWord(w,f4,ex45.mark));;
zinv24:=Filtered(Elements(Centre(ex45.group)),z->Order(z)=2);;
if Set(tauimages24)<>Set(zinv24) or Length(zinv24)<>3 then Error
("Complete central involution list");fi;
ex45quot24:=[];;
for i in [1..3] do
 A:=Build24(Concatenation("Example4.5_quotient",String(i)),F4,Co
ncatenation(ex45rel,[tauwords24[i]]),4);;
 if A.order<>1024 then Error("Quotient raw order");fi;Add(ex45qu
ot24,A);
od;
if List(ex45quot24,A->List([1..3],i->Order(A.mark[i]*A.mark[i+1]
)))<>[[4,4,8],[4,4,8],[4,4,4]] then Error("Quotient smoothness")
;fi;
if Same24(ex45quot24[1],ex45quot24[2]) then Error("Distinct smoo
th quotients");fi;
# Primary Atlas identities for the five source order-512 names.

for i in [1..5] do
 A:=atlas24[i];B:=rec(group:=Group(A.gens),mark:=A.gens);
 if i=1 then target:=q44;else target:=base24[i-1];fi;
 if not Same24(target,B) then Error("Primary Atlas base label",A
.label);fi;
 if not ForAll(A.words,w->Product(List(w,j->A.gens[j]))=One(B.gr
oup)) then Error("Primary Atlas defining word",A.label);fi;
od;
Print("PASS five exact primary Atlas order-512 marked labels\n")
;
# Exact facet identities for Examples4.3/4.4/4.5 and the two smo
```

```
oth quotients.

exampleFacets24:=[];;
for entry in [[ex43,["{4,4}*32","{4,4}*128"]],[ex44,["{4,4}*32",
"{4,4}*256"]],[ex45,["{4,4}*64","{4,8}*128a"]],[ex45quot24[1],["
{4,4}*64","{4,8}*128a"]],[ex45quot24[2],["{4,4}*64","{4,8}*128a"
]]] do
 A:=entry[1];
 for j in [1,2] do
  ar:=First(atlas22,x->x.label=entry[2][j]);B:=rec(group:=Group(
ar.gens),mark:=ar.gens);
  mark:=A.mark{[j..j+2]};target:=rec(group:=Subgroup(A.group,mar
k),mark:=mark);
  if not Same24(target,B) then Error("Example facet label",A.nam
e,entry[2][j]);fi;
 od;
 Add(exampleFacets24,[A.name,entry[2]]);
od;
Print("PASS ten exact primary Atlas example facet labels\n");
# Match the eight printed/Table3 target names to the independent
 unfiltered census.
if LoadPackage("nq")=fail then Error("NQ required for unfiltered
 census adapter");fi;
Print("SOFTWARE GAP=",GAPInfo.Version," NQ=",PackageInfo("nq")[1
].Version,"\n");
accepted24:=EvalString(StringFile("phase78_d3_k5.accepted.json")
);;
F3:=FreeGroup(3);v:=GeneratorsOfGroup(F3);P:=F3/Concatenation(Li
st(v,x->x^2),[Comm(v[1],v[3])]);;
ep:=NqEpimorphismNilpotentQuotient(P,7);is:=IsomorphismPcGroup(I
mage(ep));U:=Image(is);;
rr:=List(GeneratorsOfGroup(P),x->Image(is,Image(ep,x)));pc24:=Pc
gs(U);;
if not ForAll(RelativeOrders(pc24),x->x=2) then Error("Frozen bi
nary pc coordinates");fi;
Decode24:=function(n) local e,i;e:=[];for i in [1..Length(pc24)]
 do Add(e,n mod 2);n:=QuoInt(n,2);od;if n<>0 then Error("Coordin
ate overflow");fi;return PcElementByExponents(pc24,e);end;;
censusMatches24:=[];;
for entry in Filtered(accepted24,x->x[2]=[4,8]) do
 N:=Subgroup(U,List(entry[1],Decode24));
 if not IsNormal(U,N) or Index(U,N)<>1024 then Error("Frozen cen
sus reconstruction");fi;
 pi:=NaturalHomomorphismByNormalSubgroup(U,N);mark:=List(rr,x->I
mage(pi,x));target:=rec(group:=Image(pi),mark:=mark);
 matches:=Filtered([1..8],j->Same24(refs24[j],target));
 if Length(matches)<>1 then Error("Exact census match",matches);
fi;
 Add(censusMatches24,[entry[1],matches[1]]);
od;
if Set(List(censusMatches24,x->x[2]))<>[1..8] or Length(censusMa
tches24)<>8 then Error("All eight census targets");fi;
```

```
Print("PASS all eight type-{4,8} classes matched to new unfilter
ed census\n");
PrintTo("phase78_n009_integration_example_facets.json",exampleFa
cets24,"\n");
PrintTo("phase78_n009_integration_census_matches.json",censusMat
ches24,"\n");
PrintTo("phase78_n009_integration_examples_results.json",report2
4,"\n");
PrintTo("phase78_n009_integration_table3_matches.json",table3mat
ches24,"\n");
PrintTo("phase78_n009_integration_table4_literal_matches.json",l
iteral24,"\n");

# Reconstruct both higher-rank catalogues from the new complete
kernel census.
facetRows78:=[];;tableResults78:=[];;universal78:=[];;exceptionT
argets78:=[];;
for d78 in [4,5] do
 excess78:=11-2*d78;
 accepted78:=EvalString(StringFile(Concatenation("phase78_d",Str
ing(d78),"_k",String(excess78),".accepted.json")));
 F:=FreeGroup(d78);f:=GeneratorsOfGroup(F);rel:=List(f,x->x^2);
 for i in [1..d78] do for j in [i+1..d78] do if j-i>1 then Add(r
el,Comm(f[i],f[j]));fi;od;od;
 P:=F/rel;ep:=NqEpimorphismNilpotentQuotient(P,excess78+2);is:=I
somorphismPcGroup(Image(ep));U:=Image(is);
 rr:=List(GeneratorsOfGroup(P),x->Image(is,Image(ep,x)));pc24:=P
cgs(U);
 if not ForAll(RelativeOrders(pc24),x->x=2) then Error("Binary c
ensus coordinates");fi;
 if d78=4 then library78:=atlas22;else library78:=atlas4_22;fi;
 for entry in accepted78 do
  N:=Subgroup(U,List(entry[1],Decode24));
  if not IsNormal(U,N) or Index(U,N)<>1024 then Error("Census re
construction");fi;
  pi:=NaturalHomomorphismByNormalSubgroup(U,N);H:=Image(pi);mark
:=List(rr,x->Image(pi,x));
  labels78:=[Identify78(Subgroup(H,mark{[1..d78-1]}),mark{[1..d7
8-1]},library78),Identify78(Subgroup(H,mark{[2..d78]}),mark{[2..
d78]},library78)];
  Add(facetRows78,[entry[1],entry[2],labels78,entry[5]]);
  if labels78=["{4,4}*128","{4,4}*64"] then Add(exceptionTargets
78,rec(group:=H,mark:=mark));fi;
 od;
od;
if Length(facetRows78)<>214 or Length(exceptionTargets78)<>2 the
n Error("Facet/exceptions cardinality");fi;
if Length(Set(List(rows78,r->Canon78(r[1]))))<>107 then Error("D
istinct Table6 pairs");fi;
for row78 in rows78 do
 pair78:=Canon78(row78[1]);hits78:=Filtered(facetRows78,r->Canon
78(r[3])=pair78);
```

```
 fixed78:=Number(hits78,r->r[4]);actual78:=(Length(hits78)+fixed
78)/2;
 if actual78<>row78[3] then Error("Exact Table6 count",row78,act
ual78);fi;
 if not ForAll(hits78,r->Minimum(r[2],Reversed(r[2]))=Minimum(ro
w78[4],Reversed(row78[4]))) then Error("Table6 type");fi;
 Add(tableResults78,[row78[1],Length(hits78),fixed78,actual78]);
 if row78[2]<>0 then
  A:=First(atlas78,x->x.label=row78[1][1]);B:=First(atlas78,x->x
.label=row78[1][2]);
  F:=FreeGroup(Length(A.typ)+2);f:=GeneratorsOfGroup(F);
  rel:=Concatenation(List(A.words,w->Product(List(w,i->f[i]))),L
ist(B.words,w->Product(List(w,i->f[i+1]))),[Comm(f[1],f[Length(f
)])]);
  model:=Build24(Concatenation("Universal_",A.label,"_",B.label)
,F,rel,Length(f));
  if model.order<>row78[2] then Error("Complete universal order"
,row78);fi;
  if [Identify78(Subgroup(model.group,model.mark{[1..Length(f)-1
]}),model.mark{[1..Length(f)-1]},atlas78),Identify78(Subgroup(mo
del.group,model.mark{[2..Length(f)]}),model.mark{[2..Length(f)]}
,atlas78)]<>row78[1] then Error("Universal exact facet identitie
s");fi;
  Add(universal78,[row78[1],model.order]);
 fi;
od;
# Exact source Theorem5.5 raw presentations; match both independ
ently found markings.
F:=FreeGroup(4);f:=GeneratorsOfGroup(F);a:=f[1];b:=f[2];c:=f[3];
d:=f[4];
common78:=Concatenation(Cox24(f,[4,4,4]),[(a*b*c*b)^4,(b*c*d)^4]
);;
extras78:=[[(d*c*b*a*b*d*c)^2],[(b*d*c*d*b*a)^2*(c*b*a*b)^2,(a*b
*c*b)^2*(b*a*b*d*c*d)^2]];;
exceptionMatches78:=[];;
for i in [1,2] do
 model:=Build24(Concatenation("Theorem5.5_",String(i)),F,Concate
nation(common78,extras78[i]),4);
 if model.order<>1024 then Error("Exceptional raw order");fi;
 matches78:=Filtered([1,2],j->Same24(model,exceptionTargets78[j]
));
 if Length(matches78)<>1 then Error("Exceptional marked isomorph
ism");fi;
 Add(exceptionMatches78,matches78[1]);
od;
if Set(exceptionMatches78)<>[1,2] then Error("Exceptional comple
teness");fi;
if Length(universal78)<>106 or Collected(List(universal78,x->x[2
]))<>[[1024,101],[2048,5]] then Error("All finite universals");f
i;
# Table5 counts, from the full rank3 census, only after generati
on is complete.
```

```
expected578:=[[[4,4],1],[[4,8],8],[[4,16],14],[[4,32],4],[[4,64]
,2],[[4,128],2],[[8,8],22],[[8,16],40],[[8,32],8],[[8,64],4],[[1
6,16],10],[[16,32],10]];;
for row78 in expected578 do
 hits78:=Filtered(accepted24,r->Minimum(r[2],Reversed(r[2]))=row
78[1]);
 if (Length(hits78)+Number(hits78,r->r[5]))/2<>row78[2] then Err
or("Table5 count");fi;
od;
PrintTo("phase78_n009_integration_facet_matches.json",facetRows7
8,"\n");
PrintTo("phase78_n009_integration_duals.json",duals78,"\n");
PrintTo("phase78_n009_integration_table6_results.json",tableResu
lts78,"\n");
PrintTo("phase78_n009_integration_universal_orders.json",univers
al78,"\n");
PrintTo("phase78_n009_integration_all_presentations.json",report
24,"\n");
Print("PASS complete_presentations=",tc24," actual_IP_meets=",me
et24," atlas=72 raw_examples=34 exceptional=2 finite_universals=
106 exact_census_facets=428 table6_rows=107 table5_rows=12\n");
QUIT;
```

phase78_n009_integration_data.g

```
# Primary Atlas marked presentations; names are definitions, not
 a completeness input.
# Each word is a string of generator indices 1,...,5, in display
ed order.
# Exact source URL and captured-primary SHA256 precede every mat
hematical record.
atlas78:=[
# https://www.abstract-polytopes.com/atlas/128/916/2.html sha256
=bbc4b3daca13f8719be87e21c622c7ac99dcc14af9044bd207705266d06a6d8
c
rec(label:="{16,4}*128a",order:=128,typ:=[16, 4],words:=["11", "
22", "33", "1313", "12321232", "23232323", "12121212121212121212
121212121212"]),
# https://www.abstract-polytopes.com/atlas/128/922/2.html sha256
=14afde20067e7c9dc86bd4a772593bf38028834a97404a92f71bf2cbb83c4ce
d
rec(label:="{16,4}*128b",order:=128,typ:=[16, 4],words:=["11", "
22", "33", "1313", "23232323", "13232121323212", "21213212121212
1213"]),
# https://www.abstract-polytopes.com/atlas/256/5090/2.html sha25
6=f2973f3db2e24e5666739bae1a3d2ce819fc6ec820b50ba042ed9d882a2705
58
rec(label:="{16,4}*256a",order:=256,typ:=[16, 4],words:=["11", "
22", "33", "1313", "23232323", "13232121323212", "12121212121212
121212121212121212"]),
# https://www.abstract-polytopes.com/atlas/256/5102/2.html sha25
6=128af9dd98fb01072c66c15eac3626111f77a5e048205af30f4820690bbca7
```

```
a7
rec(label:="{16,4}*256b",order:=256,typ:=[16, 4],words:=["11", "
22", "33", "1313", "23232323", "1212123212121232", "123212321232
1232", "21212312132132132121"]),
# https://www.abstract-polytopes.com/atlas/256/5298/2.html sha25
6=12667f4e841c06bc1d5de167d44bbb1715341b0fd1e6d5f56b49546cf32102
61
rec(label:="{16,8}*256a",order:=256,typ:=[16, 8],words:=["11", "
22", "33", "1313", "1231212312", "2323232323232323"]),
# https://www.abstract-polytopes.com/atlas/256/5302/2.html sha25
6=e00e7a2a8ed96a988890fc6f29c9814f9a5349e7890d4d09e4304c236208b1
8d
rec(label:="{16,8}*256b",order:=256,typ:=[16, 8],words:=["11", "
22", "33", "1313", "123231212312312", "1212321232121212", "232323
2323232323"]),
# https://www.abstract-polytopes.com/atlas/256/5304/3.html sha25
6=4743105a3e8e20e4c2b955c9e7911a126c5373dfac19d2320cdec344e043a8
b4
rec(label:="{16,8}*256c",order:=256,typ:=[16, 8],words:=["11", "
22", "33", "1313", "3123231232", "1212123212121232", "1212121212
121212121212121212121212"]),
# https://www.abstract-polytopes.com/atlas/256/5304/4.html sha25
6=5deb92ae5dbe5f671e64d42d32b92cfa265e743b216a0eb57e549915834dec
e9
rec(label:="{16,8}*256d",order:=256,typ:=[16, 8],words:=["11", "
22", "33", "1313", "12321232", "2323232323232323", "121212121212
121212121212121212121212"]),
# https://www.abstract-polytopes.com/atlas/256/5312/3.html sha25
6=6f43330ff2fff0e07165f2569c79d54863a6b240bd4d7015dfa2d749261cf1
f9
rec(label:="{16,8}*256e",order:=256,typ:=[16, 8],words:=["11", "
22", "33", "1313", "123231212312312", "1212123212121232", "232323
2323232323", "121212312121321321321"]),
# https://www.abstract-polytopes.com/atlas/256/5312/4.html sha25
6=102ea219f7655cc4799aa7cb9f9241d64ec47b43ed42cb11125072ebcff0c2
fa
rec(label:="{16,8}*256f",order:=256,typ:=[16, 8],words:=["11", "
22", "33", "1313", "123231212312312", "1212123212121232", "232323
2323232323", "121231212132121212"]),
# https://www.abstract-polytopes.com/atlas/256/6649/2.html sha25
6=a5fc58ebc37b74bce00039923b101bdbe470f26d3bea3f2e1633caf25e5dd8
c7
rec(label:="{32,4}*256a",order:=256,typ:=[32, 4],words:=["11", "
22", "33", "1313", "12321232", "23232323", "121212121212121212121
121212121212121212121212121212121212121212121212121212121212121
212121212121212121212121212121212121212121212121212121212121212
121212121212121212121212121212121212121212121212"]),
# https://www.abstract-polytopes.com/atlas/256/6655/2.html sha25
6=ba344f7162834a36afb89c517e1095b4c1167155691f799663b105fc3d463d
f2
rec(label:="{32,4}*256b",order:=256,typ:=[32, 4],words:=["11", "
22", "33", "1313", "23232323", "13232121323212", "12121212121212
121212121213212121212"]),
# https://www.abstract-polytopes.com/atlas/128/916/1.html sha256
```

```
=6575789b0dad2f7052958a92844a1e40811a8ea8c7e6063a537f400e797d396
a
rec(label:="{4,16}*128a",order:=128,typ:=[4, 16],words:=["11", "
22", "33", "1313", "12121212", "12321232", "23232323232323232323
232323232323"]),
# https://www.abstract-polytopes.com/atlas/128/922/1.html sha256
=31d8b42bd323b4508c6f01770cf3fef3135b245389b828b6e5a2af09856e41e
6
rec(label:="{4,16}*128b",order:=128,typ:=[4, 16],words:=["11", "
22", "33", "1313", "12121212", "23123212132321", "23232323231232
1323"]),
# https://www.abstract-polytopes.com/atlas/256/5090/1.html sha25
6=37b1aba9f879c0870525f62d5c0b421125db0803208f2eba372fbaa4463080
70
rec(label:="{4,16}*256a",order:=256,typ:=[4, 16],words:=["11", "
22", "33", "1313", "12121212", "23123212132321", "23232323232323
2323232323232323"]),
# https://www.abstract-polytopes.com/atlas/256/5102/1.html sha25
6=b497484ccb716e3f5bba6d8c12d900f6c82f8d194ffb5f35c812d260a6c179
76
rec(label:="{4,16}*256b",order:=256,typ:=[4, 16],words:=["11", "
22", "33", "1313", "12121212", "1232123212321232", "123232321232
3232", "32323232323123123121"]),
# https://www.abstract-polytopes.com/atlas/256/6649/1.html sha25
6=3ded1a80606ade68594e27a072f3126e0aaa7e144cfade829aea52ccdea5cd
ff
rec(label:="{4,32}*256a",order:=256,typ:=[4, 32],words:=["11", "
22", "33", "1313", "12121212", "12321232", "23232323232323232323
232323232323232323232323232323232323232323232323"]),
# https://www.abstract-polytopes.com/atlas/256/6655/1.html sha25
6=c9d7ce96ad5d81606c20e396fd33b97182cf567e9c54bd8ce573e7625573e4
d0
rec(label:="{4,32}*256b",order:=256,typ:=[4, 32],words:=["11", "
22", "33", "1313", "12121212", "23123212132321", "23232323232323
232323232312321323"]),
# https://www.abstract-polytopes.com/atlas/32/27/1.html sha256=2
26ed85be2487dec1482158e430263c1703dc5f7d0cbf4f2b42c15555a419917
rec(label:="{4,4}*32",order:=32,typ:=[4, 4],words:=["11", "22",
"33", "1313", "12121212", "12321232", "23232323"]),
# https://www.abstract-polytopes.com/atlas/64/138/1.html sha256=
721fd35e9c6c2fe05e072e385d34b96d81b982dd24b682fafec424ba0abdb7a9
rec(label:="{4,4}*64",order:=64,typ:=[4, 4],words:=["11", "22",
"33", "1313", "12121212", "23232323", "312312312312"]),
# https://www.abstract-polytopes.com/atlas/128/928/1.html sha256
=5617f7b523b391005e4a66081507f2200c05ce7efb11c863c922b0ccd41b612
4
rec(label:="{4,4}*128",order:=128,typ:=[4, 4],words:=["11", "22"
, "33", "1313", "12121212", "23232323", "1232123212321232"]),
# https://www.abstract-polytopes.com/atlas/256/6661/1.html sha25
6=f4b60a47adcd02659cb6c50d07b7e8b6f0aac5c89fad085705aa07494062bf
a1
rec(label:="{4,4}*256",order:=256,typ:=[4, 4],words:=["11", "22"
```

```
, "33", "1313", "12121212", "23232323", "31231231231231231231231
2"]),
# https://www.abstract-polytopes.com/atlas/64/128/1.html sha256=
77108260ce92b7a0404d306dc9364d333e2631dae3685d1a6c5019f5df39f0a4
rec(label:="{4,8}*64a",order:=64,typ:=[4, 8],words:=["11", "22",
 "33", "1313", "12121212", "12321232", "2323232323232323"]),
# https://www.abstract-polytopes.com/atlas/64/134/1.html sha256=
b94066b08eed5d66eb5ad7a45fb4172001c90020a35c5db0bed342a9a9c45540
rec(label:="{4,8}*64b",order:=64,typ:=[4, 8],words:=["11", "22",
 "33", "1313", "12121212", "3123231232"]),
# https://www.abstract-polytopes.com/atlas/128/327/1.html sha256
=970502a4caafbc0061c31ba75d59e3ac5b41b2cd368fd090557eb18bb06c886
7
rec(label:="{4,8}*128a",order:=128,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "23123212132321", "2323232323232323
"]),
# https://www.abstract-polytopes.com/atlas/128/928/2.html sha256
=ed51a0dd6a642ef5a81becd3a71b939014b736cd25a5fe225822906c1b8c716
3
rec(label:="{4,8}*128b",order:=128,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "312312312312", "1232123212321232"]
),
# https://www.abstract-polytopes.com/atlas/256/5078/1.html sha25
6=0cb05fa3ed7cd8c41bf9b102a465a74e246022d6741148586789ed63d22d4a
68
rec(label:="{4,8}*256a",order:=256,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "1232123212321232", "12323232123232
32", "2323232323232323"]),
# https://www.abstract-polytopes.com/atlas/256/6661/2.html sha25
6=63ddd954ed300765714181e7f3647a389f5f2a79b6fea9693790533e07dbc6
e9
rec(label:="{4,8}*256b",order:=256,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "312312312312", "2323232323232323"]
),
# https://www.abstract-polytopes.com/atlas/256/6665/1.html sha25
6=912e4da58975ece856295b1e2412e3e3cb0f1abddb3e0b2c0d84fb6793d564
5d
rec(label:="{4,8}*256c",order:=256,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "1232323212323232", "23232323232323
23", "2123212323212321323"]),
# https://www.abstract-polytopes.com/atlas/256/6665/2.html sha25
6=5303148e89fbfaacce9527c327be2b33abaf4583706c5880d62168da796a07
06
rec(label:="{4,8}*256d",order:=256,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "2323232323232323", "23121232321323
2321"]),
# https://www.abstract-polytopes.com/atlas/256/5298/1.html sha25
6=ef8854fa3725517ddfaa02d62d2736ffd67cd0ce490519ea4d3dbb2cade4e7
cc
rec(label:="{8,16}*256a",order:=256,typ:=[8, 16],words:=["11", "
22", "33", "1313", "3123231232", "1212121212121212"]),
# https://www.abstract-polytopes.com/atlas/256/5302/1.html sha25
```

```
6=8661ee43431710cc6aefa692b3ec91a21be818a1615f0bcfd599532621af02
73
rec(label:="{8,16}*256b",order:=256,typ:=[8, 16],words:=["11", "
22", "33", "1313", "12323121232312", "3232121232121232", "121212
1212121212"]),
# https://www.abstract-polytopes.com/atlas/256/5304/1.html sha25
6=6edea98497d1c6494e32ab14d8ace26db2416282f64b268331649f29a8d25b
b3
rec(label:="{8,16}*256c",order:=256,typ:=[8, 16],words:=["11", "
22", "33", "1313", "1231212312", "12121212121121212", "2323232323
2323232323232323232323"]),
# https://www.abstract-polytopes.com/atlas/256/5304/2.html sha25
6=430077132c84a2ccf8f84de03be5b3f29c0e5114715f3e7f231be8bbb1bf7c
4c
rec(label:="{8,16}*256d",order:=256,typ:=[8, 16],words:=["11", "
22", "33", "1313", "12321232", "12121212121121212", "232323232323
2323232323232323232323"]),
# https://www.abstract-polytopes.com/atlas/256/5312/1.html sha25
6=13554272549558a51c9a354dc1795650cb1a408cf82fd2e17b050de7a95c7d
9d
rec(label:="{8,16}*256e",order:=256,typ:=[8, 16],words:=["11", "
22", "33", "1313", "12323121232312", "12121212121121212", "121212
3212121232", "232312312312323123"]),
# https://www.abstract-polytopes.com/atlas/256/5312/2.html sha25
6=8a007116bdaaecf77719281c3e5851a9b84b673fa3cd5f268a6bfa04bcaf38
52
rec(label:="{8,16}*256f",order:=256,typ:=[8, 16],words:=["11", "
22", "33", "1313", "12323121232312", "12121212121121212", "121212
3212121232", "321232323232332132"]),
# https://www.abstract-polytopes.com/atlas/64/128/2.html sha256=
509ded6ad933dd1a54a042f1c09e8f71f3dc5e5a9e402ee20b3b18ccc2883734
rec(label:="{8,4}*64a",order:=64,typ:=[8, 4],words:=["11", "22",
 "33", "1313", "12321232", "23232323", "12121212121121212"]),
# https://www.abstract-polytopes.com/atlas/64/134/2.html sha256=
854acdb58b686b8dde68a01a9d7cfa9ea1d65ab400dd88d9de3f4807ff831b7f
rec(label:="{8,4}*64b",order:=64,typ:=[8, 4],words:=["11", "22",
 "33", "1313", "23232323", "1231212312"]),
# https://www.abstract-polytopes.com/atlas/128/327/2.html sha256
=d8dae6e472703dd52a96115f53adad40b3da0cd1632eb03ca0b046abc33f81d
1
rec(label:="{8,4}*128a",order:=128,typ:=[8, 4],words:=["11", "22
", "33", "1313", "23232323", "13232121323212", "12121212121121212
"]),
# https://www.abstract-polytopes.com/atlas/128/928/3.html sha256
=233f8cb33aee7120eda4d789888b904940d3e0299260c881cb78b3a1a88790e
d
rec(label:="{8,4}*128b",order:=128,typ:=[8, 4],words:=["11", "22
", "33", "1313", "23232323", "312312312312", "12121212121121212",
 "12121232121121232"]),
# https://www.abstract-polytopes.com/atlas/256/5078/2.html sha25
6=33ecd235667dcf98b5c56c35c2b4b2cd919026688d8d15944d92d3ec6ea1fb
76
```

```
rec(label:="{8,4}*256a",order:=256,typ:=[8, 4],words:=["11", "22
", "33", "1313", "23232323", "1212121212121212", "12121232121212
32", "1232123212321232"]),
# https://www.abstract-polytopes.com/atlas/256/6661/3.html sha25
6=592afee9ae5f06cc0c7af768bde8779f4f57c2d8ee43db9f187fdcd2d6ec6e
06
rec(label:="{8,4}*256b",order:=256,typ:=[8, 4],words:=["11", "22
", "33", "1313", "23232323", "312312312312", "1212121212121212"]
),
# https://www.abstract-polytopes.com/atlas/256/6665/3.html sha25
6=0ef16ce13d96eee9b01e8a528bf0e61b54ee738ce67d068c8e534ac5621a92
50
rec(label:="{8,4}*256c",order:=256,typ:=[8, 4],words:=["11", "22
", "33", "1313", "23232323", "1212121212121212", "12121232121212
32", "212132123212321213"]),
# https://www.abstract-polytopes.com/atlas/256/6665/4.html sha25
6=03ec97313d17c0b0f812dac3670b7c301eb9860d03bfeb5702bdfb120ff5a4
04
rec(label:="{8,4}*256d",order:=256,typ:=[8, 4],words:=["11", "22
", "33", "1313", "23232323", "1212121212121212", "12123231232121
2132"]),
# https://www.abstract-polytopes.com/atlas/128/351/1.html sha256
=9093e3087b2c8ce6dc31c2c27f5f5fc1ca0b6cd89d6779384f0cd58b05d48d0
9
rec(label:="{8,8}*128a",order:=128,typ:=[8, 8],words:=["11", "22
", "33", "1313", "1231212312", "1212121212121212", "232323232323
2323"]),
# https://www.abstract-polytopes.com/atlas/128/351/2.html sha256
=ce4c748a19b93e23922aefb13fc12d10fc09b54dc12cb699a4fd23ad7261f3b
b
rec(label:="{8,8}*128b",order:=128,typ:=[8, 8],words:=["11", "22
", "33", "1313", "12321232", "1212121212121212", "23232323232323
23"]),
# https://www.abstract-polytopes.com/atlas/128/351/3.html sha256
=25f2d45c51102a966553d5e68c0d476cf1c15cc8854a800b23c204ed0e67bcf
2
rec(label:="{8,8}*128c",order:=128,typ:=[8, 8],words:=["11", "22
", "33", "1313", "3123231232", "1212121212121212", "121212321212
1232"]),
# https://www.abstract-polytopes.com/atlas/128/387/1.html sha256
=bfaa09d371f973f383ee656cba4b756aec032d8287df9d2054ad1edcfda13de
9
rec(label:="{8,8}*128d",order:=128,typ:=[8, 8],words:=["11", "22
", "33", "1313", "12123232132132"]),
# https://www.abstract-polytopes.com/atlas/256/722/1.html sha256
=c2a84665c8de0bd59c1488cfd2e8d63aadcde7ed3f030535d1cdec8b282a9b4
e
rec(label:="{8,8}*256a",order:=256,typ:=[8, 8],words:=["11", "22
", "33", "1313", "12323121232312", "1212121212121212", "12121232
12121232", "2323232323232323"]),
# https://www.abstract-polytopes.com/atlas/256/5078/3.html sha25
6=0ce9f52877b6f9aeb536ddbdaaf8fecacb889ba0feff8689dbf2a9dfeeb09b
```

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74
rec(label:="{8,8}*256b",order:=256,typ:=[8, 8],words:=["11", "22
", "33", "1313", "312312312312", "1212121212121212", "1212123212
121232", "1232123212321232"]),
# https://www.abstract-polytopes.com/atlas/256/5084/1.html sha25
6=d522856bb0abd2b6689900197829fe0c2f0ee73c63acd0c1184857f69cedc0
b6
rec(label:="{8,8}*256c",order:=256,typ:=[8, 8],words:=["11", "22
", "33", "1313", "12323212132132", "1212121212121212", "12121232
12121232", "2323232323232323"]),
# https://www.abstract-polytopes.com/atlas/256/5084/2.html sha25
6=0d6eb5f80fcf1fe0e5d5c9b74138941ab9515338e39525be01909daf25f907
49
rec(label:="{8,8}*256d",order:=256,typ:=[8, 8],words:=["11", "22
", "33", "1313", "12323123123212", "1212121212121212", "12321232
12321232"]),
# https://www.abstract-polytopes.com/atlas/256/6665/5.html sha25
6=d806f95edac2ad28c0c4f3eba6a85e212c40983a119c1a742a4f2eed305101
06
rec(label:="{8,8}*256e",order:=256,typ:=[8, 8],words:=["11", "22
", "33", "1313", "312312312312", "1212121212121212", "1212123212
121232", "2323232323232323", "2121321232123212321213"]),
# https://www.abstract-polytopes.com/atlas/256/6665/6.html sha25
6=9608cf64536502e461675a5d051d9b051bbeda388b1add28e7f2e86ee224e1
e4
rec(label:="{8,8}*256f",order:=256,typ:=[8, 8],words:=["11", "22
", "33", "1313", "312312312312", "1212121212121212", "1212323212
123232", "1232323212323232", "1213212123212323212"]),
# https://www.abstract-polytopes.com/atlas/256/6669/1.html sha25
6=fb23aa0d50ac8258fe9b00b0c06d34dd89f69b78024ffe7361be86b6225dde
36
rec(label:="{8,8}*256g",order:=256,typ:=[8, 8],words:=["11", "22
", "33", "1313", "12323123123212", "1212121212121212", "23232323
23232323", "2312323212132123232"]),
# https://www.abstract-polytopes.com/atlas/256/6669/2.html sha25
6=ac242b8b3d59337a8720f9016349f204865ba4de430b946ce37868a6cf931c
0c
rec(label:="{8,8}*256h",order:=256,typ:=[8, 8],words:=["11", "22
", "33", "1313", "12323212132132", "1212121212121212", "23232323
23232323", "1321212321323212121"]),
# https://www.abstract-polytopes.com/atlas/128/1135/1.html sha25
6=285dc01b8d5cbdcf365adb217fa994e9efb1459dd1b8a22b233a5026d4a6ed
e8
rec(label:="{4,4,4}*128",order:=128,typ:=[4, 4, 4],words:=["11",
 "22", "33", "44", "1313", "1414", "2424", "12121212", "12321232
", "23232323", "23432343", "34343434"]),
# https://www.abstract-polytopes.com/atlas/256/16888/1.html sha2
56=6a3cedc26c36e029c1c80d03ffe6535b618db988a503279f27f0d2245cce9
1bc
rec(label:="{4,4,4}*256a",order:=256,typ:=[4, 4, 4],words:=["11"
, "22", "33", "44", "1313", "1414", "2424", "12121212", "1232123
2", "23232323", "34343434", "423423423423"]),
```

```
# https://www.abstract-polytopes.com/atlas/256/16888/2.html sha2
56=752c60d899773e711f4e65a4acf5e35c11ee8d0d083f36a9b4ac8564e3a31
605
rec(label:="{4,4,4}*256b",order:=256,typ:=[4, 4, 4],words:=["11"
, "22", "33", "44", "1313", "1414", "2424", "12121212", "2323232
3", "23432343", "34343434", "312312312312"]),
# https://www.abstract-polytopes.com/atlas/256/16836/1.html sha2
56=54306bcecafd7878e378e462deb55e6c379f491ab2d015e2c61cd9308828c
8f1
rec(label:="{4,4,8}*256a",order:=256,typ:=[4, 4, 8],words:=["11"
, "22", "33", "44", "1313", "1414", "2424", "12121212", "1232123
2", "23232323", "23432343", "34343434343434"]),
# https://www.abstract-polytopes.com/atlas/256/16864/1.html sha2
56=493fab45cc43ae4d632e1b41f2fed4b226a5ed5d9af819ca969385b0ecde0
28a
rec(label:="{4,4,8}*256b",order:=256,typ:=[4, 4, 8],words:=["11"
, "22", "33", "44", "1313", "1414", "2424", "12121212", "1232123
2", "23232323", "4234342343"]),
# https://www.abstract-polytopes.com/atlas/256/16885/1.html sha2
56=75bdbc71fbb33b76a9ded0cbc81f95b2b4dddab587bda987ae7fa48651b79
a90
rec(label:="{4,8,4}*256a",order:=256,typ:=[4, 8, 4],words:=["11"
, "22", "33", "44", "1313", "1414", "2424", "12121212", "3434343
4", "3123231232", "2342323423"]),
# https://www.abstract-polytopes.com/atlas/256/16894/1.html sha2
56=dc6945485b1af225af4698c12a5268e293ef9f1a702b44362c38a3c99f488
6f5
rec(label:="{4,8,4}*256b",order:=256,typ:=[4, 8, 4],words:=["11"
, "22", "33", "44", "1313", "1414", "2424", "12121212", "1232123
2", "34343434", "2342323423"]),
# https://www.abstract-polytopes.com/atlas/256/16894/2.html sha2
56=2f0772c9b0aa496076cfa6b4509dcbc275923ef20729e26aef559442d9a08
a36
rec(label:="{4,8,4}*256c",order:=256,typ:=[4, 8, 4],words:=["11"
, "22", "33", "44", "1313", "1414", "2424", "12121212", "2343234
3", "34343434", "3123231232"]),
# https://www.abstract-polytopes.com/atlas/256/16906/1.html sha2
56=6d7d45f6cd5eedb1e52a6fd3dd7003422fc627bf60769c8859dd98be3b53b
7fc
rec(label:="{4,8,4}*256d",order:=256,typ:=[4, 8, 4],words:=["11"
, "22", "33", "44", "1313", "1414", "2424", "12121212", "1232123
2", "23432343", "34343434", "2323232323232323"]),
# https://www.abstract-polytopes.com/atlas/256/16836/2.html sha2
56=96aab1dd903b3b3e04822d9e156ddecdad6f6a8d6442f739fb92c1bb30a1b
82d
rec(label:="{8,4,4}*256a",order:=256,typ:=[8, 4, 4],words:=["11"
, "22", "33", "44", "1313", "1414", "2424", "12321232", "2323232
3", "23432343", "34343434", "1212121212121212"]),
# https://www.abstract-polytopes.com/atlas/256/16864/2.html sha2
56=2b2dddf86816e09641d75f64c9dd7ea5b7415933b7876556c51e0eb92a428
c27
rec(label:="{8,4,4}*256b",order:=256,typ:=[8, 4, 4],words:=["11"
```

```
, "22", "33", "44", "1313", "1414", "2424", "23232323", "2343234
3", "34343434", "1231212312"]),
# https://www.abstract-polytopes.com/atlas/512/60809/1.html sha2
56=78d50057c869f47e122b5191495d50807507c38e092f4b9e3299f4eb6310e
23e
rec(label:="{4,4}*512",order:=512,typ:=[4, 4],words:=["11", "22"
, "33", "1313", "12121212", "23232323", "123212321232123212321232123
212321232"]),
# https://www.abstract-polytopes.com/atlas/512/58326/1.html sha2
56=30d77168174f8d730b82156ff65e01b5374c545f8181b7b6774f773f18cad
1c2
rec(label:="{4,8}*512a",order:=512,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "1232123212321232", "23232323232323
23", "31232323123123232312"]),
# https://www.abstract-polytopes.com/atlas/512/58338/1.html sha2
56=2862c62ea84ae95380f7f859d1fa10b8c752e5c5b192bc9c351e2c7e0685c
891
rec(label:="{4,8}*512b",order:=512,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "1232323212323232", "23232323232323
23", "31231231231231231231231231"]),
# https://www.abstract-polytopes.com/atlas/512/58338/2.html sha2
56=fd7c33c3acc540ebbfa59631f0cab485affbe2428e59ce7da8e38e76e2377
3eb
rec(label:="{4,8}*512c",order:=512,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "2323232323232323", "12323123212323
1232", "31232323123123232312"]),
# https://www.abstract-polytopes.com/atlas/512/60821/1.html sha2
56=6783aec8b02b2b7e05f05e3e25443915e56b5eec7e7ac7d0c0354651fec4a
f65
rec(label:="{4,8}*512d",order:=512,typ:=[4, 8],words:=["11", "22
", "33", "1313", "12121212", "1232323212323232", "23232323232323
23", "23123212132312312312312321"])
];;
# Table6: oriented labels, raw universal order (0 denotes the re
tained infinite row), class count.
# Exactly four printed right-hand {4,16} types are corrected to
their {16,4} duals.
rows78:=[
[["{4,4}*32", "{4,4}*256"], 1024, 1, [4, 4, 4]],
[["{4,4}*64", "{4,4}*64"], 1024, 1, [4, 4, 4]],
[["{4,4}*128", "{4,4}*64"], 0, 2, [4, 4, 4]],
[["{4,4}*32", "{4,8}*256a"], 1024, 1, [4, 4, 8]],
[["{4,4}*32", "{4,8}*256b"], 1024, 1, [4, 4, 8]],
[["{4,4}*32", "{4,8}*256c"], 1024, 1, [4, 4, 8]],
[["{4,4}*32", "{4,8}*256d"], 1024, 1, [4, 4, 8]],
[["{4,4}*64", "{4,8}*128a"], 2048, 2, [4, 4, 8]],
[["{4,4}*64", "{4,8}*128b"], 2048, 2, [4, 4, 8]],
[["{4,4}*128", "{4,8}*64a"], 1024, 1, [4, 4, 8]],
[["{4,4}*128", "{4,8}*64b"], 1024, 1, [4, 4, 8]],
[["{4,4}*32", "{4,16}*256a"], 1024, 1, [4, 4, 16]],
[["{4,4}*32", "{4,16}*256b"], 1024, 1, [4, 4, 16]],
[["{4,4}*64", "{4,16}*128a"], 1024, 1, [4, 4, 16]],
```

```
[["{4,4}*64", "{4,16}*128b"], 1024, 1, [4, 4, 16]],
[["{4,4}*32", "{4,32}*256a"], 1024, 1, [4, 4, 32]],
[["{4,4}*32", "{4,32}*256b"], 1024, 1, [4, 4, 32]],
[["{4,8}*64a", "{8,4}*256a"], 1024, 1, [4, 8, 4]],
[["{4,8}*64a", "{8,4}*256b"], 1024, 1, [4, 8, 4]],
[["{4,8}*64a", "{8,4}*256c"], 1024, 1, [4, 8, 4]],
[["{4,8}*64a", "{8,4}*256d"], 1024, 1, [4, 8, 4]],
[["{4,8}*64b", "{8,4}*256a"], 1024, 1, [4, 8, 4]],
[["{4,8}*64b", "{8,4}*256b"], 1024, 1, [4, 8, 4]],
[["{4,8}*64b", "{8,4}*256c"], 1024, 1, [4, 8, 4]],
[["{4,8}*64b", "{8,4}*256d"], 1024, 1, [4, 8, 4]],
[["{4,8}*128a", "{8,4}*128a"], 2048, 2, [4, 8, 4]],
[["{4,8}*128a", "{8,4}*128b"], 2048, 2, [4, 8, 4]],
[["{4,8}*128b", "{8,4}*128b"], 2048, 2, [4, 8, 4]],
[["{4,8}*64a", "{8,8}*256a"], 1024, 1, [4, 8, 8]],
[["{4,8}*64a", "{8,8}*256b"], 1024, 1, [4, 8, 8]],
[["{4,8}*64a", "{8,8}*256c"], 1024, 1, [4, 8, 8]],
[["{4,8}*64a", "{8,8}*256d"], 1024, 1, [4, 8, 8]],
[["{4,8}*64a", "{8,8}*256e"], 1024, 1, [4, 8, 8]],
[["{4,8}*64a", "{8,8}*256f"], 1024, 1, [4, 8, 8]],
[["{4,8}*64a", "{8,8}*256g"], 1024, 1, [4, 8, 8]],
[["{4,8}*64a", "{8,8}*256h"], 1024, 1, [4, 8, 8]],
[["{4,8}*64b", "{8,8}*256a"], 1024, 1, [4, 8, 8]],
[["{4,8}*64b", "{8,8}*256b"], 1024, 1, [4, 8, 8]],
[["{4,8}*64b", "{8,8}*256c"], 1024, 1, [4, 8, 8]],
[["{4,8}*64b", "{8,8}*256d"], 1024, 1, [4, 8, 8]],
[["{4,8}*64b", "{8,8}*256e"], 1024, 1, [4, 8, 8]],
[["{4,8}*64b", "{8,8}*256f"], 1024, 1, [4, 8, 8]],
[["{4,8}*64b", "{8,8}*256g"], 1024, 1, [4, 8, 8]],
[["{4,8}*64b", "{8,8}*256h"], 1024, 1, [4, 8, 8]],
[["{4,8}*128a", "{8,8}*128a"], 1024, 1, [4, 8, 8]],
[["{4,8}*128a", "{8,8}*128b"], 1024, 1, [4, 8, 8]],
[["{4,8}*128a", "{8,8}*128c"], 1024, 1, [4, 8, 8]],
[["{4,8}*128a", "{8,8}*128d"], 1024, 1, [4, 8, 8]],
[["{4,8}*128b", "{8,8}*128a"], 1024, 1, [4, 8, 8]],
[["{4,8}*128b", "{8,8}*128b"], 1024, 1, [4, 8, 8]],
[["{4,8}*128b", "{8,8}*128c"], 1024, 1, [4, 8, 8]],
[["{4,8}*128b", "{8,8}*128d"], 1024, 1, [4, 8, 8]],
[["{4,8}*64a", "{8,16}*256a"], 1024, 1, [4, 8, 16]],
[["{4,8}*64a", "{8,16}*256b"], 1024, 1, [4, 8, 16]],
[["{4,8}*64a", "{8,16}*256c"], 1024, 1, [4, 8, 16]],
[["{4,8}*64a", "{8,16}*256d"], 1024, 1, [4, 8, 16]],
[["{4,8}*64a", "{8,16}*256e"], 1024, 1, [4, 8, 16]],
[["{4,8}*64a", "{8,16}*256f"], 1024, 1, [4, 8, 16]],
[["{4,8}*64b", "{8,16}*256c"], 1024, 1, [4, 8, 16]],
[["{4,8}*64b", "{8,16}*256d"], 1024, 1, [4, 8, 16]],
[["{4,8}*64b", "{8,16}*256e"], 1024, 1, [4, 8, 16]],
[["{4,8}*64b", "{8,16}*256f"], 1024, 1, [4, 8, 16]],
[["{4,16}*128a", "{16,4}*256a"], 1024, 1, [4, 16, 4]],
[["{4,16}*128a", "{16,4}*256b"], 1024, 1, [4, 16, 4]],
[["{4,16}*128b", "{16,4}*256a"], 1024, 1, [4, 16, 4]],
[["{4,16}*128b", "{16,4}*256b"], 1024, 1, [4, 16, 4]],
```

```
[["{4,16}*128a", "{16,8}*256a"], 1024, 1, [4, 16, 8]],
[["{4,16}*128a", "{16,8}*256b"], 1024, 1, [4, 16, 8]],
[["{4,16}*128a", "{16,8}*256c"], 1024, 1, [4, 16, 8]],
[["{4,16}*128a", "{16,8}*256d"], 1024, 1, [4, 16, 8]],
[["{4,16}*128a", "{16,8}*256e"], 1024, 1, [4, 16, 8]],
[["{4,16}*128a", "{16,8}*256f"], 1024, 1, [4, 16, 8]],
[["{4,16}*128b", "{16,8}*256a"], 1024, 1, [4, 16, 8]],
[["{4,16}*128b", "{16,8}*256b"], 1024, 1, [4, 16, 8]],
[["{4,16}*128b", "{16,8}*256c"], 1024, 1, [4, 16, 8]],
[["{4,16}*128b", "{16,8}*256d"], 1024, 1, [4, 16, 8]],
[["{4,16}*128b", "{16,8}*256e"], 1024, 1, [4, 16, 8]],
[["{4,16}*128b", "{16,8}*256f"], 1024, 1, [4, 16, 8]],
[["{4,32}*256a", "{32,4}*256a"], 1024, 1, [4, 32, 4]],
[["{4,32}*256a", "{32,4}*256b"], 1024, 1, [4, 32, 4]],
[["{4,32}*256b", "{32,4}*256b"], 1024, 1, [4, 32, 4]],
[["{8,4}*64a", "{4,8}*128a"], 1024, 1, [8, 4, 8]],
[["{8,4}*64a", "{4,8}*128b"], 1024, 1, [8, 4, 8]],
[["{8,4}*64b", "{4,8}*128a"], 1024, 1, [8, 4, 8]],
[["{8,4}*64b", "{4,8}*128b"], 1024, 1, [8, 4, 8]],
[["{8,4}*64a", "{4,16}*128a"], 1024, 1, [8, 4, 16]],
[["{8,4}*64a", "{4,16}*128b"], 1024, 1, [8, 4, 16]],
[["{8,4}*64b", "{4,16}*128a"], 1024, 1, [8, 4, 16]],
[["{8,4}*64b", "{4,16}*128b"], 1024, 1, [8, 4, 16]],
[["{8,8}*128a", "{8,8}*128a"], 1024, 1, [8, 8, 8]],
[["{8,8}*128a", "{8,8}*128b"], 1024, 1, [8, 8, 8]],
[["{8,8}*128a", "{8,8}*128c"], 1024, 1, [8, 8, 8]],
[["{8,8}*128a", "{8,8}*128d"], 1024, 1, [8, 8, 8]],
[["{8,8}*128b", "{8,8}*128a"], 1024, 1, [8, 8, 8]],
[["{8,8}*128b", "{8,8}*128b"], 1024, 1, [8, 8, 8]],
[["{8,8}*128b", "{8,8}*128d"], 1024, 1, [8, 8, 8]],
[["{8,8}*128c", "{8,8}*128a"], 1024, 1, [8, 8, 8]],
[["{8,8}*128c", "{8,8}*128d"], 1024, 1, [8, 8, 8]],
[["{8,8}*128d", "{8,8}*128d"], 1024, 1, [8, 8, 8]],
[["{4,4,4}*128", "{4,4,4}*256a"], 1024, 1, [4, 4, 4, 4]],
[["{4,4,4}*256a", "{4,4,4}*256b"], 1024, 1, [4, 4, 4, 4]],
[["{4,4,4}*128", "{4,4,8}*256a"], 1024, 1, [4, 4, 4, 8]],
[["{4,4,4}*128", "{4,4,8}*256b"], 1024, 1, [4, 4, 4, 8]],
[["{4,4,8}*256a", "{4,8,4}*256b"], 1024, 1, [4, 4, 8, 4]],
[["{4,4,8}*256a", "{4,8,4}*256d"], 1024, 1, [4, 4, 8, 4]],
[["{4,4,8}*256b", "{4,8,4}*256a"], 1024, 1, [4, 4, 8, 4]],
[["{4,4,8}*256b", "{4,8,4}*256c"], 1024, 1, [4, 4, 8, 4]]
];;
```

phase78_n009_integration_run.py

```
#!/usr/bin/env python3
"""Fresh exhaustive kernel census and complete-presentation reco
gnition.

Inputs are exactly three printed GAP/data programs; no saved cen
sus, private
primary text, Atlas permutation array or SmallGroups catalogue i
```

```
s read.
The output directory must not exist. GAP/NQ remain trusted exact
 algorithms.
"""
import argparse
import hashlib
import json
from pathlib import Path
import re
import shutil
import subprocess
import time

ROOT = Path(__file__).resolve().parent
INPUTS = ["phase78_n009_integration_census.g", "phase78_n009_int
egration_finite.g", "phase78_n009_integration_data.g"]
CASES = [(3, 3, 61, 37, 3, 20, 3904, [7, 23, 61]), (3, 5, 401, 2
35, 15, 125, 25664, [7, 23, 61, 147, 401]), (4, 3, 1483, 199, 11
, 105, 379648, [31, 267, 1483]), (5, 1, 127, 15, 1, 8, 130048, [
127])]
FINAL = "PASS complete_presentations=214 actual_IP_meets=43840 a
tlas=72 raw_examples=34 exceptional=2 finite_universals=106 exac
t_census_facets=428 table6_rows=107 table5_rows=12"


def require(condition, message):
    if not condition:
        raise RuntimeError(message)


def sha(path):
    return hashlib.sha256(path.read_bytes()).hexdigest()


def main():
    ap = argparse.ArgumentParser(description=__doc__)
    ap.add_argument("--gap", default="gap")
    ap.add_argument("--output-dir", type=Path, required=True)
    ap.add_argument("--timeout", type=int, default=1200)
    args = ap.parse_args()
    out = args.output_dir.resolve()
    require(not out.exists(), "Refusing existing output director
y")
    bindings = [{"path": n, "sha256": sha(ROOT / n)} for n in IN
PUTS]
    out.mkdir(parents=True)
    start = time.monotonic()
    for n in INPUTS:
        shutil.copy2(ROOT / n, out / n)
    timings = {}
    for name, program in [("census", INPUTS[0]), ("finite", INPU
TS[1])]:
```

```
        t = time.monotonic()
        run = subprocess.run([args.gap, "-q", "-T", program], cw
d=out, text=True, stdout=subprocess.PIPE, stderr=subprocess.PIPE
, timeout=args.timeout, check=False)
        (out / (name + ".log")).write_text(run.stdout)
        (out / (name + ".err")).write_text(run.stderr)
        require(run.returncode == 0 and not run.stderr.strip(),
f"{name}: execution error or warning")
        require(not re.search(r"(^|\n)(Error,|Syntax error:|brk>
)", run.stdout), f"{name}: GAP failure")
        timings[name] = round(time.monotonic() - t, 6)
    census = (out / "census.log").read_text()
    finite = (out / "finite.log").read_text()
    require(census.count("PASS: complete invariant-kernel ladder
") == 4 and FINAL in finite, "Incomplete terminal markers")
    observed = re.findall(r"RESULT rank=(\d+) excess=(\d+) candi
dates=(\d+) ordered=(\d+) selfdual=(\d+) up_to_reversal=(\d+) ac
tual_meets=(\d+)", census)
    require([tuple(map(int, r)) for r in observed] == [x[:7] for
 x in CASES], "Census totals differ")
    for d, k, candidates, ordered, fixed, classes, meets, levels
 in CASES:
        for j, n in enumerate(levels, 1):
            data = json.loads((out / f"phase78_d{d}_k{k}.level{j
}.json").read_text())
            require(len(data) == n and len({tuple(x) for x in da
ta}) == n, "Ladder cardinality or duplicate key")
        accepted = json.loads((out / f"phase78_d{d}_k{k}.accepte
d.json").read_text())
        require(len(accepted) == ordered and sum(x[4] for x in a
ccepted) == fixed, "Accepted cardinality/fixed count")
    seeds = json.loads((out / "phase78_d3_k3.seed.json").read_te
xt())
    require(len(seeds) == 4 and sorted(x[1] for x in seeds) == [
1, 2, 3, 4], "All four distinct seed rows")
    read = lambda suffix: json.loads((out / ("phase78_n009_integ
ration_" + suffix + ".json")).read_text())
    require(len(read("all_presentations")) == 214, "Complete pre
sentation count")
    require(read("table4_literal_matches") == [[1, 1], [2, 2], [
3, 3], [4, 4], [5, 3], [6, 1], [7, 7], [8, 3]], "Literal correct
ion map")
    require(len(read("table3_matches")) == 12, "Table3 cells")
    require(len(read("facet_matches")) == 214 and len(read("tabl
e6_results")) == 107, "Facet/table6 rows")
    require(len(read("universal_orders")) == 106, "Finite univer
sals")
    require(len(read("census_matches")) == 8 and sorted(x[1] for
 x in read("census_matches")) == list(range(1, 9)), "Eight named
 source targets")
    versions = re.findall(r"VERSION GAP=(\S+) NQ=(\S+)", census)
    require(len(versions) == 4 and len(set(versions)) == 1, "Sof
```

```
tware identity")
    for row in bindings:
        require(sha(ROOT / row["path"]) == row["sha256"] == sha(
out / row["path"]), "Mathematical input changed during replay")
    files = [{"path": p.name, "sha256": sha(p)} for p in sorted(
out.iterdir()) if p.is_file()]
    report = dict(status="PASS", software=dict(zip(["GAP", "NQ"]
, versions[0])), elapsed_seconds=round(time.monotonic()-start, 6
), timings=timings,
        cases=[dict(zip(["rank", "excess", "terminal_kernels", "
ordered", "selfdual", "classes", "actual_meets", "ladder_sizes"]
, c)) for c in CASES],
        total_census_meets=539264, complete_presentations=214, p
resentation_IP_meets=43840,
        source_table6_rows=107, source_table5_rows=12, named_cen
sus_facet_isomorphisms=428,
        source_statement_scope="Corrected principal contract wit
h explicitly retained geometric/square/infinite inputs; no histo
rical novelty or Lean verification.",
        candidate_generation_uses_source_totals=False, SmallGrou
ps_dependency=False, old_census_input=False, private_primary_inp
ut=False,
        inputs=bindings, runner=dict(path=Path(__file__).name, s
ha256=sha(Path(__file__))), outputs=files)
    (out / "phase78_n009_integration_run.json").write_text(json.
dumps(report, indent=2)+"\n")
    print(json.dumps({k: report[k] for k in ["status", "software
", "elapsed_seconds", "total_census_meets", "complete_presentati
ons", "presentation_IP_meets"]}))


if __name__ == "__main__":
    main()
```